

Uncovering Insights of Compound Flooding with Data-Driven AI

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Abstract

Compound flooding, driven by nonlinear interactions between multiple hydrometeorological factors, poses a significant challenge to hazard prevention. Existing forecasting approaches, whether physics-based or data-driven, often emphasize temporal patterns while underexploring how multiple interacting factors jointly shape flood dynamics. To address this problem, we conduct a large-scale data-driven analysis of compound flooding in South Florida, a typical area for compound flooding, by integrating tidal conditions, rainfall, groundwater stage, and human water management activities. Our analysis reveals three key findings: (i) models that capture temporal dynamics alone fail to represent multi-factor interactions during compound events; (ii) subsurface saturation, as reflected by groundwater levels, emerges as a dominant predictor of flood severity, often outweighing immediate rainfall intensity in this porous coastal region; and (iii) the spatial state of surrounding monitoring stations within a finite effective radius provides critical causal context for flooding, while extending temporal history yields diminishing returns during extreme events. These findings suggest that compound flooding is governed more by spatially coupled system states than by long-term temporal dependencies, challenging rain-centric and sequence-dominated forecasting paradigms. By framing data-driven models as tools for scientific inquiry rather than prediction alone, this study offers new insights into the mechanisms of compound flooding and informs the design of more physically grounded early-warning systems for coastal environments. Our dataset and code are publicly available at <https://github.com/AslanDing/SFBench>.

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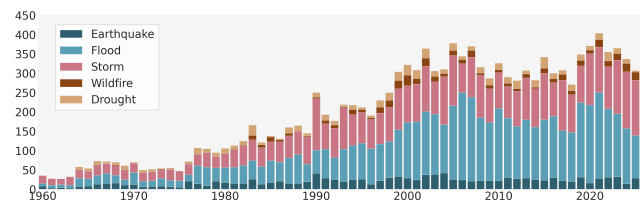


Figure 1: Global historical statistics for 5 types of natural disaster events. Data source: EM-DAT [17]. Flood is the most common disaster.

Keywords

Compound Flood, Machine Learning, Data-driven Modeling

1 Introduction

Floods are among the most frequent and destructive natural hazards, causing severe environmental damage [78], catastrophic loss of life [30], and economic disruption [8], as shown in Figure 1. Compared with single-driver flood events, such as stream overflows or flash floods [21], compound floods arise from the concurrence of multiple interacting drivers across meteorological, hydrological, and oceanographic processes [60]. According to previous research [6], compound flooding is projected to increase globally by over 25% by 2100 relative to present under a high-emissions scenario. These events are particularly difficult to predict, as nonlinear interactions among drivers can amplify flood severity beyond what any single factor would suggest. Recent studies indicate that both the frequency and spatial extent of compound floods are increasing under climate change [25, 63, 69], underscoring the urgency of understanding their underlying mechanisms.

Despite their importance, the causal structure of compound flooding remains poorly understood. Classical physics-based methods simulate flood dynamics by solving partial differential equations that encode hydrodynamic processes [52, 79], such as the Hydrologic Engineering Center's River Analysis System (HEC-RAS) [9]. Despite their accuracy and explainability, their application is constrained by extensive data requirements, including high-resolution terrain data, canal networks, and river geometries [59, 81]. As a

result, physics-based models are often difficult to deploy consistently across regions, limiting their utility for large-scale comparative analysis of compound flood drivers. In parallel, data-driven approaches based on machine learning have emerged as an alternative for flood forecasting, leveraging advances in deep learning to model complex, nonlinear relationships directly from observations. Prior studies have applied Convolutional Neural Networks (CNN) [38], Recurrent Neural Networks (RNN) [26], Graph Neural Networks (GNN) [32], and Transformers [62] to capture temporal and spatial dependencies in flood processes. However, most existing models emphasize a single driving force and treat the compound flooding forecasting task as a temporal analysis task, overlooking the interaction of various factors. For example, Google researchers use Long Short-Term Memory networks (LSTMs) [26] to forecast extreme floods with watershed streamflow data [48]. In [36], the authors build a model to capture the relationship between rainfall and discharge of catchments. These methods lead to oversimplified scientific insights, compressing the complex interplay among multiple driving factors into a sequential dependency.

A fundamental obstacle to resolving these questions is the lack of integrated observational frameworks that support systematic, multi-factor analysis of compound flooding. Existing datasets [31, 35, 57, 61] are often limited in geographic scope or omit key drivers such as groundwater dynamics and human water management operations. Consequently, previous studies have largely examined compound floods through isolated case studies or restricted regional settings [74], making it difficult to unravel the relative importance of interacting factors or to assess how spatial context influences the causality of floods.

In this study, to bridge the gap between existing methods and real-world scenarios, we leverage data-driven models not merely as forecasting tools but as analytical probes to investigate the mechanisms underlying compound flooding. We select South Florida as a typical area for the analysis, a low-lying coastal region that serves as a natural laboratory for compound flood analysis due to its porous geology, dense water management infrastructure, and frequent exposure to hurricanes and extreme precipitation [27]. Flooding in this region is shaped by the interaction of multiple drivers, including rainfall, tide, groundwater saturation, and human-controlled hydraulic structures, providing a uniquely rich setting for studying multi-factor flood dynamics, as shown in Figure 2. To enable this analysis, we assemble a comprehensive observational dataset spanning 1985 to 2024, integrating water stage, tide, subsurface water(groundwater) stage, rainfall, and operational data from gates and pumps across 2,452 monitoring stations. Rather than treating this dataset as a benchmark alone, we use it to support controlled analyses of how different modeling paradigms respond to variations in temporal history, spatial context, and driver availability.

With a diverse set of learning architectures, including Multilayer Perceptrons (MLPs), RNNs, CNNs, GNNs, Transformers, and Large Language Models (LLM)-based approaches, we systematically examine model behavior under feature ablation, spatial perturbation, and extreme-event evaluation. The analysis yields three central findings: (i) models that depend on temporal patterns struggle to represent critical multi-factor interactions during compound flood events; (ii) subsurface saturation consistently emerges as a dominant driver

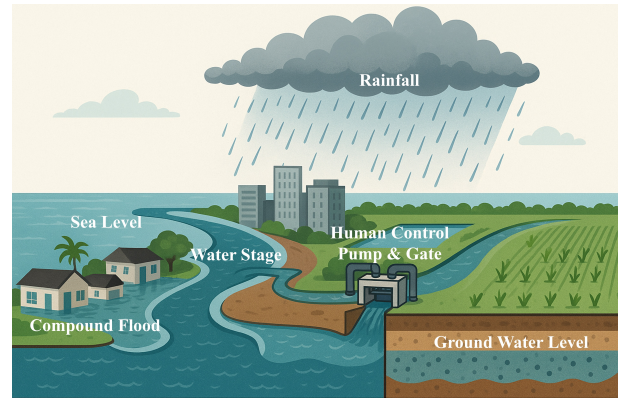


Figure 2: Schematic diagram of multiple factors.

of flood severity in South Florida, frequently exceeding the influence of short-term rainfall; and (iii) spatially localized system states within a finite effective radius provide essential causal context for flooding, while extending temporal history offers limited benefit during extremes.

While demonstrated in South Florida, these results reveal fundamental insights that compound flood dynamics emerge not from isolated temporal sequences but from the spatial entanglement of multi-hazard drivers. This work reframes data-driven models as generalizable tools for scientific discovery, offering a universal blueprint for understanding compound flood mechanisms and developing flood forecasting systems, especially for coastal regions.

2 Related Work

Flood Datasets. Monitoring floods presents a significant challenge due to unpredictable nature and potentially devastating consequences. Existing flood datasets can be broadly categorized into satellite image datasets [7, 18, 47, 53, 56, 75] and time series monitoring datasets [2, 12, 19, 31, 35, 57, 74]. Satellite image datasets utilize remote sensing to capture the extent of surface water [4]. While effective in delineating flood-affected areas, this type of dataset [7, 75] often lacks temporal dynamics and information on the underlying hydrological and meteorological factors driving flood formation. Time series monitoring datasets typically utilize fixed monitoring stations to record hydrological-related data such as soil moisture and temperature. A prominent example within this category is the CAMELS-x family of datasets [1, 3, 12, 16, 19]. For instance, CAMELS-BR [12] encompasses data from 3,679 gauges across Brazil. LamaH-CE [35] provides daily and hourly time series data from 882 gauges, including runoff and catchment attributes. In [34], LamaH-CE is used as a benchmark for flood forecasting, primarily focusing on water stream data across temporal and spatial aspects. In summary, these datasets primarily focus on general hydrological modeling. Analyzing compound floods, as highlighted in [27], necessitates detailed data on rainfall, water levels, and groundwater, which are often limited in existing time series datasets.

Data-Driven Methods for Forecasting. High computational costs often constrain traditional physical models [9, 52, 79]. In contrast, data-driven AI excels at extracting complex, non-linear patterns directly from high-dimensional data, providing a scalable framework that can efficiently decode the spatial entanglement of multi-hazard drivers across diverse regions. Deep learning methodologies

have shown promise in this domain. Based on their core architectural designs, these methods applied to time series forecasting can be categorized as follows: *MLP-based models* [14, 39, 40, 67, 82] leverage the capabilities of multilayer perceptrons for analyzing temporal sequences. *RNN-based methods* [28, 37, 54, 58, 65] are widely adopted in forecasting due to their inherent ability to model temporal dependencies within sequential data. *CNN-based methods* [15, 46, 64, 66, 70] employ convolutional operations to extract hierarchical features from time series data, enabling effective learning of underlying patterns and trends. *GNN-based methods* [10, 11, 43, 72, 73, 77] utilize graph structures to model intricate relationships between different time series variables, enhancing forecasting accuracy. *Transformer-based methods* [41, 42, 49, 68, 71, 83, 85] have demonstrated remarkable performance in capturing long-range dependencies and complex temporal dynamics within time series data. *LLM-based methods* [22, 29, 51, 86] explore the application of prompting and reprogramming techniques to align time series data with text embeddings for forecasting tasks.

3 Methods

To enable systematic analysis of these interactions, we first construct an observational data framework, SF²Bench, that integrates heterogeneous data sources with explicit physical meaning into a coherent spatiotemporal representation. Then, we perform spatiotemporal alignment to construct a coherent representation. After that, we apply data-driven AI analysis to reveal the underlying dependencies among various processes.

3.1 Data Sources and Physical Interpretation

SF²Bench is an observational time series dataset that characterizes the evolving state of a managed coastal hydrological system in South Florida. It aggregates measurements from 2,452 monitoring stations operated by the South Florida Water Management District (SFWMD)¹, covering a spatial extent of 67,349 km² and a temporal range from 1985 to 2024. Because station availability varies over time, the dataset is organized into eight temporal splits that reflect the effective coverage period of individual monitoring locations.

The dataset integrates multiple physical and anthropogenic variables that jointly govern the emergence of compound flooding events [27]. These variables include surface water stage, rainfall, groundwater stage, coastal water levels, and records of human-controlled hydraulic operations. Together, they define the instantaneous state of the regional flood system by capturing external forcings, boundary conditions, internal storage, and active system controls. Rainfall serves as the primary external forcing driving surface runoff and water accumulation. Groundwater stage represents antecedent subsurface storage and reflects the land's capacity to absorb incoming precipitation, which critically modulates flood severity during extreme rainfall events. Coastal water levels, including tidal influences, act as dynamic boundary conditions for inland drainage. At several monitoring stations, these coastal effects are inherently embedded in the observed water stage due to direct hydraulic connectivity with the ocean. In addition to natural drivers, SF²Bench explicitly incorporates records of human-controlled hydraulic operations, including pumps and gates. Pumps enable

water transfer from lower to higher elevations, while gates regulate downstream flow from higher to lower elevations. In South Florida, where an extensive network of canals, lakes, storage areas, and control structures actively manages water distribution, such operational interventions directly alter system dynamics and flood propagation pathways. Representing these controls is therefore essential for capturing the coupled human–natural processes underlying compound flood behavior.

A comparison between SF²Bench and existing hydrological datasets is provided in Table 1. Compared to CAMELS-x [1, 3, 12, 16, 19], which emphasizes large-sample hydrology across diverse regions, SF²Bench focuses on a flood-prone coastal area where compound flooding is a dominant hazard. In contrast to BangladeshFlood [57], SF²Bench incorporates a broader set of interacting drivers, capturing both temporal evolution and spatial heterogeneity relevant to compound flood processes.

3.2 Spatiotemporal Alignment

To facilitate the construction of AI-ready datasets, we provide detailed information on aligning heterogeneous observational records.

The Observational Network. The raw observations underlying SF²Bench are sourced from DBHYDRO², an environmental data repository maintained by SFWMD. The raw data stream, originally encompassing 3,731 sensors, underwent a rigorous spatiotemporal density filtration protocol. We prioritized stations with long-term data continuity and high relevance to cross-boundary hydraulic interactions, resulting in a finalized network of 2,452 high-fidelity nodes. This multi-modal array comprises water stage ($n = 993$), precipitation ($n = 349$), and groundwater levels ($n = 582$), integrated with the active control signals of 99 pump stations and 429 hydraulic gates. By synthesizing these disparate sensing modalities, the repository provides a comprehensive digital pulse of the region's hydrological state. Each station provides localized observations of distinct system components. Together, these measurements form a distributed sensing network that captures external forcings, internal storage states, boundary conditions, and human-controlled interventions within the regional flood system. In our framework, water stage, rather than streamflow, is used here because streamflow approaches zero and is hard to measure in many catchments, such as lakes and wetlands. A summary of data types and their physical interpretations is provided in Table 2.

Temporal Alignment. Raw observations in the DBHYDRO are recorded at native breakpoint frequencies, which vary substantially across stations and variables, ranging from sub-hourly to irregular event-based recordings. While this high-resolution logging preserves fine-scale dynamics, it results in heterogeneous and asynchronous time series that are not directly comparable across space or variables. Moreover, observational gaps and uneven station lifespans introduce additional temporal inconsistency. To construct a unified system-level representation, we align all observations to an hourly time grid. For each station and variable, measurements within each hourly interval are aggregated using their mean value, yielding a temporally consistent snapshot. This hourly resolution balances the need to capture flood-relevant dynamics with the requirement for long-term temporal coverage across stations. Due to

¹<https://www.sfwmd.gov/>

²<https://apps.sfwmd.gov/dbhydroInsights/%23/homepage>

Table 1: Comparison with other datasets for flood forecasting. N/A indicates Not Available. * marks datasets with Other Attributions. Climatic indices capture climate statistics (e.g., evapotranspiration). Land cover attributes describe surface materials. Soil attributes include properties such as porosity, while geological attributes refer to subsurface features. Anthropogenic influences cover human activities (e.g., discharges). Other catchment attributes include location, area, and topography.

Dataset	Time Span	Interval	Type	Gauges	Area(km ²)	Public	Other Attributions
DarlingFlood	1900-2018	Daily	Flow	12	3.5×10^4	No	Rainfall
SekongFlood	1981-2013	Daily	Flow	8	2.8×10^4	No	Rainfall
BangladeshFlood	1979-2013	Daily	Stage	24	1.5×10^5	No	N/A
Qi River	1979-2020	Hour	Flow	7	7.1×10^3	No	Rainfall
Tunxi basins	1981-2007	Hour	Flow	12	N/A	No	Rainfall
CAMELS*	1989-2009	Daily	Flow	671	1.0×10^4	Yes	Climatic Indices
CAMELS-CL*	1913-2018	Daily	Flow	516	N/A	Yes	Land Cover Attributes
CAMELS-GB*	1970-2015	Daily	Flow	671	2.1×10^5	Yes	Soil Attributes
CAMELS-BR*	1925-2024	Daily	Flow	4,025	N/A	Yes	Geological Attributes
CAMELS-AUS*	1951-2014	Daily	Flow	107	6.9×10^5	Yes	Anthropogenic Influences
LamaH-CE*	1951-2014	Daily & Hour	Flow	859	1.7×10^5	Yes	Other Catchment Attributes
SF ² Bench	1985-2024	Hour	Stage	2,452	6.7×10^4	Yes	Rainfall, Groundwater, Human Control

heterogeneous data availability, the aligned time series are organized into eight contiguous temporal splits³, each corresponding to a contiguous period with sufficient station coverage.

Handling Missing Observations. Temporal alignment inevitably introduces missing values where observations are unavailable. To ensure the integrity of the physical signals during this transition, we applied differentiated interpolation strategies based on the variables' nature: continuous field and event-based sparse representation. Surface water stage and groundwater level are treated as continuous state variables that evolve smoothly over time, and missing values are filled using linear interpolation. In contrast, rainfall and hydraulic control records (pumps and gates) represent stochastic external events; missing entries are therefore interpreted as the absence of events and filled with zero.

3.3 Data-Driven AI Analysis

We investigate the extent to which data-driven models can capture the coupled dynamics of a human-regulated hydrological system using large-scale observational data. By reframing water stage forecasting from a purely predictive task into a system identification problem, we aim to reveal the underlying spatiotemporal dependencies linking surface water dynamics, subsurface processes, meteorological forcing, and human interventions.

Scientific Objective. We examine whether AI models can learn hydrological dynamics and forecast water stages from multi-factor observational data. Specifically, we aim to understand how surface water levels evolve in response to meteorological forcing, subsurface groundwater interactions, and human-operated hydraulic controls, such as pumps and gates.

We formulate the hydrological system as a spatio-temporal coupled network driven by both natural processes and human control signals. Given water stage time series data $X \in \mathbb{R}^{N \times L}$ from N water monitoring stations, the forecasting objective is to predict the water stage values for the next T time steps, denoted as $Y \in \mathbb{R}^{N \times T}$, using a fixed look-back window of length L . In addition to the surface water stage, the system also has access to additional time

Table 2: The summary of the data type in SF²Bench.

Type	Stations	Unit	Description
Water	993	Feet	Water Stage
Groundwater	582	Feet	Stage of Groundwater
Rainfall	349	Inches	Rainfall
Pump	99	RPM	Rotational Speed
Gate	429	Feet	Opening Level

series information, including groundwater levels $X_w \in \mathbb{R}^{N_w \times L}$ (from N_w stations), rainfall $X_r \in \mathbb{R}^{N_r \times L}$ (from N_r stations), pump control data $X_p \in \mathbb{R}^{N_p \times L}$ (from N_p stations), gate control data $X_g \in \mathbb{R}^{N_g \times L}$ (from N_g stations), and location information for the monitoring stations. From the data-driven model perspective, this formulation treats water stage evolution as the outcome of interacting latent processes operating across space and time. For the primary benchmark experiments aimed at fair comparison across different modeling approaches, we focus on water stage forecasting as the supervised prediction target. The additional modalities are regarded as external drivers that can potentially enrich system understanding and improve predictive performance, and their integration is explored in extended experimental settings.

Data-Driven Models. To cover a broad range of inductive biases, we benchmark six categories of time series forecasting architectures, including MLP, CNN, RNN, GNN, Transformer, and LLM-based models. We select two advanced methods as representative examples for each of these categories. Additionally, for the first four classical architectures, we also implement a basic foundational architecture. The specific advanced methods we evaluate are: 1. MLP-Based Models: NLinear [82], TSMixer [14], 2. CNN-Based Models: ModernTCN [46], TimesNet [70], 3. RNN-Based Models: DeepAR [58], DilatedRNN [13], 4. GNN-Based Models: FourierGNN [77], StemGNN [11], 5. Transformer-Based Models: PatchTST [49], iTransformer [42], 6. LLM-Based Models: GPT4TS [86], AutoTimes [44]. We follow the source code of NeuralForecast⁴ for the implementation of these approaches⁵.

⁴<https://github.com/Nixtla/neuralforecast>

⁵The summary of these methods can be found in Appendix D.

³Detailed split information is reported in Appendix Table 5.

3.4 Experimental Design

We design a series of controlled experiments to examine whether existing machine learning models can capture the spatiotemporal and multi-factor characteristics of compound flooding systems. Rather than optimizing for a single benchmark score, our experimental design aims to answer three scientific questions: (i) *whether current AI models can effectively learn spatiotemporal dependencies under multi-factor factors*, (ii) *which input factors are most informative for compound flooding prediction*, and (iii) *the relative importance of temporal versus spatial context in forecasting performance*.

Spatial Scope and Computational Constraints. Due to the substantial memory and training costs associated with large-scale models, particularly LLM-based approaches, we adopt a two-level evaluation strategy. First, we conduct comprehensive benchmarking within three hydrologically relevant regions selected based on observed flood occurrences. The three areas are selected based on the flood observation data presented in Figure 3a, which visualizes these flood locations alongside the selected areas. These regions represent areas where compound flooding effects are most pronounced. Second, for the full dataset, we report results for four foundational methods to assess large-scale generalization while maintaining computational feasibility.

Factor and Context Ablations. To isolate the contribution of different factors, we conduct targeted ablation studies. These include removing or selectively incorporating groundwater, rainfall, and human control signals, as well as varying the spatial extent of input stations and the temporal length of historical context. By systematically controlling these factors, we assess how different models leverage temporal accumulation, spatial coupling, and cross-modal information in learning compound flooding dynamics.

Temporal Splits and Forecasting Horizons. To evaluate generalization across hydrological regimes, we apply temporally disjoint data splits, using the last year for testing, the second-to-last year for validation, and all preceding data for training. A fixed look-back window of two days is used for all models, while prediction horizons of one, three, five, and seven days are evaluated to assess short- and medium-term forecasting capability.

3.5 Evaluation Criteria

The evaluation is designed to reflect both overall predictive fidelity and sensitivity to hydrologically critical extreme events. To quantify general predictive accuracy, we follow standard time series forecasting practices to report the Mean Absolute Error (MAE) and Mean Squared Error (MSE), which measure deviations between predicted and observed water stage values across all time steps.

In addition, to explicitly evaluate performance on extreme flood-related events, we also employ the Symmetric Extremal Dependence Index (SEDI) [23, 76], following prior work in flood and climate forecasting [24]. The formulation of SEDI is as follows:

$$\text{SEDI}(p) = \frac{|\hat{Y} < V_{1-\frac{p}{2}} \& Y < V_{1-\frac{p}{2}}| + |\hat{Y} > V_{\frac{p}{2}} \& Y > V_{\frac{p}{2}}|}{|Y < V_{1-\frac{p}{2}}| + |Y > V_{\frac{p}{2}}|}, \quad (1)$$

where $|\cdot|$ means the number of the true values satisfying the condition, $\hat{Y}$ is the forecasting results, p is the quantile of the threshold, and $V_{1-\frac{p}{2}}, V_{\frac{p}{2}}$ are the lower and upper threshold of top and worst $\frac{p}{2}$

percent, respectively. SEDI focuses on the co-occurrence of extreme predictions and observations, thereby emphasizing the model's responsiveness to tail events. Specifically, extreme conditions are defined using quantile-based thresholds (e.g., the 95th and 5th percentiles of the observed values), and each time step is classified as either normal or extreme. A higher SEDI score indicates stronger dependence between predicted and observed extremes, reflecting improved capability in capturing flood-relevant dynamics.

3.6 Experimental Setup

In this paper, the default look-back window size is two days, and the forecast window length is a list of [1, 3, 5, 7] days. For the SEDI metric, we report the average performance across four forecast settings. Except for Table 3, MAE and MSE results are reported based on the average of four forecast window lengths.

4 Results

4.1 Dataset Analysis & Visualization

To establish system-level intuition and verify the physical consistency of SF²Bench, we conduct a comprehensive series of visualization-based analyses from both spatial and temporal perspectives. These analyses serve as sanity checks to examine whether the observed patterns are coherent across space and time and consistent with established hydrological processes.

Spatial Distribution. Figure 3a presents the geographic distribution of all monitoring stations, which are predominantly deployed along the complex river network of South Florida and radiate outward from Lake Okeechobee. The staggered spatial layout of hydrological, groundwater, and rainfall stations enables joint spatial reasoning across different components of the hydrological system. For instance, water level variations at a given site are expected to be influenced by nearby precipitation patterns as well as local groundwater conditions, which reflect subsurface water storage capacity. To further contextualize the dataset, we overlay observed flood events recorded in 2008 and from 2020 to 2023 in Figure 3b, based on reports from the South Florida Water Management District (SFWMD) [50]. Notably, these flood events are primarily concentrated in urbanized regions, highlighting the interaction between natural hydrological processes and human-modified landscapes.

Temporal Pattern. Figure 3 illustrates the average annual temporal patterns across all monitoring stations, using data from split S5 and S7 as a representative example. From these patterns, we observe a strong correlation between groundwater level and water level data. The water stage generally rises from approximately the 150th to the 300th day of the year, followed by a gradual decrease. According to the National Weather Service (NWS), Florida's climate is characterized by distinct dry and rainy seasons, with the latter typically spanning from May to October. This aligns well with the observed data patterns in our dataset. Taking rainfall as a reference, we note that increases in water level tend to correspond with rainfall events, such as the rainfall peak after the 300th day and the subsequent rise in water level. In addition, human control activities on hydraulic structures appear to influence water level changes in response to rainfall. We can infer that human intervention has played a role in mitigating potential flooding.

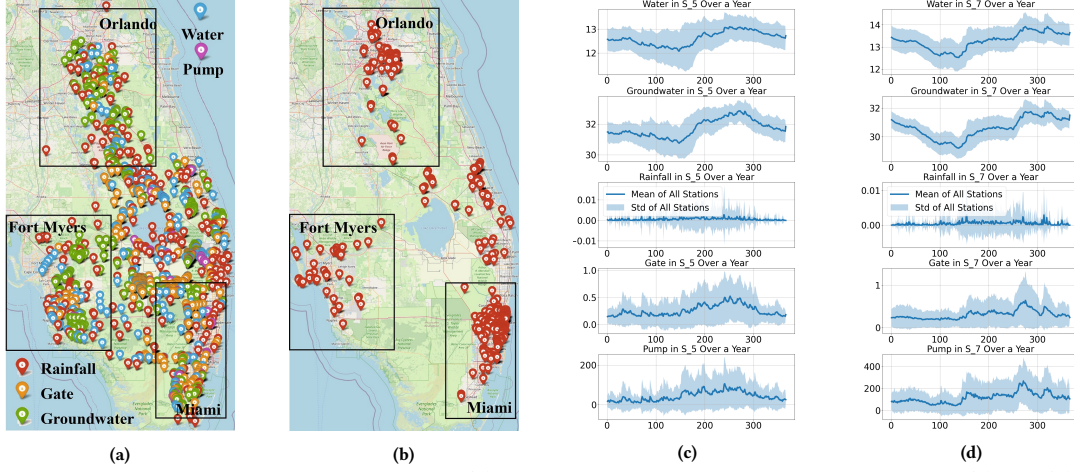


Figure 3: (a). Location distribution of monitor stations. (b). Flood observation location distribution. In (a) and (b), we highlight three interest parts: Orlando, Fort Myers, and Miami. (c) and (d). Temporal patterns of key features over a year. The x-axis is the number of days in one year.

4.2 Observations & Discovery

We organize our observations around the core scientific questions concerning the modeling of compound flooding systems.

Q1: Can existing machine learning models effectively capture multi-factor spatiotemporal dynamics?

Across all benchmark experiments, we observe that current machine learning models exhibit uneven capability in capturing the multi-factor and spatiotemporal structure of the hydrological system. Table 3 reports the average MSE and MAE results across three regions and eight data splits. The top-performing methods include PatchTST, iTransformer, MLP, NLinear, and AutoTimes. PatchTST and iTransformer achieve the best MAE, while MLP, NLinear, and AutoTimes perform best under MSE. In general, a small MAE but large MSE suggests high accuracy for most points with occasional large errors, whereas the opposite indicates greater stability but lower pointwise accuracy. Table 3 also shows model performance on extreme cases; in contrast, GNN-based models, particularly FourierGNN, demonstrate competitive performance on extreme-event metrics. We also observe similar results for performance across all stations in Appendix F.1.

From our observations, the MLP achieves near-best performance across almost all experiments, despite its relatively concise model structure. This suggests that existing methods that emphasize temporal dependency may fail to adequately capture the interactions among multiple driving factors. Furthermore, these results reveal a trade-off in model biases that certain architectures favor over all system averages, whereas others are specifically tuned to the patterns of correlated extremes.

Q2: Do average accuracy metrics reflect performance on extreme events (e.g., floods)?

We find that improvements in MAE and MSE do not fully correspond to improved extreme-event prediction. Models such as FourierGNN achieve strong SEDI performance despite weaker average accuracy, indicating a decoupling between mean behavior and tail dynamics. This highlights the necessity of evaluating compound flooding models using metrics that explicitly target extreme events,

as average error measures alone may obscure critical system-level behaviors relevant to flood risk assessment.

Q3: Which factors are most informative for compound flooding prediction?

To quantify the contribution of diverse drivers to compound flooding predictability, we conducted a systematic ablation study across five representative architectures. These models are categorized into two paradigms: channel-independent (NLinear, PatchTST) and channel-dependent (iTransformer, TimesNet, TSMixer). While the former treats all variables as a unified input pool, the latter variants designate water stage as the primary supervised target, treating auxiliary factors as exogenous covariates.

By systematically isolating factors including rainfall (R), anthropogenic control signals (C), and groundwater (G), we reveal a non-uniform contribution of auxiliary inputs that is deeply tied to model inductive biases. As shown in Table 4, a distinct bifurcation emerges that, while NLinear and TSMixer reach a performance plateau using water stage data alone, architectures like iTransformer and PatchTST demonstrate a superior capacity for multimodal synergy, showing substantial error reductions when the input space is enriched. This suggests that transformer-based architectures are more adept at capturing the cross-channel dependencies essential for simulating complex hydrological responses. Beyond architectural differences, the ablation reveals a critical physical insight regarding the hierarchy of environmental drivers. Across the most responsive models, groundwater consistently emerges as the most informative auxiliary factor, far outweighing the predictive contribution of immediate rainfall or control data. The exclusion of groundwater (denoted as “w/o G ” or “w/o GR”) leads to a significantly higher degradation in MSE and MAE compared to the removal of other variables, such as “w/o R ” or “w/o C ”. Notably, for iTransformer and TimesNet, providing groundwater as the sole auxiliary input obtains the highest performance gains among all ablation settings. This suggests that subsurface water storage serves as a dominant latent state that governs surface water dynamics during compound events, effectively acting as a long-term memory component of the hydrological cycle.

Table 3: Benchmark of average MAE, MSE, and SEDI results on three interest areas across 8 splits. The best and second results are shown in bold font and underlined, respectively. The best performances are in **bold, and the second-best method is underlined.**

Methods	MAE↓					MSE↓					SEDI↑		
	1D	3D	5D	7D	Avg.	1D	3D	5D	7D	Avg.	10%	5%	1%
MLP-Based Models													
MLP	0.0788	0.1351	0.1764	0.2063	0.1492	<u>0.0521</u>	0.1029	0.1432	0.1707	0.1172	0.6897	0.5834	0.3666
TSMixer	0.0928	0.1442	0.1816	0.2126	0.1578	0.0648	0.1132	0.1566	0.1903	0.1283	0.6144	0.4942	0.2480
NLinear	0.0817	0.1373	0.1769	0.2082	0.1510	0.0556	<u>0.1105</u>	0.1547	0.1841	<u>0.1262</u>	0.6278	0.5111	0.2767
CNN-Based Models													
TCN	0.1792	0.2281	0.2697	0.3088	0.2465	0.3722	0.4647	0.4173	0.5386	0.4482	0.5311	0.3706	0.1387
ModernTCN	0.0798	0.1441	0.1891	0.2248	0.1594	0.0681	0.2443	0.2808	0.2723	0.2164	0.6067	0.4846	0.2512
TimesNet	0.0983	0.1528	0.1927	0.2267	0.1676	0.0704	0.1358	0.1829	0.2241	0.1533	0.5829	0.4589	0.2222
RNN-Based Models													
LSTM	0.1182	0.1821	0.2232	0.2576	0.1953	0.1339	0.1985	0.2348	0.2873	0.2136	0.6097	0.4702	0.1680
DeepAR	0.1178	0.1837	0.2247	0.2596	0.1964	0.1230	0.1954	0.2389	0.2691	0.2066	0.5989	0.4643	0.1478
DilatedRNN	0.0919	0.1573	0.2022	0.2374	0.1722	0.1033	0.1672	0.2341	0.2591	0.1909	0.6164	0.5014	0.2499
GNN-Based Models													
GCN	0.1696	0.2006	0.2504	0.2799	0.2251	7.2299	1.5645	2.1341	1.0374	2.9915	0.6179	0.5138	0.2828
StemGNN	0.1332	0.2181	0.3153	0.3570	0.2559	0.1632	0.2543	0.4189	0.4716	0.3270	0.5507	0.4270	0.1690
FourierGNN	0.0921	0.1503	0.1930	0.2280	0.1658	0.0768	0.1416	0.2071	0.2125	0.1595	<u>0.6623</u>	<u>0.5511</u>	<u>0.3217</u>
Transformer-Based Models													
PatchTST	0.0741	<u>0.1316</u>	0.1719	<u>0.2044</u>	0.1455	0.0531	0.1132	0.1566	0.1903	0.1283	0.6296	0.5086	0.2685
iTransformer	<u>0.0756</u>	0.1314	<u>0.1722</u>	0.2041	<u>0.1458</u>	0.0253	0.1112	0.1583	0.1911	0.1284	0.6286	0.5079	0.2690
LLM-Based Models													
GPT4TS	0.1256	0.1521	0.1911	0.2247	0.1734	0.0966	0.1410	0.1847	0.2245	0.1617	0.5581	0.4640	0.2268
AutoTimes	0.0846	0.1362	0.1752	0.2062	0.1505	0.0584	0.1125	<u>0.1522</u>	<u>0.1823</u>	0.1263	0.6239	0.5015	0.2568

Table 4: Input factors ablation study on S6. All, G, R, and C represent all factors: groundwater, rainfall, and human control(pump and gate). The best performances are in **bold, and the second-best method is underlined.**

	Models	w/ All	w/o G	w/o R	w/o C
MAE	iTransformer	0.1406	0.1407	0.1406	<u>0.1405</u>
	PatchTST	0.1376	0.1391	0.1378	<u>0.1377</u>
	TSMixer	0.1596	<u>0.1419</u>	0.2523	0.2111
	NLinear	0.1546	0.1577	0.1483	0.1540
	TimesNet	0.1642	0.1599	0.1662	0.1650
MSE	iTransformer	<u>0.0953</u>	0.0960	0.0949	0.0954
	PatchTST	<u>0.0917</u>	0.0932	0.0927	0.0916
	TSMixer	0.1080	0.0946	0.4061	0.2698
	NLinear	0.0992	0.1011	0.0966	0.0984
	TimesNet	0.1188	0.1154	0.1226	0.1202
	Models	w/o GR	w/o RC	w/o GC	w/o GRC
MAE	iTransformer	0.1411	0.1410	0.1402	0.1411
	PatchTST	0.1398	0.1396	0.1385	0.1421
	TSMixer	0.1418	0.2807	0.1423	0.1422
	NLinear	0.1485	<u>0.1450</u>	0.1579	0.1435
	TimesNet	0.1592	0.1656	0.1578	<u>0.1580</u>
MSE	iTransformer	0.0957	0.0956	0.0949	0.0962
	PatchTST	0.0950	0.0938	0.0923	0.0971
	TSMixer	0.0946	0.6697	<u>0.0943</u>	0.0941
	NLinear	0.0973	<u>0.0954</u>	0.1006	0.0947
	TimesNet	0.1145	0.1237	0.1111	<u>0.1123</u>

Q4: Is temporal context or spatial context more critical for accurate forecasting?

To characterize the boundary of predictability in compound flooding, we investigated the relative importance of spatial coverage and temporal receptivity across different modeling paradigms. By defining an anchor area of radius R as the forecasting target and incrementally expanding the interest area using scale factors from 1.0 to 1.8 (as illustrated in Figure 4a), we quantified the impact of regional spatial information. As shown in Figure 4b, models such as iTransformer, PatchTST, and TSMixer exhibit consistent error reductions as more surrounding monitoring stations are integrated. This performance gain underscores the value of capturing upstream-downstream hydraulic correlations and regional meteorological trends. However, these spatial benefits tend to saturate as the radius increases, suggesting that the marginal utility of geographic data diminishes once the immediate hydrological catchment or local forcing zone is fully represented.

In parallel, we examined the influence of temporal context by varying the input look-back window from 6 hours to 6 days. The results in Figure 4c reveal that temporal extension generally brings in more substantial performance improvements than spatial expansion, particularly for PatchTST and TSMixer, which achieved significant MSE reductions with longer histories. This dominance of the temporal dimension suggests that the sequential evolution of water stages contains more critical predictive information than instantaneous spatial snapshots, reflecting cumulative rainfall effects and tidal phase transitions. We also observed a performance inflection point for the iTransformer, where accuracy began to degrade

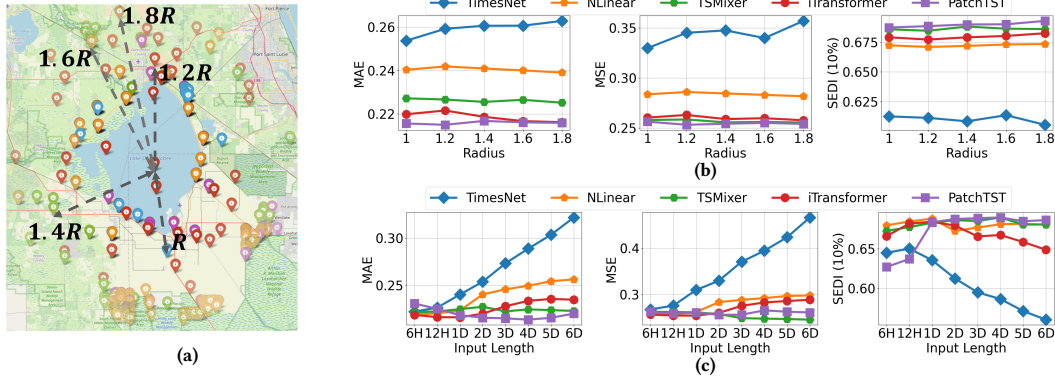


Figure 4: (a).The different interest areas with a radius for the ablation study on spatial information. (b). Study on spatial information. (c). Study on temporal information.

beyond a one-day input window. This phenomenon reflects a trade-off between the depth of the temporal context and the data density required to resolve long-range dependencies without overfitting.

In summary, our analysis indicates that while spatial information provides necessary geographical constraints, the temporal context is the primary driver of accuracy, though its benefits are ultimately constrained by the complexity of the underlying physical patterns and the limits of the available training data.

4.3 Compound Flooding Event Study

To complement the statistical evaluation, we provide an event-level case study on Hurricane Ian, which struck Florida in September 2022 and triggered widespread compound flooding. This event provides a natural stress test for AI models, as it involves intense rainfall, elevated groundwater levels, and rapid surface water response under human regulation. Evaluating model behavior during such an extreme, non-stationary period allows us to assess whether learned dynamics remain physically plausible beyond average conditions.

We focus on the middle reach of the Kissimmee River, where multiple water monitoring stations are located in close proximity to documented flood zones (Figure 5a). This region provides a representative setting for evaluating compound flooding dynamics, with high-quality observations available for key driving factors, including surface water level, groundwater level, rainfall, and human management activities. All models are trained using data before the event and tasked with forecasting water stage trajectories over the subsequent 24-hour period, enabling a direct comparison of model behavior under extreme, non-stationary conditions.

Q5. How do different AI models respond to rapidly evolving compound flood dynamics under extreme conditions? Figure 5b summarizes the 24-hour water level forecasting results at three representative monitoring stations along the Kissimmee River. Across all stations, most methods successfully capture the overall rising trend of water levels driven by sustained rainfall. However, clear differences emerge in their ability to track the timing and magnitude of flood peaks. ModernTCN and DilatedRNN exhibit delayed responses and reduced peak predictions, suggesting limited capacity to adapt to unexpected regime shifts. In contrast, transformer-based models demonstrate improved responsiveness to rapid water level increases, although iTransformer shows noticeable overestimation

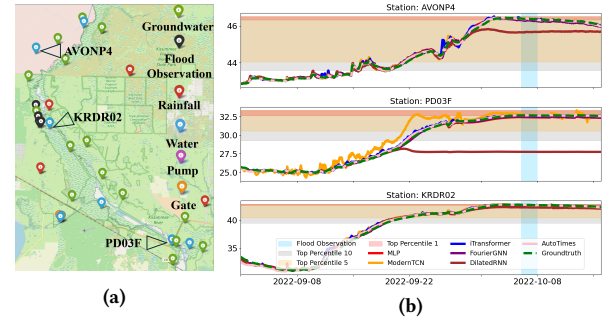


Figure 5: (a) The interest area of the case study. (b) The comparison among different methods. We mark the flood observation on the x-axis and the top percentile 10/5/1 with colors.

and vibrant behavior at certain stations, indicating potential sensitivity to noise amplification under highly non-stationary inputs. These qualitative observations are consistent with the quantitative evaluation reported in Table 3, reinforcing that accurate compound flood forecasting requires not only trend-following capability but also robust temporal alignment and stability under extreme hydrological forcing.

5 Conclusion

We investigate the use of AI-based forecasting models for compound floods forecasting. By combining standard accuracy metrics with SEDI, we provide a comprehensive assessment of model performance. Our results reveal that while AI models can capture the overall flooding trend, their reliability in predicting extreme water level responses varies substantially. The case study demonstrates that model failures often emerge during rapid regime shifts driven by the interaction of multiple factors. These findings reveal the shortcomings of existing data-driven AI methods and explore the driving factors behind compound flooding events. The future work will focus on developing more effective data-driven models capable of integrating multiple driving factors from spatiotemporal information, as well as constructing interpretable frameworks to support the prevention and management of compound floods. In particular, such frameworks should help elucidate how models balance and weigh competing drivers, such as storm surge and peak river discharge, in shaping compound flood dynamics.

GenAI Disclosure

In this paper, we leverage LLMs, including ChatGPT and Gemini, to refine sentence-level writing.

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A Time Split of SF²Bench

We provide the detailed information about the number of monitor stations in each split in Table 5. In each split, the number of water monitor stations is more than other kinds of features, which means the water level feature is the main information, and others are additional parts.

Table 5: The summary of the all splits in SF²Bench

Splits	Time Span	Interval	Water	Groundwater	Rainfall	Pump	Gate
S0	1985-1990	1 Hour	159	40	143	17	82
S1	1990-1995	1 Hour	227	36	139	18	104
S2	1995-2000	1 Hour	332	44	170	26	94
S3	2000-2005	1 Hour	402	178	227	31	107
S4	2005-2010	1 Hour	518	296	254	48	172
S5	2010-2015	1 Hour	585	333	216	65	256
S6	2015-2020	1 Hour	670	317	186	85	300
S7	2020-2024	1 Hour	716	352	194	89	329

We also provide the geographical distribution information of monitor stations in different splits in Figure 10. As time goes by, the number of monitoring stations gradually increases, and in terms of spatial distribution, the locations of monitoring stations remain consistent.

In addition, we visualize the temporal pattern of features over a year in each split. As shown in Figure 6, 7, 8, and 9, the temporal patterns in 8 splits are similar and consistent. From the view of the climate, the dry and rainy seasons are highly consistent but the intensity of rainfall in different splits is different. These patterns that are highly aligned with the actual situation demonstrate the quality of the dataset.

B Experimental Details

C Data Preprocessing

Normalization. To make the data have a zero mean and unit variance, we follow [20, 80, 84] using z-score to normalize the time series data. For the time series, whose variance is less than 1E-4, we regard its variance as one to avoid the variable overflow (inf or NaN). For forecasting tasks, all the report metrics are based on the normalized data.

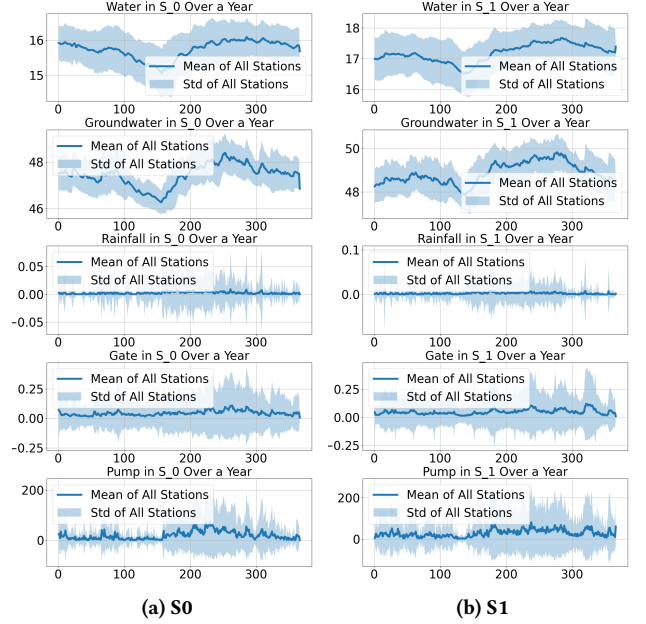


Figure 6: The temporal pattern of features of S0 and S1.

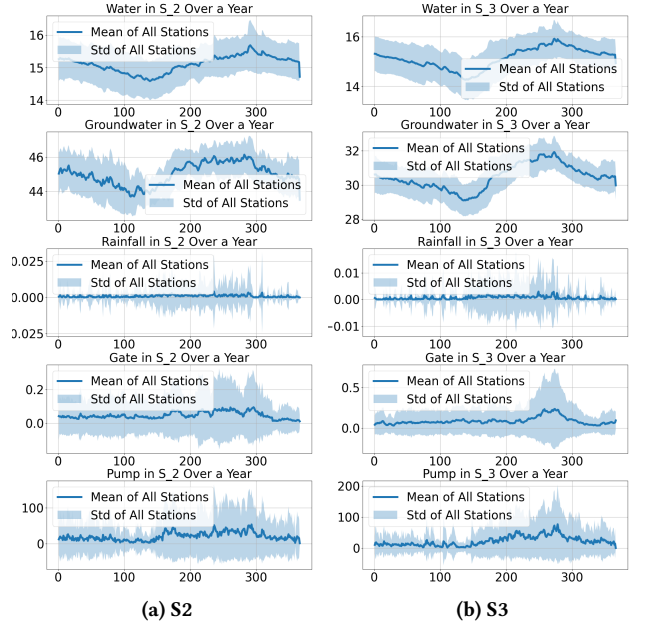


Figure 7: The temporal pattern of features of S2 and S3.

Timestamp Features. Because of some methods, such as NLinear, we do not consider extracting the timestamp feature as part of the input. For those methods that are capable of handling the timestamp feature, we ignore this part. In our code, we also provide timestamp features extracted by following [80, 84] for further works.

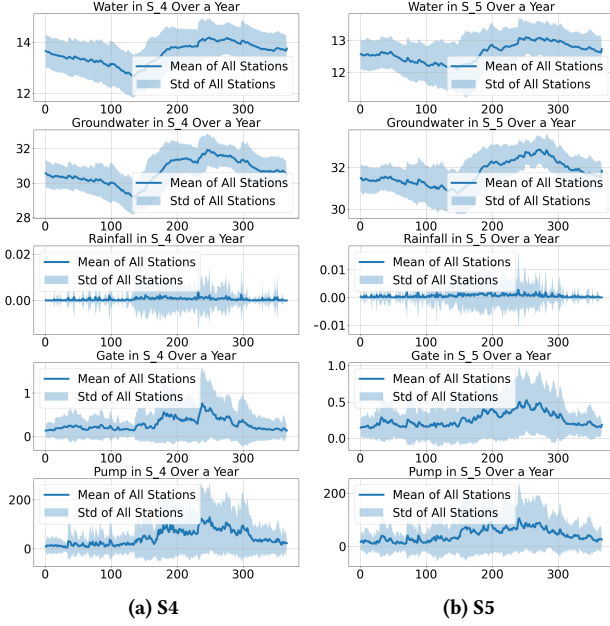


Figure 8: The temporal pattern of features of S4 and S5.

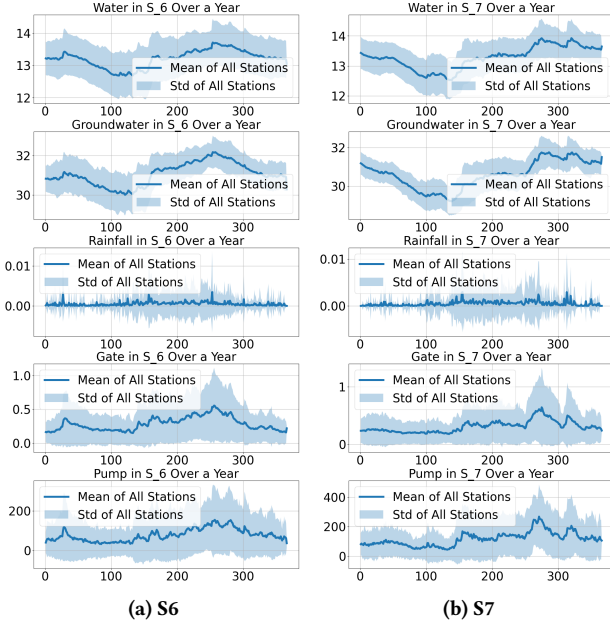


Figure 9: The temporal pattern of features of S6 and S7.

D Methods

MLP. We implement a classical three-layer MLP with ReLU as the activation function. The input layer dimension is determined by the input sequence length, and the output layer dimension corresponds to the forecast horizon. The hidden layer dimension is 64. For training, the learning rate is 1×10^{-4} , weight decay is 1×10^{-6} , and the batch size is 64. The model is trained for 15 epochs.

NLinear [82]. This is a simple linear model that treats each time series independently, modeling future values using a linear transformation of the most recent input values. Implementation is based on the NeuralForecast library⁶. The learning rate is 1×10^{-4} , weight decay is 1×10^{-6} , batch size is 64, and training is performed for 50 epochs.

TSMixer [14]. Inspired by MLP-Mixer models from vision tasks, TSMixer is a neural network architecture for time series forecasting. It alternately applies MLPs along the time and feature axes, learning dependencies across both dimensions without requiring attention mechanisms or complex sequence modeling. Implementation follows the NeuralForecast default settings. The architecture includes 2 mixing layers, and the second feed-forward layer has 64 units. The learning rate is 1×10^{-4} , batch size is 32, and the model is trained for 10 epochs.

TCN [5]. It incorporates causal convolutions, ensuring predictions depend only on current and past inputs, thus preserving temporal order. They also utilize dilated convolutions to efficiently capture long-range dependencies by expanding the receptive field without significantly increasing layers. Our implementation follows the official code⁷ using the popular channel-wise setting. It employs a three-layer backbone with a kernel size of 3 and a fixed dilation of 1. The learning rate is 1×10^{-3} , weight decay is 1×10^{-7} , batch size is 256, and training is performed for 50 epochs.

ModernTCN [46]. ModernTCN introduces a streamlined, fully convolutional architecture that aims to simplify design while enhancing performance. It incorporates components like depth-wise separable convolutions and Gated Linear Units (GLUs) to efficiently capture local and long-range temporal dependencies. Implementation follows the long-term forecasting settings from the source code⁸, using the Weather dataset hyperparameters as defaults. The learning rate is 1×10^{-4} , batch size is 256, and the model is trained for 100 epochs.

TimesNet [70]. TimesNet models temporal variations in a two-dimensional space by reshaping time series data into a pseudo-image format and applying 2D convolutional techniques. This enables it to capture both short-term dynamics and long-term dependencies. Implementation uses the Time-Series-Library⁹, with default hyperparameters from the long-term forecasting setting for the Weather dataset. The learning rate is 1×10^{-4} , batch size is 32, and training is performed for 10 epochs.

LSTM [26]. Our Long Short-Term Memory (LSTM) implementation is a two-layer model with a hidden dimension of 32. The learning rate is 1×10^{-3} , weight decay is 1×10^{-6} , batch size is 64, and the model is trained for 50 epochs.

DeepAR [58]. DeepAR is a global model trained on multiple related time series, which aids generalization, especially for series with limited history. It employs an RNN architecture to predict future values by modeling the conditional distribution of the next value given past observations. Implementation is based on the NeuralForecast library. The learning rate is 1×10^{-3} , batch size is 64, and training is performed for 20 epochs.

⁶<https://github.com/Nixtla/neuralforecast>

⁷<https://github.com/locuslab/TCN/tree/master>

⁸<https://github.com/luodh/hh/ModernTCN>

⁹<https://github.com/thuml/Time-Series-Library>

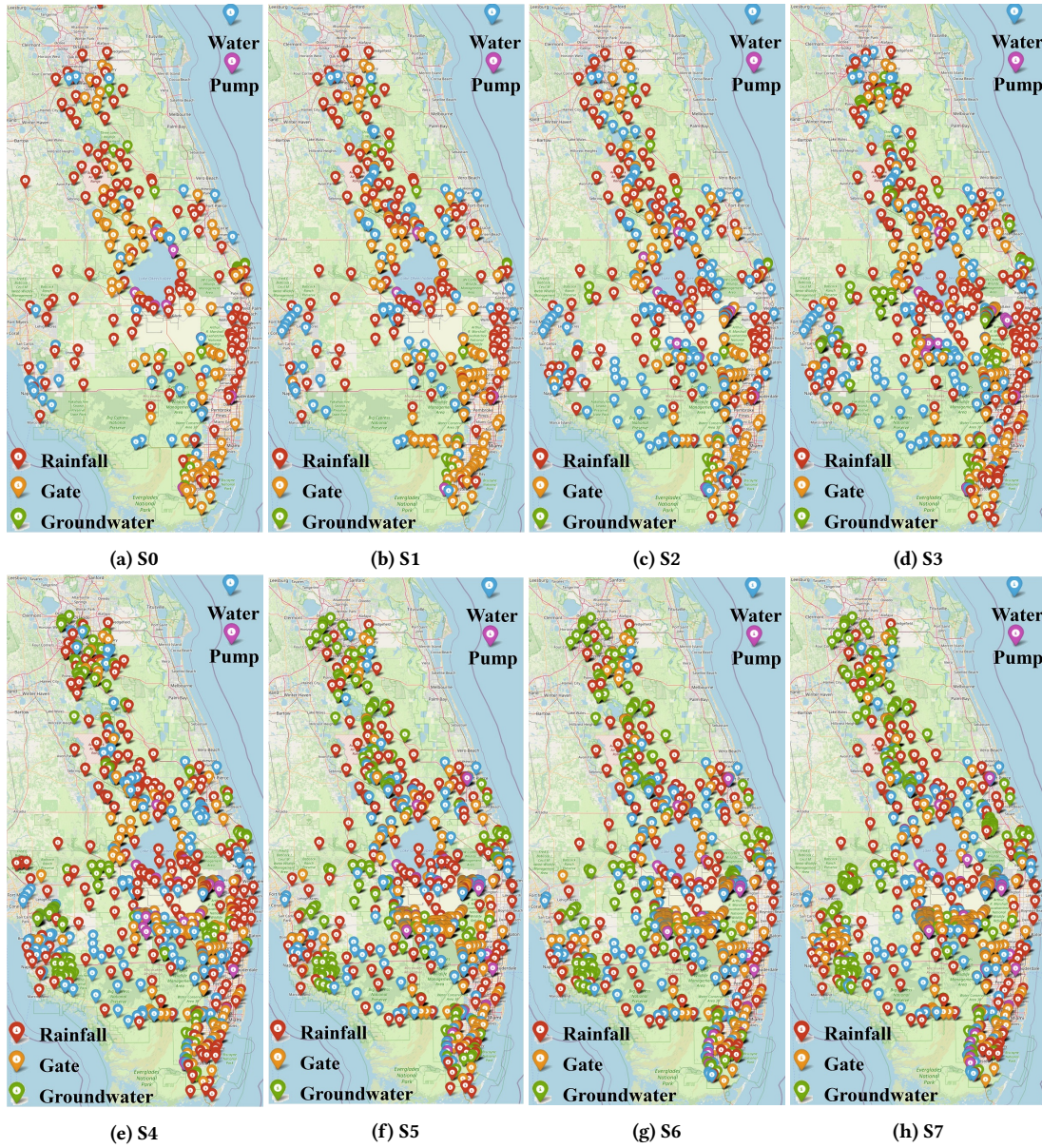


Figure 10: The location distribution in each split.

DilatedRNN [13]. Dilated RNNs exponentially expand their receptive field by stacking layers with different dilation factors, allowing efficient capture of short- and long-range patterns without a drastic increase in parameters. This makes them suitable for forecasting tasks with wide-ranging temporal dependencies. Implementation is based on the NeuralForecast library. The learning rate is 1×10^{-3} , batch size is 64, and training is performed for 40 epochs.

GCN [33]. The GCN architecture consists of two layers with a hidden dimension of 32. The graph topology is derived from location information using Delaunay triangulation¹⁰. The learning rate is 1×10^{-4} , weight decay is 1×10^{-5} , batch size is 32, and the model

¹⁰<https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.Delaunay.html>

is trained for 50 epochs.

FourierGNN [77]. FourierGNN leverages graph neural networks and Fourier transforms to capture temporal and inter-variable dependencies. Time series variables are treated as graph nodes, with edges representing their relationships. Fourier transforms project data into the frequency domain to model periodic and long-range dependencies. Implementation follows the source code¹¹. The learning rate is 1×10^{-5} , batch size is 32, and training is performed for 100 epochs.

StemGNN [11]. StemGNN is designed to capture both temporal (via temporal convolutions) and spatial (via spectral graph convolutions)

¹¹<https://github.com/aikunyi/FourierGNN>

dependencies in time-series data, learning smooth representations over the graph structure and dynamic patterns. Implementation follows the source code¹². The learning rate is 1×10^{-4} , batch size is 32, and the model is trained for 50 epochs.

iTransformer [42]. The iTransformer uses an encoder-decoder structure where the encoder processes the sequence in reverse order. This allows the decoder to predict future values based on this processed representation, enhancing focus on relevant temporal sequences while mitigating the computational cost of traditional transformers. Implementation is based on the NeuralForecast library. The learning rate is 1×10^{-4} , batch size is 32, and training is performed for 10 epochs.

PatchTST [49]. PatchTST is a Transformer-based architecture employing patching and channel independence. Time series are divided into patches, which are transformed into tokens and processed by a transformer model to capture local and global dependencies via self-attention. This is particularly useful for long-term forecasting. Implementation is based on the NeuralForecast library. The learning rate is 1×10^{-4} , batch size is 128, and training is performed for 100 epochs.

GPT4TS [86]. GPT4TS treats time series as a language, leveraging pretrained language models (LLMs) to learn temporal patterns. Time series data is tokenized for LLM processing, enabling zero-shot or few-shot generalization. Implementation follows the source code¹³, using long-term forecasting settings for the Weather dataset as default hyperparameters. The default language model is GPT-2 [55]. The learning rate is 1×10^{-4} , batch size is 64, and training is performed for 10 epochs.

AutoTimes [44]. AutoTimes projects time series segments into the embedding space of language tokens, leveraging the autoregressive capabilities of LLMs for forecasting. By training the model to predict subsequent time series segments given preceding ones, AutoTimes generates multi-step forecasts. Implementation follows the source code¹⁴, with GPT-2 [55] as the language model. The learning rate is 5×10^{-4} , batch size is 64, and training is performed for 10 epochs.

For all benchmark experiments, AdamW [45] is used as the optimizer, and the loss function is Mean Squared Error (MSE).

E Platform

All experiments are conducted on two Linux machines, one with 8 NVIDIA A100 GPUs, each with 40GB of memory, and another with 4 RTX 4090 GPUs. We used Python 3.12.9 and Pytorch 2.6.0 to construct our project.

F Detailed Results

F.1 Results on All Stations

For the entire dataset, we only report the results of the basic foundational techniques. We provide the results in Table 6. As the results show, MLP achieves the best performance, and GCN is the second-best, Consistent with previous results in Section 4.2.

¹²<https://github.com/microsoft/StemGNN>

¹³<https://github.com/DAMO-DI-ML/NeurIPS2023-One-Fits-All>

¹⁴<https://github.com/thuml/AutoTimes/tree/main>

Table 6: Average results of basic methods on the whole dataset. The best and second results are shown in bold font and underlined, respectively. The best performances are in bold, and the second-best method is underlined.

Metric	MLP	LSTM	TCN	GCN
↓MAE	<u>0.2328</u>	0.3302	0.3491	<u>0.3053</u>
↓MSE	<u>0.3902</u>	2.2772	<u>0.6807</u>	1.0673
↑SEDI(10%)	0.6853	0.5751	0.5167	<u>0.6137</u>
↑SEDI(5%)	<u>0.5970</u>	0.4516	0.3904	<u>0.5213</u>
↑SEDI(1%)	<u>0.4107</u>	0.1828	0.1842	<u>0.3134</u>

F.2 Detailed Benchmark Results

In this section, we report the detailed results, including MSE, MAE, and three SEDI values on different prediction lengths, on three interest areas(Orlando, Miami, and Fort Myers) from Table 7 to 36. The best average results are presented in bold font, while the second-best are underlined. Furthermore, from Table 37 to 41, we report the detailed results of basic methods on the whole dataset. To provide qualitative analysis, from Figure 11 to 16, we demonstrate the case visualization of each method on the S7 split in the Orlando area.

F.3 Detailed Other Results

To provide more information about the ablation study, we report the detailed results of the factor ablation study in Table 42 to 44. We report the average results on three interest areas on the S6 split. For the spatial and temporal information ablation study, the results are available from Table 45 to 48.

G Limitations

While SF²Bench provides five key factors, it currently lacks explicit topological linkage information between monitoring stations due to the intricate nature of South Florida’s water system. Although we provide some flood observation data, the locations of these observations may not directly correspond to our monitoring stations. Therefore, this information is provided in a separate file rather than being integrated into the time series data.

Table 7: Benchmark(MAE) on part 1(Orlando area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
MLP	1D	0.0915	0.1229	0.0679	0.0810	0.0716	0.0691	0.0598	0.0846	0.0810	
	3D	0.1424	0.1928	0.1167	0.1437	0.1174	0.1154	0.1086	0.1314	0.1335	
	5D	0.1746	0.2472	0.1607	0.1844	0.1503	0.1529	0.1418	0.1718	0.1730	
	7D	0.1919	0.2899	0.1848	0.2244	0.1750	0.1793	0.1753	0.1970	0.2022	
	Avg.	0.1501	0.2132	0.1325	0.1584	0.1286	0.1292	0.1214	0.1462	0.1474	
MLP	TSMixer	1D	0.1046	0.1355	0.0812	0.0846	0.0767	0.0778	0.1184	0.1015	0.0975
	3D	0.1494	0.2140	0.1252	0.1378	0.1185	0.1219	0.1500	0.1458	0.1453	
	5D	0.1769	0.2673	0.1595	0.1783	0.1496	0.1547	0.1751	0.1798	0.1802	
	7D	0.1938	0.3093	0.1892	0.2120	0.1765	0.1819	0.2098	0.2096	0.2103	
	Avg.	0.1562	0.2315	0.1388	0.1532	0.1303	0.1341	0.1633	0.1592	0.1583	
NLinear	1D	0.0980	0.1279	0.0777	0.0820	0.0771	0.0741	0.0640	0.0929	0.0867	
	3D	0.1475	0.2109	0.1234	0.1390	0.1207	0.1211	0.1079	0.1415	0.1390	
	5D	0.1835	0.2663	0.1584	0.1814	0.1527	0.1541	0.1422	0.1763	0.1769	
	7D	0.1964	0.3085	0.1891	0.2161	0.1803	0.1814	0.1711	0.2065	0.2062	
	Avg.	0.1563	0.2284	0.1371	0.1546	0.1327	0.1327	0.1213	0.1543	0.1522	
TCN	1D	0.1887	0.1417	0.2145	0.3830	0.1322	0.0889	0.1164	0.1913	0.1821	
	3D	0.1588	0.2302	0.1857	0.4058	0.1968	0.1359	0.1471	0.2127	0.2091	
	5D	0.2110	0.3028	0.2139	0.4483	0.2128	0.1928	0.2024	0.2579	0.2552	
	7D	0.2135	0.3379	0.2605	0.5715	0.2686	0.2052	0.2102	0.2695	0.2921	
	Avg.	0.1930	0.2532	0.2186	0.4522	0.2026	0.1557	0.1690	0.2329	0.2346	
CNN	ModernTCN	1D	0.0893	0.1238	0.0721	0.0760	0.0704	0.0661	0.0588	0.0802	0.0796
	3D	0.1486	0.2113	0.1366	0.1450	0.1212	0.1162	0.1073	0.1324	0.1398	
	5D	0.1886	0.2685	0.1798	0.1887	0.1632	0.1537	0.1426	0.1697	0.1818	
	7D	0.2140	0.3228	0.2240	0.2310	0.1936	0.1833	0.1728	0.2002	0.2177	
	Avg.	0.1601	0.2316	0.1531	0.1602	0.1371	0.1298	0.1204	0.1456	0.1547	
TimesNet	1D	0.1068	0.1436	0.0805	0.0909	0.0919	0.0925	0.0787	0.1051	0.0987	
	3D	0.1558	0.2190	0.1252	0.1515	0.1382	0.1402	0.1254	0.1582	0.1517	
	5D	0.1813	0.2745	0.1637	0.1933	0.1735	0.1766	0.1578	0.1958	0.1896	
	7D	0.2034	0.3222	0.2025	0.2309	0.2102	0.2146	0.1886	0.2263	0.2248	
	Avg.	0.1618	0.2398	0.1430	0.1666	0.1534	0.1560	0.1376	0.1714	0.1662	
Transformer	iTransformer	1D	0.0865	0.1227	0.0645	0.0730	0.0710	0.0700	0.0629	0.0889	0.0799
	3D	0.1395	0.1963	0.1103	0.1313	0.1193	0.1195	0.1058	0.1376	0.1325	
	5D	0.1717	0.2546	0.1471	0.1723	0.1534	0.1528	0.1401	0.1775	0.1712	
	7D	0.1889	0.2962	0.1801	0.2068	0.1833	0.1830	0.1708	0.2048	0.2017	
	Avg.	0.1466	0.2175	0.1255	0.1459	0.1317	0.1313	0.1199	0.1522	0.1463	
	PatchTST	1D	0.0843	0.1148	0.0638	0.0678	0.0711	0.0693	0.0617	0.0858	0.0773
	3D	0.1365	0.1922	0.1113	0.1269	0.1149	0.1199	0.1046	0.1362	0.1303	
	5D	0.1677	0.2451	0.1471	0.1728	0.1489	0.1569	0.1409	0.1718	0.1689	
	7D	0.1866	0.2927	0.1787	0.2054	0.1786	0.1859	0.1679	0.2019	0.1997	
	Avg.	0.1438	0.2112	0.1252	0.1432	0.1284	0.1330	0.1188	0.1489	0.1441	

Table 8: Benchmark(MAE) on part 1(Orlando area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
LSTM	1D	0.1275	0.1790	0.0935	0.1714	0.1044	0.1032	0.0865	0.1211	0.1233	
	3D	0.1652	0.2751	0.1431	0.2458	0.1518	0.1528	0.1481	0.1647	0.1808	
	5D	0.1972	0.3297	0.1801	0.2977	0.1871	0.1891	0.1784	0.1985	0.2197	
	7D	0.2139	0.3517	0.2277	0.3351	0.2198	0.2143	0.1978	0.2266	0.2484	
	Avg.	0.1759	0.2839	0.1611	0.2625	0.1658	0.1648	0.1527	0.1777	0.1931	
RNN	DeepAR	1D	0.1209	0.1855	0.0946	0.1726	0.1024	0.0971	0.0881	0.1171	0.1223
		3D	0.1667	0.2677	0.1488	0.2465	0.1534	0.1620	0.1421	0.1736	0.1826
		5D	0.1942	0.3110	0.1907	0.3237	0.1889	0.1828	0.1770	0.2045	0.2216
		7D	0.2146	0.3522	0.2210	0.3619	0.2122	0.2197	0.2123	0.2317	0.2532
		Avg.	0.1741	0.2791	0.1638	0.2762	0.1642	0.1654	0.1548	0.1817	0.1949
DilatedRNN	1D	0.0939	0.1317	0.0700	0.1552	0.0836	0.0748	0.0672	0.1006	0.0971	
	3D	0.1499	0.2318	0.1249	0.2031	0.1339	0.1318	0.1224	0.1457	0.1554	
	5D	0.1794	0.2793	0.1627	0.2897	0.1678	0.1635	0.1520	0.1809	0.1969	
	7D	0.2110	0.3329	0.1959	0.3061	0.1990	0.1945	0.1931	0.1478	0.2225	
	Avg.	0.1586	0.2439	0.1384	0.2385	0.1461	0.1411	0.1337	0.1437	0.1680	
GCN	1D	0.1003	0.2139	0.1034	0.3710	0.1030	0.1005	0.0775	0.0962	0.1457	
	3D	0.1543	0.3137	0.1411	0.3384	0.1524	0.1517	0.1208	0.1392	0.1889	
	5D	0.1862	0.3741	0.1665	0.4935	0.1893	0.1818	0.1610	0.1732	0.2407	
	7D	0.2034	0.4156	0.1912	0.5967	0.2153	0.2073	0.1927	0.2024	0.2781	
	Avg.	0.1610	0.3293	0.1505	0.4499	0.1650	0.1603	0.1380	0.1528	0.2134	
GNN	FourierGNN	1D	0.0985	0.1324	0.0796	0.1337	0.0863	0.0929	0.0672	0.0927	0.0979
		3D	0.1445	0.2034	0.1178	0.2733	0.1295	0.1246	0.1125	0.1392	0.1556
		5D	0.1746	0.2644	0.1524	0.3672	0.1712	0.1640	0.1513	0.1719	0.2021
		7D	0.1918	0.3028	0.1936	0.4107	0.1992	0.2019	0.1871	0.2068	0.2367
		Avg.	0.1523	0.2257	0.1358	0.2962	0.1466	0.1459	0.1295	0.1527	0.1731
StemGNN	1D	0.1270	0.1782	0.0786	0.6896	0.1005	0.0833	0.0741	0.0969	0.1785	
	3D	0.1717	0.3602	0.1536	0.6742	0.1778	0.1720	0.1498	0.1715	0.2539	
	5D	0.2081	0.4895	0.2141	1.4591	0.2144	0.2729	0.2258	0.2412	0.4156	
	7D	0.2477	0.5122	0.2577	1.4988	0.2552	0.3307	0.2486	0.2419	0.4491	
	Avg.	0.1886	0.3850	0.1760	1.0804	0.1870	0.2147	0.1746	0.1879	0.3243	
LLM	GPT4TS	1D	0.1050	0.1372	0.0833	0.0934	0.0905	0.0898	0.0778	0.1106	0.0984
		3D	0.1530	0.2169	0.1221	0.1502	0.1335	0.1380	0.1222	0.1547	0.1488
		5D	0.1817	0.2650	0.1565	0.1928	0.1666	0.1723	0.1559	0.1948	0.1857
		7D	0.1967	0.3105	0.1930	0.2294	0.1969	0.2054	0.1962	0.2224	0.2188
		Avg.	0.1591	0.2324	0.1387	0.1664	0.1469	0.1514	0.1380	0.1706	0.1629
AutoTimes	1D	0.0967	0.1369	0.0747	0.0770	0.0768	0.0784	0.0660	0.0963	0.0878	
	3D	0.1453	0.2057	0.1196	0.1296	0.1164	0.1203	0.1099	0.1409	0.1360	
	5D	0.1733	0.2664	0.1499	0.1751	0.1498	0.1534	0.1411	0.1739	0.1729	
	7D	0.1902	0.3029	0.1814	0.2113	0.1780	0.1823	0.1691	0.2032	0.2023	
	Avg.	0.1514	0.2280	0.1314	0.1482	0.1302	0.1336	0.1215	0.1536	0.1497	

Table 9: Benchmark(MSE) on part 1(Orlando area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
MLP	1D	0.1472	0.1254	0.0376	0.0610	0.0379	0.0275	0.0248	0.0523	0.0642	
	3D	0.2661	0.2014	0.0621	0.1232	0.0681	0.0536	0.0465	0.0913	0.1140	
	5D	0.3569	0.2609	0.0875	0.1714	0.0999	0.0772	0.0670	0.1226	0.1554	
	7D	0.2963	0.3124	0.1097	0.2215	0.1199	0.0975	0.0871	0.1498	0.1743	
	Avg.	0.2666	0.2250	0.0742	0.1443	0.0814	0.0640	0.0564	0.1040	0.1270	
MLP	1D	0.1649	0.1283	0.0423	0.0696	0.0380	0.0312	0.0661	0.0633	0.0755	
	3D	0.2915	0.2214	0.0679	0.1255	0.0694	0.0575	0.0862	0.1000	0.1274	
	5D	0.1769	0.2903	0.0936	0.1673	0.0981	0.0812	0.1010	0.1325	0.1626	
	7D	0.2817	0.3469	0.1190	0.2048	0.1248	0.1031	0.1384	0.1644	0.1854	
	Avg.	0.2688	0.2467	0.0807	0.1418	0.0826	0.0682	0.0979	0.1150	0.1377	
NLinear	1D	0.1468	0.1294	0.0449	0.0678	0.0405	0.0308	0.0291	0.0580	0.0684	
	3D	0.2830	0.2223	0.0699	0.1257	0.0720	0.0575	0.0510	0.0974	0.1223	
	5D	0.4103	0.2915	0.0950	0.1689	0.1013	0.0812	0.0722	0.1300	0.1688	
	7D	0.3001	0.3481	0.1204	0.2080	0.1291	0.1029	0.0924	0.1617	0.1828	
	Avg.	0.2851	0.2478	0.0825	0.1426	0.0857	0.0681	0.0612	0.1118	0.1356	
TCN	1D	0.3257	0.1416	0.1462	0.8044	0.1152	0.0344	0.0567	0.1260	0.2188	
	3D	0.2475	0.2295	0.1342	0.9118	0.1694	0.0597	0.0731	0.1423	0.2459	
	5D	0.2906	0.3102	0.1367	0.6579	0.1763	0.1011	0.1119	0.1884	0.2466	
	7D	0.3160	0.3587	0.1805	1.3613	0.2556	0.1105	0.1087	0.2119	0.3629	
	Avg.	0.2950	0.2600	0.1494	0.9338	0.1791	0.0764	0.0876	0.1671	0.2686	
CNN	1D	0.1391	0.1432	0.0407	0.0778	0.0392	0.0279	0.0255	0.0520	0.0682	
	3D	0.2746	0.2966	0.1185	0.1814	0.0799	0.0575	0.0505	0.0921	0.1439	
	5D	0.3575	0.3305	0.1278	0.2059	0.1303	0.0850	0.0723	0.1257	0.1794	
	7D	0.3067	0.4557	0.1821	0.2844	0.1657	0.1096	0.0950	0.1579	0.2196	
	Avg.	0.2695	0.3065	0.1173	0.1874	0.1038	0.0700	0.0608	0.1069	0.1528	
TimesNet	1D	0.1770	0.1473	0.0429	0.0742	0.0546	0.0449	0.0394	0.0761	0.0821	
	3D	0.3474	0.2545	0.0758	0.1626	0.1049	0.0829	0.0737	0.1365	0.1548	
	5D	0.3436	0.3207	0.1069	0.2093	0.1475	0.1193	0.0939	0.1907	0.1915	
	7D	0.3053	0.4134	0.1426	0.2522	0.2020	0.1648	0.1283	0.2267	0.2294	
	Avg.	0.2933	0.2840	0.0921	0.1746	0.1273	0.1030	0.0838	0.1575	0.1644	
Transformer	iTransformer	1D	0.1277	0.1337	0.0357	0.0674	0.0382	0.0314	0.0295	0.0619	0.0657
		3D	0.2760	0.2269	0.0688	0.1343	0.0755	0.0627	0.0529	0.1057	0.1254
		5D	0.3581	0.3195	0.1004	0.1846	0.1081	0.0855	0.0769	0.1483	0.1727
		7D	0.2999	0.3668	0.1288	0.2222	0.1384	0.1129	0.1003	0.1810	0.1938
		Avg.	0.2654	0.2617	0.0834	0.1521	0.0901	0.0731	0.0649	0.1242	0.1394
	PatchTST	1D	0.1315	0.1368	0.0370	0.0608	0.0385	0.0325	0.0321	0.0593	0.0661
		3D	0.3008	0.2298	0.0845	0.1276	0.0713	0.0645	0.0535	0.1041	0.1295
		5D	0.3504	0.2951	0.1222	0.1736	0.1042	0.0946	0.0777	0.1368	0.1693
		7D	0.3145	0.3605	0.1365	0.2170	0.1412	0.1198	0.0968	0.1708	0.1946
		Avg.	0.2743	0.2555	0.0951	0.1448	0.0888	0.0778	0.0650	0.1177	0.1399

Table 10: Benchmark(MSE) on part 1(Orlando area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
LSTM	1D	0.2408	0.2451	0.0505	0.2788	0.0655	0.0496	0.0435	0.0846	0.1323	
	3D	0.2765	0.3219	0.0784	0.3653	0.1097	0.0768	0.0837	0.1162	0.1786	
	5D	0.3053	0.3853	0.1085	0.4750	0.1569	0.1011	0.0953	0.1525	0.2225	
	7D	0.3058	0.4057	0.1427	0.5133	0.1855	0.1226	0.1103	0.1818	0.2459	
	Avg.	0.2821	0.3395	0.0950	0.4081	0.1294	0.0875	0.0832	0.1338	0.1948	
RNN	DeepAR	1D	0.2406	0.2367	0.0480	0.2190	0.0615	0.0435	0.0386	0.0782	0.1208
		3D	0.2720	0.3088	0.0793	0.4084	0.1117	0.0783	0.0652	0.1212	0.1806
		5D	0.3033	0.3441	0.1072	0.5899	0.1537	0.1003	0.0883	0.1564	0.2304
		7D	0.3046	0.4056	0.1356	0.5250	0.1639	0.1221	0.1070	0.1800	0.2430
		Avg.	0.2801	0.3238	0.0925	0.4356	0.1227	0.0860	0.0748	0.1340	0.1937
DilatedRNN	1D	0.1818	0.1370	0.0372	0.3297	0.0496	0.0356	0.0304	0.0860	0.1109	
	3D	0.2517	0.2674	0.0689	0.3124	0.0887	0.0645	0.0722	0.1150	0.1551	
	5D	0.2822	0.3308	0.1004	0.6438	0.1338	0.0915	0.0916	0.1423	0.2270	
	7D	0.3125	0.3989	0.1242	0.5068	0.1702	0.1159	0.1257	0.0714	0.2282	
	Avg.	0.2571	0.2835	0.0827	0.4482	0.1106	0.0769	0.0800	0.1037	0.1803	
GCN	1D	0.1465	0.5157	0.0443	1.1053	0.0519	0.0353	0.0275	0.0518	0.2473	
	3D	0.2526	1.6031	0.0656	0.7137	0.0944	0.0654	0.0494	0.0891	0.3667	
	5D	0.3055	1.9044	0.0867	1.1121	0.1266	0.0895	0.0730	0.1184	0.4770	
	7D	0.2687	2.2856	0.1071	1.4113	0.1651	0.1087	0.0928	0.1471	0.5733	
	Avg.	0.2433	1.5772	0.0759	1.0856	0.1095	0.0747	0.0607	0.1016	0.4161	
GNN	FourierGNN	1D	0.1619	0.1324	0.0412	0.0901	0.0442	0.0326	0.0275	0.0575	0.0734
		3D	0.2714	0.2058	0.0649	0.2944	0.0746	0.0560	0.0489	0.0941	0.1388
		5D	0.3066	0.2721	0.0891	0.4409	0.1106	0.0806	0.0705	0.1238	0.1868
		7D	0.2766	0.3139	0.1154	0.4880	0.1381	0.1065	0.0919	0.1528	0.2104
		Avg.	0.2541	0.2310	0.0777	0.3283	0.0919	0.0689	0.0597	0.1071	0.1523
StemGNN	1D	0.2290	0.1703	0.0403	1.1688	0.0652	0.0410	0.0325	0.0723	0.2274	
	3D	0.2703	0.4534	0.0903	1.2569	0.1323	0.0896	0.0743	0.1263	0.3117	
	5D	0.3017	0.7711	0.1305	3.3143	0.1663	0.1663	0.1417	0.2080	0.6500	
	7D	0.3225	0.7901	0.1768	3.4285	0.1979	0.2323	0.1489	0.1864	0.6854	
	Avg.	0.2809	0.5462	0.1095	2.2921	0.1404	0.1323	0.0994	0.1483	0.4686	
LLM	GPT4TS	1D	0.1984	0.1399	0.0456	0.0882	0.0542	0.0431	0.0387	0.0935	0.0877
		3D	0.3883	0.2385	0.0811	0.1648	0.0986	0.0788	0.0690	0.1343	0.1567
		5D	0.3668	0.3102	0.0997	0.2037	0.1304	0.1091	0.0870	0.1876	0.1868
		7D	0.2928	0.3700	0.1378	0.2503	0.1622	0.1494	0.1319	0.2233	0.2147
		Avg.	0.3116	0.2646	0.0911	0.1768	0.1114	0.0951	0.0816	0.1597	0.1615
AutoTimes	1D	0.1531	0.1507	0.0401	0.0671	0.0390	0.0310	0.0287	0.0623	0.0715	
	3D	0.3072	0.2333	0.0683	0.1278	0.0686	0.0572	0.0508	0.0978	0.1264	
	5D	0.3527	0.2975	0.0916	0.1700	0.0992	0.0811	0.0715	0.1319	0.1619	
	7D	0.2913	0.3428	0.1181	0.2061	0.1265	0.1031	0.0916	0.1612	0.1801	
	Avg.	0.2761	0.2561	0.0795	0.1427	0.0833	0.0681	0.0606	0.1133	0.1350	

Table 11: Benchmark(SEDI(10%)) on part 1(Orlando area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
MLP	1D	0.8143	0.9334	0.9011	0.8780	0.6133	0.7659	0.9351	0.7772	0.8273	
	3D	0.7599	0.8673	0.8592	0.9286	0.5201	0.7173	0.9040	0.7097	0.7833	
	5D	0.6435	0.8095	0.8142	0.8702	0.4517	0.6956	0.8571	0.6573	0.7249	
	7D	0.6022	0.8056	0.7759	0.8158	0.4198	0.6622	0.8516	0.5969	0.6912	
	Avg.	0.7050	0.8540	0.8376	0.8731	0.5012	0.7103	0.8870	0.6853	0.7567	
MLP	1D	0.7483	0.8690	0.8917	0.9141	0.5971	0.7394	0.7901	0.7191	0.7836	
	3D	0.5937	0.7493	0.8103	0.8420	0.4798	0.6766	0.7626	0.6334	0.6935	
	5D	0.5167	0.6755	0.7538	0.7783	0.4139	0.6214	0.7289	0.5685	0.6321	
	7D	0.4706	0.6241	0.7042	0.7236	0.3734	0.5854	0.6798	0.5107	0.5840	
	Avg.	0.5823	0.7295	0.7900	0.8145	0.4660	0.6557	0.7403	0.6079	0.6733	
NLinear	1D	0.7729	0.8803	0.8951	0.9171	0.5984	0.7844	0.9259	0.7302	0.8130	
	3D	0.6007	0.7581	0.8166	0.8390	0.4784	0.6961	0.8550	0.6410	0.7106	
	5D	0.5198	0.6821	0.7576	0.7721	0.4129	0.6535	0.7956	0.5750	0.6461	
	7D	0.4731	0.6290	0.7063	0.7159	0.3701	0.6155	0.7512	0.5173	0.5973	
	Avg.	0.5916	0.7374	0.7939	0.8110	0.4649	0.6873	0.8319	0.6159	0.6918	
TCN	1D	0.5634	0.8939	0.7243	0.7307	0.4343	0.7156	0.8132	0.5039	0.6724	
	3D	0.5462	0.8010	0.6694	0.7131	0.3092	0.7072	0.7778	0.3687	0.6116	
	5D	0.4606	0.7638	0.6291	0.6572	0.2804	0.6123	0.7136	0.3351	0.5565	
	7D	0.4598	0.7014	0.6092	0.5674	0.2201	0.5762	0.5692	0.3620	0.5082	
	Avg.	0.5075	0.7900	0.6580	0.6671	0.3110	0.6528	0.7184	0.3924	0.5872	
CNN	1D	0.7912	0.8737	0.8857	0.8915	0.6023	0.7692	0.9240	0.7395	0.8096	
	3D	0.6031	0.7637	0.7893	0.8041	0.4970	0.6746	0.8469	0.6370	0.7020	
	5D	0.4960	0.6839	0.7201	0.7403	0.4277	0.6142	0.7898	0.5668	0.6298	
	7D	0.4357	0.6233	0.6667	0.6818	0.3869	0.5726	0.7499	0.5076	0.5781	
	Avg.	0.5815	0.7362	0.7654	0.7794	0.4785	0.6576	0.8277	0.6127	0.6799	
TimesNet	1D	0.7224	0.8584	0.8701	0.8789	0.5727	0.7000	0.9025	0.7044	0.7762	
	3D	0.5820	0.7493	0.7919	0.7959	0.4368	0.6345	0.7953	0.5956	0.6727	
	5D	0.4866	0.6771	0.7336	0.7370	0.3854	0.5657	0.7453	0.5191	0.6062	
	7D	0.4083	0.6298	0.6863	0.6808	0.3325	0.5181	0.7137	0.4662	0.5545	
	Avg.	0.5498	0.7286	0.7705	0.7732	0.4318	0.6045	0.7892	0.5713	0.6524	
Transformer	iTransformer	1D	0.8049	0.8881	0.9039	0.9057	0.5997	0.7663	0.9234	0.7186	0.8138
		3D	0.6433	0.7820	0.8294	0.8350	0.4807	0.6836	0.8500	0.6270	0.7164
		5D	0.5540	0.7202	0.7733	0.7744	0.4121	0.6302	0.7956	0.5481	0.6510
		7D	0.5043	0.6631	0.7177	0.7288	0.3558	0.5843	0.7486	0.4997	0.6003
		Avg.	0.6266	0.7634	0.8061	0.8110	0.4621	0.6661	0.8294	0.5983	0.6954
	PatchTST	1D	0.8082	0.8872	0.8947	0.9082	0.5937	0.7683	0.9141	0.7252	0.8125
		3D	0.6607	0.7851	0.8203	0.8352	0.4994	0.6789	0.8537	0.6293	0.7203
		5D	0.5672	0.7188	0.7626	0.7659	0.4175	0.6272	0.7914	0.5625	0.6516
		7D	0.5081	0.6574	0.7167	0.7186	0.3708	0.5887	0.7506	0.5049	0.6020
		Avg.	0.6361	0.7621	0.7986	0.8070	0.4704	0.6658	0.8274	0.6055	0.6966

Table 12: Benchmark(SED(10%)) on part 1(Orlando area) stations(RNN, GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.7709	0.8270	0.8825	0.8781	0.5134	0.7182	0.9039	0.7302	0.7780
	3D	0.6027	0.8070	0.8162	0.8833	0.4448	0.5963	0.8482	0.6473	0.7057
	5D	0.5247	0.7396	0.7646	0.7848	0.3667	0.6529	0.7852	0.5842	0.6503
	7D	0.4409	0.6678	0.7459	0.7427	0.3836	0.6063	0.7591	0.4874	0.6042
	Avg.	0.5848	0.7604	0.8023	0.8222	0.4271	0.6434	0.8241	0.6123	0.6846
DeepAR	1D	0.7411	0.8286	0.9006	0.9469	0.5732	0.7033	0.9113	0.7681	0.7966
	3D	0.6168	0.7816	0.7859	0.7710	0.4175	0.6046	0.8704	0.6535	0.6877
	5D	0.5086	0.7014	0.7872	0.8217	0.3716	0.6180	0.7106	0.6096	0.6411
	7D	0.4686	0.7151	0.7627	0.8162	0.3346	0.6131	0.6993	0.5065	0.6145
	Avg.	0.5838	0.7567	0.8091	0.8390	0.4242	0.6348	0.7979	0.6344	0.6850
DilatedRNN	1D	0.8033	0.9139	0.8972	0.9067	0.6086	0.7706	0.9528	0.7437	0.8246
	3D	0.6053	0.8119	0.7907	0.8376	0.4752	0.5884	0.8004	0.6515	0.6951
	5D	0.5254	0.7325	0.7443	0.8017	0.4259	0.6454	0.8208	0.6270	0.6654
	7D	0.4664	0.7255	0.7337	0.7261	0.4031	0.6275	0.7647	0.5970	0.6305
	Avg.	0.6001	0.7960	0.7915	0.8180	0.4782	0.6580	0.8347	0.6548	0.7039
GCN	1D	0.8018	0.7777	0.8496	0.7072	0.5680	0.7274	0.9103	0.7172	0.7574
	3D	0.6240	0.7415	0.8169	0.6969	0.5115	0.6977	0.8412	0.6788	0.7011
	5D	0.5324	0.7147	0.7394	0.6223	0.4217	0.6488	0.8378	0.6075	0.6406
	7D	0.4949	0.6795	0.6887	0.5567	0.3824	0.6114	0.7941	0.5809	0.5986
	Avg.	0.6133	0.7283	0.7737	0.6458	0.4709	0.6713	0.8458	0.6461	0.6744
FourierGNN	1D	0.8505	0.8952	0.9035	0.8991	0.5899	0.7981	0.9211	0.7834	0.8301
	3D	0.6929	0.8059	0.8177	0.7987	0.5029	0.6747	0.8521	0.6853	0.7288
	5D	0.6012	0.7985	0.7753	0.7296	0.4659	0.6222	0.8349	0.6308	0.6823
	7D	0.5372	0.7731	0.7619	0.6927	0.4257	0.6830	0.8133	0.5499	0.6546
	Avg.	0.6705	0.8182	0.8146	0.7800	0.4961	0.6945	0.8554	0.6624	0.7240
StemGNN	1D	0.6812	0.8950	0.8559	0.4137	0.5486	0.7748	0.9207	0.7578	0.7310
	3D	0.5934	0.7234	0.8229	0.3851	0.4289	0.6430	0.8470	0.6376	0.6352
	5D	0.5151	0.5492	0.6765	0.0829	0.3765	0.4472	0.7608	0.5663	0.4968
	7D	0.4576	0.5168	0.6141	0.0549	0.3011	0.4146	0.8487	0.5028	0.4638
	Avg.	0.5618	0.6711	0.7423	0.2341	0.4138	0.5699	0.8443	0.6161	0.5817
GPT4TS	1D	0.7588	0.8697	0.8790	0.8738	0.5594	0.7173	0.9066	0.6863	0.7814
	3D	0.5816	0.7515	0.8012	0.7983	0.4484	0.6322	0.8277	0.5967	0.6797
	5D	0.4917	0.6840	0.7480	0.7384	0.3970	0.5830	0.7591	0.5241	0.6157
	7D	0.4388	0.6309	0.6941	0.6878	0.3582	0.5430	0.7099	0.4672	0.5662
	Avg.	0.5677	0.7340	0.7805	0.7746	0.4408	0.6189	0.8008	0.5686	0.6607
AutoTimes	1D	0.7957	0.8638	0.8896	0.9167	0.5985	0.7343	0.9218	0.7225	0.8054
	3D	0.6299	0.7622	0.8190	0.8520	0.4892	0.6712	0.8516	0.6389	0.7142
	5D	0.5392	0.6780	0.7622	0.7817	0.4135	0.6243	0.7982	0.5804	0.6472
	7D	0.4895	0.6363	0.7204	0.7309	0.3710	0.5879	0.7553	0.5180	0.6012
	Avg.	0.6136	0.7351	0.7978	0.8203	0.4680	0.6544	0.8317	0.6149	0.6920

Table 13: Benchmark(SED(5%)) on part 1(Orlando area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.5128	0.9250	0.7735	0.8437	0.4047	0.7105	0.8731	0.6329	0.7095
	3D	0.5032	0.8575	0.7227	0.9190	0.3281	0.6590	0.8035	0.5460	0.6674
	5D	0.4580	0.7990	0.6804	0.8465	0.2779	0.6314	0.7573	0.5308	0.6227
	7D	0.4174	0.8105	0.6355	0.7808	0.2684	0.5818	0.7227	0.4304	0.5809
	Avg.	0.4729	0.8480	0.7030	0.8475	0.3198	0.6457	0.7892	0.5350	0.6451
MLP	1D	0.5204	0.8556	0.7619	0.8916	0.3911	0.7094	0.7193	0.6167	0.6833
	3D	0.3386	0.7182	0.6688	0.7902	0.3201	0.6698	0.6612	0.4533	0.5775
	5D	0.2734	0.6303	0.6004	0.7015	0.2664	0.5025	0.6127	0.3537	0.4926
	7D	0.2257	0.5757	0.5444	0.6277	0.2313	0.4536	0.5548	0.2847	0.4372
	Avg.	0.3395	0.6950	0.6439	0.7528	0.3022	0.5838	0.6370	0.4271	0.5477
NLinear	1D	0.5802	0.8689	0.7743	0.9014	0.4122	0.7091	0.8529	0.6285	0.7159
	3D	0.3752	0.7312	0.6721	0.7904	0.3451	0.6185	0.7482	0.4648	0.5932
	5D	0.3456	0.6392	0.6011	0.6967	0.2872	0.5471	0.6727	0.3603	0.5187
	7D	0.2322	0.5818	0.5408	0.6184	0.2505	0.4979	0.6121	0.2912	0.4531
	Avg.	0.3833	0.7053	0.6471	0.7517	0.3238	0.5932	0.7215	0.4362	0.5702
TCN	1D	0.3567	0.8742	0.2636	0.6239	0.2001	0.6360	0.4088	0.4724	0.4795
	3D	0.4252	0.7796	0.3238	0.6102	0.1573	0.4839	0.4081	0.3764	0.4456
	5D	0.2808	0.7319	0.2196	0.5901	0.1484	0.3628	0.4501	0.3326	0.3895
	7D	0.3380	0.6245	0.2753	0.4466	0.1223	0.3825	0.2879	0.2875	0.3456
	Avg.	0.3502	0.7526	0.2706	0.5677	0.1570	0.4663	0.3887	0.3672	0.4150
CNN	1D	0.5274	0.8480	0.7496	0.8700	0.3762	0.7081	0.8590	0.6339	0.6965
	3D	0.3525	0.7260	0.6128	0.7416	0.3183	0.5902	0.7544	0.4769	0.5716
	5D	0.2805	0.6333	0.5298	0.6535	0.2708	0.5110	0.6824	0.3742	0.4919
	7D	0.2334	0.5539	0.4603	0.5819	0.2352	0.4597	0.6208	0.3082	0.4317
	Avg.	0.3484	0.6903	0.5881	0.7118	0.3001	0.5673	0.7292	0.4483	0.5479
TimesNet	1D	0.4528	0.8341	0.7355	0.8507	0.3561	0.5952	0.8295	0.5725	0.6533
	3D	0.3185	0.7133	0.6279	0.7271	0.2782	0.5261	0.7186	0.4477	0.5447
	5D	0.2639	0.6133	0.5616	0.6561	0.2305	0.4305	0.6461	0.3400	0.4678
	7D	0.2184	0.5629	0.4993	0.5816	0.1851	0.3748	0.5961	0.2652	0.4104
	Avg.	0.3134	0.6809	0.6061	0.7039	0.2625	0.4816	0.6976	0.4064	0.5190
Transformer	iTransformer	1D	0.5441	0.8814	0.7818	0.8940	0.3864	0.7065	0.8510	0.5962
		3D	0.3974	0.7637	0.6710	0.7924	0.3104	0.5943	0.7502	0.4656
		5D	0.3080	0.6876	0.6000	0.7083	0.2453	0.5257	0.6786	0.3554
		7D	0.2514	0.6234	0.5337	0.6420	0.2047	0.4701	0.6208	0.2896
		Avg.	0.3752	0.7390	0.6466	0.7592	0.2867	0.5742	0.7252	0.4267
	PatchTST	1D	0.5491	0.8715	0.7655	0.8953	0.3878	0.7004	0.8402	0.6004
		3D	0.3800	0.7584	0.6663	0.7948	0.3210	0.5907	0.7519	0.4659
		5D	0.3052	0.6788	0.5937	0.6959	0.2631	0.5219	0.6719	0.3734
		7D	0.2502	0.6018	0.5339	0.6317	0.2196	0.4734	0.6156	0.3054
		Avg.	0.3711	0.7276	0.6399	0.7544	0.2979	0.5716	0.7199	0.4363

Table 14: Benchmark(SEDI(5%)) on part 1(Orlando area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.4017	0.8107	0.7419	0.8659	0.3198	0.6701	0.8120	0.5639	0.6483
	3D	0.3042	0.7513	0.6543	0.8644	0.2781	0.5208	0.6626	0.3957	0.5539
	5D	0.2563	0.7351	0.5817	0.7538	0.2252	0.4925	0.6287	0.3346	0.5010
	7D	0.2557	0.6068	0.5389	0.7051	0.2231	0.4831	0.5848	0.2670	0.4581
	Avg.	0.3045	0.7260	0.6292	0.7973	0.2615	0.5416	0.6720	0.3903	0.5403
DeepAR	1D	0.5253	0.8201	0.7607	0.9462	0.3488	0.6711	0.8081	0.5904	0.6838
	3D	0.3304	0.7414	0.6407	0.7018	0.2825	0.5496	0.6188	0.4553	0.5401
	5D	0.2763	0.6725	0.5825	0.8152	0.2105	0.5123	0.5172	0.3650	0.4939
	7D	0.2556	0.6896	0.5741	0.8037	0.2025	0.4243	0.4710	0.3234	0.4680
	Avg.	0.3469	0.7309	0.6395	0.8167	0.2611	0.5393	0.6038	0.4335	0.5465
DilatedRNN	1D	0.5340	0.8917	0.7632	0.8782	0.3685	0.7112	0.8792	0.6419	0.7085
	3D	0.3663	0.8079	0.6105	0.7865	0.2975	0.5017	0.6871	0.4803	0.5672
	5D	0.3059	0.6907	0.5751	0.7343	0.2560	0.5750	0.6966	0.4549	0.5361
	7D	0.3052	0.6703	0.5421	0.6376	0.2480	0.5455	0.6476	0.4271	0.5029
	Avg.	0.3779	0.7652	0.6227	0.7591	0.2925	0.5833	0.7276	0.5010	0.5787
GCN	1D	0.5137	0.7241	0.7046	0.6410	0.3511	0.7351	0.8413	0.5751	0.6357
	3D	0.3818	0.6910	0.6424	0.6378	0.3101	0.7135	0.7400	0.4640	0.5726
	5D	0.3292	0.6557	0.5711	0.5477	0.2499	0.6674	0.7444	0.4156	0.5226
	7D	0.3021	0.6169	0.5142	0.4740	0.2346	0.5327	0.6955	0.4517	0.4777
	Avg.	0.3817	0.6719	0.6081	0.5751	0.2864	0.6622	0.7553	0.4766	0.5522
FourierGNN	1D	0.5676	0.8926	0.7800	0.8434	0.3851	0.7068	0.8439	0.6727	0.7115
	3D	0.4629	0.7868	0.6564	0.7028	0.3275	0.5714	0.7651	0.5707	0.6054
	5D	0.3950	0.7889	0.6028	0.6333	0.2748	0.5038	0.7352	0.5055	0.5549
	7D	0.3873	0.7530	0.6115	0.5815	0.2572	0.5880	0.7235	0.4659	0.5460
	Avg.	0.4532	0.8053	0.6626	0.6903	0.3112	0.5925	0.7669	0.5537	0.6045
StemGNN	1D	0.3886	0.8590	0.7176	0.2562	0.3499	0.7175	0.8579	0.6320	0.5973
	3D	0.3247	0.6710	0.6642	0.1607	0.2578	0.5487	0.7490	0.5132	0.4862
	5D	0.2763	0.4268	0.4857	0.0234	0.2251	0.3075	0.6028	0.3641	0.3390
	7D	0.2189	0.3938	0.3786	0.0456	0.1851	0.2623	0.7196	0.3385	0.3178
	Avg.	0.3021	0.5877	0.5615	0.1215	0.2545	0.4590	0.7324	0.4619	0.4351
GPT4TS	1D	0.4904	0.8517	0.7383	0.8528	0.3567	0.6286	0.8333	0.5737	0.6657
	3D	0.3418	0.7131	0.6424	0.7412	0.2897	0.5298	0.7352	0.4407	0.5542
	5D	0.2811	0.6474	0.5588	0.6647	0.2451	0.4542	0.6619	0.3371	0.4813
	7D	0.2508	0.5786	0.4827	0.6002	0.2173	0.4100	0.5996	0.2547	0.4243
	Avg.	0.3410	0.6977	0.6055	0.7147	0.2772	0.5056	0.7075	0.4016	0.5314
AutoTimes	1D	0.5235	0.8448	0.7499	0.9034	0.3786	0.6736	0.8310	0.5961	0.6876
	3D	0.3733	0.7387	0.6568	0.8177	0.3245	0.5828	0.7471	0.4660	0.5884
	5D	0.3023	0.6336	0.5923	0.7174	0.2638	0.5145	0.6738	0.3625	0.5075
	7D	0.2343	0.5856	0.5382	0.6401	0.2301	0.4620	0.6092	0.3000	0.4499
	Avg.	0.3584	0.7007	0.6343	0.7696	0.2993	0.5582	0.7153	0.4311	0.5584

Table 15: Benchmark(SEDI(1%)) on part 1(Orlando area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.2398	0.8330	0.3903	0.7439	0.2199	0.2365	0.4548	0.1809	0.4124
	3D	0.1994	0.7910	0.2967	0.7730	0.2051	0.1198	0.3723	0.1528	0.3638
	5D	0.1508	0.7647	0.2900	0.7580	0.1927	0.1346	0.3785	0.1380	0.3509
	7D	0.1877	0.7879	0.2579	0.6627	0.1935	0.1097	0.3831	0.1107	0.3367
	Avg.	0.1944	0.7942	0.3087	0.7344	0.2028	0.1502	0.3972	0.1456	0.3660
MLP	1D	0.2154	0.7569	0.3670	0.7701	0.2092	0.1228	0.2195	0.1553	0.3520
	3D	0.1462	0.5898	0.2661	0.5455	0.1401	0.0936	0.1884	0.0939	0.2580
	5D	0.1104	0.4818	0.1939	0.4161	0.1069	0.0697	0.1647	0.0636	0.2009
	7D	0.1078	0.3947	0.1464	0.3318	0.0871	0.0548	0.1533	0.0456	0.1652
	Avg.	0.1450	0.5558	0.2433	0.5159	0.1358	0.0852	0.1815	0.0896	0.2440
NLinear	1D	0.2205	0.7772	0.3813	0.7807	0.2302	0.2270	0.3975	0.1560	0.3963
	3D	0.1508	0.6056	0.2785	0.5478	0.1460	0.1937	0.3198	0.0888	0.2914
	5D	0.1156	0.4949	0.2062	0.4135	0.1131	0.1686	0.2773	0.0607	0.2313
	7D	0.1104	0.4048	0.1576	0.3231	0.0919	0.1536	0.2289	0.0453	0.1895
	Avg.	0.1493	0.5706	0.2559	0.5163	0.1453	0.1857	0.3059	0.0877	0.2771
TCN	1D	0.1520	0.7471	0.1639	0.2160	0.0801	0.0385	0.0000	0.0404	0.1797
	3D	0.1481	0.6252	0.1179	0.3218	0.0634	0.0303	0.0232	0.0397	0.1712
	5D	0.1935	0.5904	0.0941	0.2732	0.0516	0.0161	0.0496	0.0063	0.1593
	7D	0.1863	0.3808	0.1512	0.1816	0.0468	0.0121	0.0149	0.0185	0.1240
	Avg.	0.1700	0.5859	0.1318	0.2481	0.0605	0.0243	0.0219	0.0262	0.1586
CNN	1D	0.1937	0.7645	0.3466	0.7558	0.2153	0.2077	0.3743	0.1827	0.3801
	3D	0.1202	0.6078	0.2371	0.4858	0.1331	0.1410	0.2984	0.1105	0.2667
	5D	0.0829	0.5308	0.1742	0.3875	0.1018	0.1315	0.2742	0.0778	0.2201
	7D	0.0804	0.4283	0.1325	0.3070	0.0795	0.0623	0.2074	0.0606	0.1698
	Avg.	0.1193	0.5829	0.2226	0.4840	0.1324	0.1356	0.2886	0.1079	0.2592
TimesNet	1D	0.1451	0.7329	0.3521	0.6753	0.1455	0.1175	0.3185	0.1265	0.3267
	3D	0.0951	0.5803	0.2451	0.4443	0.1134	0.0911	0.2359	0.0766	0.2352
	5D	0.0651	0.4805	0.1894	0.3933	0.0900	0.0482	0.2077	0.0577	0.1915
	7D	0.0372	0.3977	0.1499	0.2984	0.0631	0.0399	0.1878	0.0400	0.1518
	Avg.	0.0856	0.5479	0.2341	0.4528	0.1030	0.0742	0.2375	0.0752	0.2263
Transformer	1D	0.2066	0.7969	0.3789	0.7930	0.2251	0.1355	0.3441	0.1508	0.3789
	3D	0.1275	0.6825	0.2843	0.5750	0.1318	0.0913	0.2617	0.0926	0.2808
	5D	0.0961	0.5919	0.2149	0.4326	0.0930	0.0727	0.2305	0.0576	0.2237
	7D	0.0714	0.4919	0.1664	0.3504	0.0721	0.0542	0.1959	0.0448	0.1809
	Avg.	0.1254	0.6408	0.2611	0.5378	0.1305	0.0884	0.2581	0.0864	0.2661
PatchTST	1D	0.2149	0.7937	0.3601	0.8041	0.2110	0.1291	0.3226	0.1593	0.3744
	3D	0.1327	0.6565	0.2770	0.5851	0.1429	0.0968	0.2652	0.0913	0.2809
	5D	0.0915	0.5578	0.2143	0.4411	0.1047	0.0734	0.2243	0.0673	0.2218
	7D	0.0757	0.4552	0.1661	0.3639	0.0799	0.0570	0.1975	0.0518	0.1809
	Avg.	0.1287	0.6158	0.2544	0.5485	0.1346	0.0891	0.2524	0.0924	0.2645

Table 16: Benchmark(SEDI(1%)) on part 1(Orlando area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
LSTM	1D	0.1231	0.5677	0.3286	0.5730	0.1160	0.1235	0.2804	0.1423	0.2818	
	3D	0.0722	0.3377	0.2315	0.4083	0.1017	0.0801	0.1284	0.0580	0.1772	
	5D	0.0462	0.4222	0.1899	0.1858	0.0782	0.0618	0.2246	0.0612	0.1587	
	7D	0.0583	0.2870	0.0658	0.3472	0.0747	0.0526	0.1112	0.0411	0.1297	
	Avg.	0.0749	0.4036	0.2040	0.3786	0.0927	0.0795	0.1862	0.0756	0.1869	
RNN	DeepAR	1D	0.1138	0.6157	0.3092	0.4319	0.1276	0.1392	0.2927	0.1144	0.2681
		3D	0.0954	0.3621	0.1573	0.2978	0.0974	0.1047	0.1110	0.0291	0.1568
		5D	0.0588	0.3228	0.1519	0.2054	0.0821	0.0691	0.0985	0.0716	0.1325
		7D	0.1016	0.2944	0.0939	0.0666	0.0793	0.0690	0.1249	0.0787	0.1135
		Avg.	0.0924	0.3987	0.1781	0.2504	0.0966	0.0955	0.1568	0.0734	0.1677
DilatedRNN	1D	0.2111	0.7473	0.3720	0.7648	0.1895	0.1384	0.3324	0.2115	0.3709	
	3D	0.1417	0.6452	0.2337	0.5917	0.1539	0.0764	0.1583	0.1111	0.2640	
	5D	0.1170	0.5410	0.1941	0.4338	0.1278	0.0712	0.2353	0.1309	0.2314	
	7D	0.0923	0.4478	0.1716	0.3167	0.1039	0.0766	0.1639	0.0805	0.1817	
	Avg.	0.1405	0.5953	0.2428	0.5267	0.1438	0.0906	0.2225	0.1335	0.2620	
GCN	1D	0.2025	0.5785	0.3234	0.4610	0.2208	0.1729	0.3612	0.1966	0.3146	
	3D	0.1393	0.5720	0.2439	0.4175	0.0475	0.1439	0.3082	0.1379	0.2513	
	5D	0.1185	0.5251	0.1685	0.3352	0.0296	0.1351	0.3633	0.1096	0.2231	
	7D	0.1255	0.4695	0.1648	0.2308	0.0269	0.1249	0.3558	0.1684	0.2083	
	Avg.	0.1465	0.5363	0.2252	0.3611	0.0812	0.1442	0.3471	0.1531	0.2493	
GNN	FourierGNN	1D	0.2074	0.7982	0.3768	0.7408	0.2155	0.1250	0.3068	0.1698	0.3675
		3D	0.1567	0.7499	0.2746	0.6374	0.1849	0.1035	0.2813	0.1395	0.3160
		5D	0.1326	0.7111	0.2177	0.5194	0.1881	0.1060	0.2724	0.1089	0.2820
		7D	0.1830	0.6629	0.2348	0.4327	0.1766	0.1203	0.3300	0.1289	0.2836
		Avg.	0.1699	0.7305	0.2760	0.5826	0.1913	0.1137	0.2976	0.1368	0.3123
StemGNN	1D	0.1717	0.4968	0.3191	0.0206	0.1620	0.1239	0.2858	0.2005	0.2225	
	3D	0.1410	0.3176	0.2135	0.0176	0.1152	0.0530	0.1569	0.1344	0.1437	
	5D	0.1395	0.0319	0.1679	0.0013	0.0691	0.0051	0.1753	0.0661	0.0820	
	7D	0.1056	0.0337	0.1386	0.0003	0.0784	0.0036	0.2534	0.0487	0.0828	
	Avg.	0.1395	0.2200	0.2098	0.0100	0.1062	0.0464	0.2179	0.1124	0.1328	
LLM	GPT4TS	1D	0.1732	0.7775	0.3626	0.7129	0.1662	0.1221	0.3125	0.1272	0.3443
		3D	0.1047	0.6083	0.2679	0.4717	0.1214	0.0922	0.2501	0.0771	0.2492
		5D	0.0698	0.5106	0.1923	0.3910	0.0886	0.0498	0.2192	0.0558	0.1971
		7D	0.0579	0.4151	0.1366	0.3045	0.0733	0.0356	0.2041	0.0334	0.1576
		Avg.	0.1014	0.5779	0.2399	0.4700	0.1124	0.0749	0.2465	0.0734	0.2370
AutoTimes	1D	0.1985	0.7427	0.3622	0.8053	0.2213	0.1268	0.3335	0.1362	0.3658	
	3D	0.1179	0.6341	0.2763	0.5863	0.1497	0.0970	0.2677	0.0801	0.2761	
	5D	0.0788	0.5089	0.2057	0.4334	0.1039	0.0723	0.2216	0.0601	0.2106	
	7D	0.0714	0.4392	0.1579	0.3310	0.0873	0.0559	0.1895	0.0444	0.1721	
	Avg.	0.1167	0.5812	0.2505	0.5390	0.1406	0.0880	0.2531	0.0802	0.2562	

Table 17: Benchmark(MAE) on part 2(Miami area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
MLP	1D	0.0739	0.0969	0.1538	0.1234	0.1172	0.1069	0.1168	0.1548	0.1180	
	3D	0.1588	0.1621	0.2502	0.2129	0.1952	0.1638	0.1855	0.2459	0.1968	
	5D	0.2147	0.2079	0.3080	0.2637	0.2568	0.1984	0.2268	0.3070	0.2479	
	7D	0.2650	0.2436	0.3517	0.3046	0.2989	0.2236	0.2595	0.3454	0.2865	
	Avg.	0.1781	0.1776	0.2659	0.2262	0.2170	0.1732	0.1971	0.2633	0.2123	
MLP	1D	0.0932	0.1147	0.1777	0.1381	0.1322	0.1152	0.1279	0.2124	0.1389	
	3D	0.1637	0.1770	0.2667	0.2123	0.2056	0.1680	0.1926	0.2856	0.2089	
	5D	0.2179	0.2209	0.3239	0.2646	0.2615	0.2014	0.2349	0.3376	0.2578	
	7D	0.2621	0.2558	0.3663	0.3055	0.3081	0.2258	0.2663	0.3779	0.2960	
	Avg.	0.1842	0.1921	0.2836	0.2301	0.2269	0.1776	0.2054	0.3034	0.2254	
NLinear	1D	0.0857	0.1055	0.1672	0.1274	0.1207	0.1085	0.1191	0.1484	0.1228	
	3D	0.1603	0.1709	0.2606	0.2059	0.1988	0.1642	0.1882	0.2414	0.1988	
	5D	0.2153	0.2167	0.3190	0.2608	0.2560	0.1984	0.2311	0.3010	0.2498	
	7D	0.2603	0.2526	0.3618	0.3022	0.3031	0.2237	0.2635	0.3457	0.2891	
	Avg.	0.1804	0.1864	0.2772	0.2241	0.2196	0.1737	0.2005	0.2591	0.2151	
TCN	1D	0.1747	0.1498	0.1937	0.1790	0.1882	0.1957	0.1439	0.7995	0.2531	
	3D	0.2303	0.1938	0.2971	0.2766	0.2811	0.2347	0.2049	0.9442	0.3328	
	5D	0.3770	0.2350	0.3711	0.3214	0.3427	0.2733	0.2633	0.8945	0.3848	
	7D	0.3407	0.2960	0.3925	0.3659	0.3887	0.3089	0.2950	1.0090	0.4246	
	Avg.	0.2807	0.2186	0.3136	0.2857	0.3002	0.2531	0.2268	0.9118	0.3488	
CNN	1D	0.0777	0.1049	0.1764	0.1385	0.1134	0.1034	0.1154	0.1565	0.1233	
	3D	0.1611	0.1813	0.2938	0.2362	0.1982	0.1669	0.1906	0.2810	0.2136	
	5D	0.2289	0.2449	0.3621	0.3098	0.2576	0.2038	0.2351	0.3383	0.2726	
	7D	0.2809	0.2903	0.4201	0.3565	0.3054	0.2314	0.2702	0.3758	0.3163	
	Avg.	0.1871	0.2054	0.3131	0.2603	0.2186	0.1764	0.2028	0.2879	0.2315	
TimesNet	1D	0.0962	0.1269	0.1984	0.1537	0.1505	0.1309	0.1520	0.1927	0.1502	
	3D	0.1684	0.1943	0.2871	0.2323	0.2249	0.1854	0.2122	0.2767	0.2227	
	5D	0.2212	0.2409	0.3516	0.2849	0.2861	0.2200	0.2587	0.3347	0.2748	
	7D	0.2722	0.2761	0.3937	0.3329	0.3326	0.2409	0.2915	0.3775	0.3147	
	Avg.	0.1895	0.2095	0.3077	0.2509	0.2485	0.1943	0.2286	0.2954	0.2406	
Transformer	iTransformer	1D	0.0731	0.0925	0.1477	0.1178	0.1129	0.1024	0.1104	0.1409	0.1122
		3D	0.1446	0.1603	0.2490	0.1975	0.1901	0.1593	0.1807	0.2334	0.1894
		5D	0.2057	0.2052	0.3095	0.2518	0.2498	0.1938	0.2248	0.3007	0.2427
		7D	0.2513	0.2460	0.3529	0.2933	0.2968	0.2193	0.2555	0.3480	0.2829
		Avg.	0.1687	0.1760	0.2648	0.2151	0.2124	0.1687	0.1928	0.2558	0.2068
	PatchTST	1D	0.0691	0.0926	0.1472	0.1158	0.1114	0.1012	0.1104	0.1394	0.1109
		3D	0.1419	0.1610	0.2449	0.1975	0.1927	0.1590	0.1828	0.2376	0.1897
		5D	0.2002	0.2094	0.3042	0.2524	0.2504	0.1943	0.2251	0.2999	0.2420
		7D	0.2463	0.2461	0.3499	0.2947	0.3005	0.2200	0.2574	0.3444	0.2824
		Avg.	0.1644	0.1773	0.2615	0.2151	0.2138	0.1686	0.1939	0.2553	0.2062

Table 18: Benchmark(MAE) on part 2(Miami area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.1150	0.1373	0.2165	0.1678	0.1691	0.1378	0.1531	0.2537	0.1688
	3D	0.2287	0.2081	0.3074	0.2574	0.2650	0.2002	0.2272	0.3706	0.2581
	5D	0.2646	0.2556	0.3832	0.3241	0.3190	0.2378	0.2694	0.4339	0.3109
	7D	0.3326	0.2989	0.4425	0.3728	0.3855	0.2579	0.2909	0.4743	0.3569
	Avg.	0.2352	0.2250	0.3374	0.2805	0.2847	0.2084	0.2352	0.3831	0.2737
RNN DeepAR	1D	0.1196	0.1356	0.2067	0.1678	0.1659	0.1410	0.1546	0.2557	0.1684
	3D	0.2086	0.2169	0.3113	0.2615	0.2693	0.2047	0.2301	0.3624	0.2581
	5D	0.2839	0.2586	0.3710	0.3123	0.3166	0.2299	0.2790	0.4224	0.3092
	7D	0.3404	0.2992	0.4478	0.3695	0.3657	0.2598	0.3007	0.4670	0.3562
	Avg.	0.2381	0.2276	0.3342	0.2778	0.2794	0.2088	0.2411	0.3769	0.2730
DilatedRNN	1D	0.0781	0.1134	0.1706	0.1359	0.1309	0.1155	0.1217	0.2111	0.1347
	3D	0.1724	0.1856	0.2843	0.2268	0.2393	0.1769	0.2149	0.3164	0.2271
	5D	0.2463	0.2351	0.3483	0.2967	0.3012	0.2133	0.2548	0.3866	0.2853
	7D	0.3312	0.2732	0.4052	0.3472	0.3469	0.2460	0.2917	0.4338	0.3344
	Avg.	0.2070	0.2018	0.3021	0.2516	0.2546	0.1879	0.2208	0.3370	0.2454
GCN	1D	0.0910	0.1254	0.1880	0.1761	0.1626	0.1374	0.1528	0.9648	0.2498
	3D	0.1718	0.1904	0.2843	0.2485	0.2410	0.1928	0.2173	0.5350	0.2601
	5D	0.2460	0.2340	0.3390	0.3061	0.3085	0.2319	0.2571	0.7279	0.3313
	7D	0.2998	0.2700	0.3911	0.3452	0.3508	0.2609	0.2914	0.6410	0.3563
	Avg.	0.2022	0.2050	0.3006	0.2690	0.2657	0.2057	0.2297	0.7172	0.2994
GNN FourierGNN	1D	0.0825	0.1072	0.1604	0.1365	0.1226	0.1165	0.1234	0.2444	0.1367
	3D	0.1593	0.1722	0.2527	0.2059	0.2024	0.1734	0.1925	0.3399	0.2123
	5D	0.2109	0.2189	0.3114	0.2658	0.2568	0.2123	0.2320	0.4031	0.2639
	7D	0.2896	0.2506	0.3590	0.3121	0.3067	0.2357	0.2624	0.4222	0.3048
	Avg.	0.1856	0.1872	0.2709	0.2301	0.2221	0.1845	0.2026	0.3524	0.2294
StemGNN	1D	0.0859	0.1404	0.1762	0.1382	0.1444	0.1204	0.1252	0.3656	0.1620
	3D	0.2047	0.2609	0.2958	0.2616	0.2763	0.2432	0.2227	0.4821	0.2809
	5D	0.2874	0.3173	0.3743	0.3535	0.4289	0.2839	0.2765	0.5247	0.3558
	7D	0.3603	0.3402	0.4218	0.4172	0.4253	0.3211	0.3283	0.6507	0.4081
	Avg.	0.2346	0.2647	0.3170	0.2926	0.3187	0.2422	0.2382	0.5058	0.3017
LLM GPT4TS	1D	0.0931	0.1264	0.1860	0.1431	0.1392	0.1304	0.1510	0.1974	0.1458
	3D	0.1639	0.1959	0.2833	0.2283	0.2281	0.1895	0.2159	0.2917	0.2246
	5D	0.2256	0.2429	0.3469	0.2834	0.2897	0.2195	0.2535	0.3389	0.2751
	7D	0.2728	0.2842	0.3941	0.3275	0.3342	0.2466	0.2878	0.3960	0.3179
	Avg.	0.1888	0.2124	0.3026	0.2456	0.2478	0.1965	0.2271	0.3060	0.2408
AutoTimes	1D	0.0863	0.1029	0.1665	0.1318	0.1270	0.1140	0.1312	0.1587	0.1273
	3D	0.1530	0.1663	0.2545	0.2069	0.2008	0.1668	0.1926	0.2450	0.1982
	5D	0.2065	0.2109	0.3079	0.2619	0.2556	0.2017	0.2342	0.3065	0.2481
	7D	0.2552	0.2469	0.3515	0.3009	0.3029	0.2267	0.2654	0.3482	0.2872
	Avg.	0.1752	0.1817	0.2701	0.2254	0.2216	0.1773	0.2058	0.2646	0.2152

Table 19: Benchmark(MSE) on part 2(Miami area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
MLP	1D	0.0315	0.0629	0.1405	0.0774	0.0602	0.0525	0.0787	0.1295	0.0792	
	3D	0.0808	0.1311	0.2777	0.1506	0.1289	0.0935	0.1471	0.2762	0.1607	
	5D	0.1300	0.1819	0.3613	0.2040	0.1912	0.1197	0.1905	0.3950	0.2217	
	7D	0.1783	0.2232	0.4256	0.2480	0.2456	0.1399	0.2230	0.4617	0.2682	
	Avg.	0.1052	0.1498	0.3012	0.1700	0.1565	0.1014	0.1599	0.3156	0.1824	
MLP	TSMixer	1D	0.0397	0.0747	0.1695	0.0877	0.0697	0.0578	0.0868	0.2450	0.1039
	3D	0.0932	0.1486	0.3241	0.1611	0.1419	0.0985	0.1605	0.3833	0.1889	
	5D	0.1460	0.2061	0.4276	0.2192	0.2111	0.1263	0.2114	0.4777	0.2532	
	7D	0.1968	0.2537	0.5054	0.2679	0.2786	0.1475	0.2509	0.5544	0.3069	
	Avg.	0.1189	0.1708	0.3566	0.1840	0.1753	0.1075	0.1774	0.4151	0.2132	
NLinear	1D	0.0374	0.0687	0.1605	0.0830	0.0639	0.0554	0.0816	0.1269	0.0847	
	3D	0.0924	0.1442	0.3186	0.1581	0.1376	0.0974	0.1568	0.2825	0.1734	
	5D	0.1450	0.2026	0.4224	0.2174	0.2065	0.1253	0.2083	0.3881	0.2395	
	7D	0.1962	0.2511	0.5021	0.2666	0.2743	0.1471	0.2484	0.4696	0.2944	
	Avg.	0.1177	0.1666	0.3509	0.1813	0.1706	0.1063	0.1738	0.3168	0.1980	
TCN	1D	0.0976	0.0963	0.1756	0.1090	0.1377	0.1694	0.0984	5.8002	0.8355	
	3D	0.1318	0.1480	0.3271	0.2042	0.2212	0.2113	0.1527	7.0577	1.0567	
	5D	0.3490	0.2009	0.4521	0.2532	0.3033	0.2986	0.2079	5.1319	0.8996	
	7D	0.2479	0.2662	0.4783	0.2929	0.3587	0.2911	0.2391	6.7494	1.1155	
	Avg.	0.2066	0.1779	0.3583	0.2148	0.2552	0.2426	0.1745	6.1848	0.9768	
CNN	ModernTCN	1D	0.0395	0.0838	0.2180	0.1137	0.0630	0.0539	0.0818	0.3165	0.1213
	3D	0.1168	0.1894	0.4307	0.2323	0.1464	0.1036	0.1678	2.9814	0.5460	
	5D	0.1869	0.3223	0.5611	0.3355	0.2198	0.1352	0.2221	2.7558	0.5923	
	7D	0.2499	0.4475	0.7525	0.3950	0.2877	0.1613	0.2656	1.4768	0.5045	
	Avg.	0.1483	0.2608	0.4906	0.2691	0.1792	0.1135	0.1843	1.8826	0.4411	
TimesNet	1D	0.0453	0.0899	0.1964	0.1033	0.0884	0.0700	0.1090	0.1729	0.1094	
	3D	0.1100	0.1839	0.3598	0.1995	0.1694	0.1154	0.1857	0.3311	0.2069	
	5D	0.1666	0.2634	0.4952	0.2586	0.2486	0.1484	0.2477	0.4500	0.2848	
	7D	0.2283	0.3207	0.5949	0.3303	0.3296	0.1674	0.2892	0.5382	0.3498	
	Avg.	0.1376	0.2145	0.4116	0.2230	0.2090	0.1253	0.2079	0.3731	0.2377	
Transformer	iTransformer	1D	0.0337	0.0645	0.1413	0.0798	0.0614	0.0536	0.0766	0.1210	0.0790
	3D	0.0880	0.1424	0.3086	0.1614	0.1361	0.0964	0.1512	0.2755	0.1700	
	5D	0.1486	0.2002	0.4253	0.2206	0.2052	0.1266	0.2082	0.3973	0.2415	
	7D	0.2032	0.2658	0.4903	0.2697	0.2727	0.1490	0.2408	0.5022	0.2992	
	Avg.	0.1184	0.1682	0.3414	0.1828	0.1688	0.1064	0.1692	0.3240	0.1974	
	PatchTST	1D	0.0326	0.0667	0.1392	0.0789	0.0615	0.0525	0.0781	0.1213	0.0789
	3D	0.0872	0.1459	0.2950	0.1614	0.1389	0.0963	0.1586	0.2825	0.1707	
	5D	0.1402	0.2096	0.3952	0.2208	0.2068	0.1249	0.2112	0.3960	0.2381	
	7D	0.1899	0.2624	0.4758	0.2686	0.2763	0.1473	0.2494	0.4766	0.2933	
	Avg.	0.1125	0.1711	0.3263	0.1824	0.1709	0.1053	0.1743	0.3191	0.1952	

Table 20: Benchmark(MSE) on part 2(Miami area) stations(RNN,GNN,LLM)

Method	Metric	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.0575	0.1123	0.2405	0.1158	0.1092	0.0775	0.1224	1.1111	0.2433
	3D	0.1531	0.1869	0.3854	0.2070	0.2091	0.1206	0.1944	1.4368	0.3617
	5D	0.1983	0.2390	0.4905	0.2732	0.2894	0.1479	0.2368	1.3811	0.4070
	7D	0.2734	0.2901	0.5826	0.3426	0.3981	0.1677	0.2615	1.8380	0.5192
	Avg.	0.1706	0.2071	0.4248	0.2346	0.2515	0.1284	0.2038	1.4417	0.3828
DeepAR	1D	0.0611	0.1108	0.2390	0.1248	0.1152	0.0790	0.1230	0.9006	0.2192
	3D	0.1276	0.1894	0.3864	0.2041	0.2173	0.1226	0.1953	1.3494	0.3490
	5D	0.1973	0.2385	0.4877	0.2623	0.2918	0.1463	0.2344	1.4248	0.4104
	7D	0.2820	0.3010	0.5995	0.3402	0.3564	0.1667	0.2657	1.4515	0.4704
	Avg.	0.1670	0.2099	0.4282	0.2328	0.2452	0.1286	0.2046	1.2816	0.3622
DilatedRNN	1D	0.0366	0.0808	0.1810	0.0931	0.0823	0.0635	0.0944	0.8270	0.1823
	3D	0.1137	0.1710	0.3481	0.1894	0.2205	0.1151	0.1746	1.0864	0.3023
	5D	0.1970	0.2275	0.4660	0.2738	0.3055	0.1469	0.2277	1.3925	0.4046
	7D	0.2899	0.2742	0.5486	0.3313	0.3679	0.1781	0.2746	1.3743	0.4549
	Avg.	0.1593	0.1884	0.3859	0.2219	0.2441	0.1259	0.1928	1.1701	0.3360
GCN	1D	0.0333	0.0964	0.1750	0.1817	0.1209	0.0909	0.0948	170.0996	21.3616
	3D	0.0829	0.1800	0.3166	0.2245	0.1894	0.1255	0.1616	32.4988	4.2224
	5D	0.1407	0.2346	0.4015	0.2996	0.2709	0.1500	0.2053	44.7123	5.8019
	7D	0.2021	0.2599	0.4775	0.3266	0.3373	0.1828	0.2379	17.1457	2.3962
	Avg.	0.1147	0.1927	0.3426	0.2581	0.2296	0.1373	0.1749	66.1141	8.4455
FourierGNN	1D	0.0339	0.0703	0.1442	0.0825	0.0631	0.0563	0.0809	0.6083	0.1425
	3D	0.0851	0.1359	0.2816	0.1522	0.1320	0.0988	0.1493	0.9670	0.2502
	5D	0.1331	0.1885	0.3656	0.2117	0.1916	0.1290	0.1920	1.6289	0.3800
	7D	0.2053	0.2240	0.4330	0.2525	0.2511	0.1483	0.2250	1.0897	0.3536
	Avg.	0.1144	0.1547	0.3061	0.1747	0.1594	0.1081	0.1618	1.0735	0.2816
StemGNN	1D	0.0391	0.1055	0.1724	0.0912	0.0873	0.0665	0.0898	1.2556	0.2384
	3D	0.1283	0.2207	0.3498	0.2164	0.2680	0.1644	0.1796	1.5368	0.3830
	5D	0.2199	0.2945	0.4663	0.3239	0.5381	0.2094	0.2414	1.6949	0.4985
	7D	0.3149	0.3582	0.5362	0.3986	0.4688	0.2426	0.3031	2.0180	0.5801
	Avg.	0.1755	0.2447	0.3812	0.2575	0.3406	0.1707	0.2035	1.6263	0.4250
GPT4TS	1D	0.0473	0.0949	0.1828	0.0977	0.0819	0.0735	0.1124	0.1921	0.1103
	3D	0.1163	0.1995	0.3605	0.2009	0.1762	0.1251	0.1953	0.3751	0.2186
	5D	0.1908	0.2827	0.4853	0.2619	0.2654	0.1534	0.2487	0.4691	0.2947
	7D	0.2394	0.3739	0.5955	0.3207	0.3350	0.1826	0.2946	0.5944	0.3670
	Avg.	0.1484	0.2377	0.4060	0.2203	0.2146	0.1337	0.2127	0.4077	0.2477
AutoTimes	1D	0.0375	0.0718	0.1582	0.0879	0.0687	0.0581	0.0900	0.1373	0.0887
	3D	0.0889	0.1438	0.3015	0.1598	0.1410	0.0984	0.1615	0.2986	0.1742
	5D	0.1406	0.2011	0.3952	0.2175	0.2057	0.1265	0.2115	0.4012	0.2374
	7D	0.1944	0.2496	0.4659	0.2653	0.2754	0.1485	0.2510	0.4759	0.2908
	Avg.	0.1154	0.1666	0.3302	0.1826	0.1727	0.1079	0.1785	0.3283	0.1978

Table 21: Benchmark(SEDI10) on part 2(Miami area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
MLP	MLP	1D	0.8741	0.8301	0.8503	0.8179	0.8890	0.5135	0.4864	0.7414	0.7503
		3D	0.8258	0.7763	0.8076	0.7682	0.8627	0.4220	0.4150	0.6270	0.6881
		5D	0.7627	0.7409	0.7604	0.7023	0.8228	0.3802	0.3853	0.6126	0.6459
		7D	0.7108	0.7069	0.7572	0.6765	0.8028	0.3550	0.3867	0.5522	0.6185
		Avg.	0.7933	0.7636	0.7939	0.7412	0.8443	0.4177	0.4183	0.6333	0.6757
	TSMixer	1D	0.8506	0.7672	0.8000	0.7572	0.8709	0.4446	0.4225	0.5891	0.6878
		3D	0.7622	0.6598	0.7016	0.6497	0.7979	0.3370	0.3233	0.4826	0.5893
		5D	0.6958	0.5955	0.6414	0.5820	0.7421	0.2694	0.2707	0.4073	0.5255
		7D	0.6433	0.5499	0.5962	0.5360	0.6957	0.2335	0.2379	0.3550	0.4810
		Avg.	0.7380	0.6431	0.6848	0.6312	0.7767	0.3211	0.3136	0.4585	0.5709
	NLinear	1D	0.8486	0.7786	0.8039	0.7660	0.8697	0.4931	0.4452	0.7005	0.7132
		3D	0.7606	0.6705	0.7059	0.6597	0.7969	0.3753	0.3424	0.5532	0.6081
		5D	0.6961	0.6019	0.6452	0.5928	0.7418	0.3265	0.2868	0.4605	0.5439
		7D	0.6445	0.5544	0.6000	0.5465	0.6962	0.2868	0.2522	0.3980	0.4973
		Avg.	0.7374	0.6514	0.6887	0.6412	0.7761	0.3704	0.3316	0.5280	0.5906
CNN	TCN	1D	0.7593	0.7900	0.8204	0.6993	0.7827	0.2702	0.4436	0.2651	0.6038
		3D	0.7316	0.6984	0.7447	0.5739	0.7257	0.1869	0.3902	0.1932	0.5306
		5D	0.5618	0.6040	0.6806	0.5003	0.6732	0.1566	0.3651	0.1776	0.4649
		7D	0.6365	0.5666	0.6778	0.4988	0.5719	0.1185	0.2944	0.1443	0.4386
		Avg.	0.6723	0.6647	0.7309	0.5681	0.6884	0.1831	0.3733	0.1951	0.5095
	ModernTCN	1D	0.8450	0.7649	0.7755	0.7226	0.8710	0.4506	0.4383	0.6989	0.6959
		3D	0.7464	0.6562	0.6524	0.5939	0.7859	0.3206	0.3270	0.5369	0.5774
		5D	0.6737	0.5673	0.5828	0.5158	0.7293	0.2661	0.2691	0.4358	0.5050
		7D	0.6180	0.5240	0.5309	0.4738	0.6825	0.2327	0.2278	0.3797	0.4587
		Avg.	0.7208	0.6281	0.6354	0.5766	0.7672	0.3175	0.3156	0.5128	0.5592
	TimesNet	1D	0.8231	0.7314	0.7478	0.7043	0.8409	0.3910	0.3584	0.6258	0.6528
		3D	0.7335	0.6302	0.6612	0.6004	0.7609	0.2651	0.2711	0.4914	0.5517
		5D	0.6745	0.5599	0.5876	0.5382	0.7072	0.2241	0.2204	0.4053	0.4896
		7D	0.6158	0.5245	0.5430	0.4907	0.6482	0.1888	0.1921	0.3477	0.4438
		Avg.	0.7117	0.6115	0.6349	0.5834	0.7393	0.2672	0.2605	0.4675	0.5345
Transformer	iTransformer	1D	0.8637	0.8038	0.8207	0.7756	0.8762	0.4669	0.4545	0.7247	0.7233
		3D	0.7686	0.7074	0.7133	0.6568	0.8054	0.3441	0.3459	0.5810	0.6153
		5D	0.6971	0.6270	0.6584	0.5889	0.7435	0.2774	0.2809	0.4665	0.5424
		7D	0.6451	0.5751	0.6083	0.5437	0.6964	0.2396	0.2535	0.3979	0.4949
		Avg.	0.7436	0.6783	0.7002	0.6412	0.7803	0.3320	0.3337	0.5425	0.5940
	PatchTST	1D	0.8595	0.7994	0.8269	0.7725	0.8746	0.4772	0.4750	0.7220	0.7259
		3D	0.7678	0.6914	0.7225	0.6534	0.7956	0.3561	0.3503	0.5633	0.6125
		5D	0.7022	0.6194	0.6605	0.5880	0.7365	0.3018	0.2951	0.4659	0.5462
		7D	0.6513	0.5667	0.6103	0.5407	0.6906	0.2540	0.2564	0.4046	0.4968
		Avg.	0.7452	0.6692	0.7051	0.6386	0.7743	0.3473	0.3442	0.5389	0.5954

Table 22: Benchmark(SED10) on part 2(Miami area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
LSTM	1D	0.8360	0.7782	0.8055	0.7590	0.8905	0.4015	0.3896	0.6608	0.6901	
	3D	0.7519	0.6710	0.6999	0.6089	0.8233	0.2542	0.2837	0.5149	0.5760	
	5D	0.6692	0.6430	0.6611	0.5801	0.7650	0.2298	0.2401	0.3879	0.5220	
	7D	0.6402	0.5991	0.6269	0.4682	0.7060	0.1930	0.2364	0.3669	0.4796	
	Avg.	0.7243	0.6728	0.6984	0.6041	0.7962	0.2696	0.2874	0.4826	0.5669	
RNN	DeepAR	1D	0.8152	0.7886	0.7825	0.7384	0.8582	0.3910	0.4268	0.6741	0.6843
		3D	0.7450	0.6511	0.7208	0.6443	0.8068	0.2397	0.2744	0.5008	0.5729
		5D	0.6943	0.6334	0.6518	0.5570	0.7574	0.2019	0.2405	0.4519	0.5235
		7D	0.6155	0.5923	0.6195	0.5072	0.7319	0.2241	0.2245	0.3572	0.4840
		Avg.	0.7175	0.6664	0.6936	0.6117	0.7886	0.2642	0.2916	0.4960	0.5662
DilatedRNN	1D	0.8692	0.7880	0.8197	0.7892	0.8793	0.4684	0.4362	0.6942	0.7180	
	3D	0.7796	0.7056	0.7306	0.6319	0.7575	0.2916	0.3150	0.5490	0.5951	
	5D	0.6999	0.6585	0.6645	0.5788	0.7220	0.2488	0.2843	0.4543	0.5389	
	7D	0.6469	0.5807	0.5970	0.5429	0.6725	0.2202	0.2308	0.3934	0.4856	
	Avg.	0.7489	0.6832	0.7029	0.6357	0.7578	0.3072	0.3166	0.5227	0.5844	
GCN	1D	0.8394	0.7752	0.8086	0.7437	0.8631	0.4585	0.4685	0.6737	0.7038	
	3D	0.7913	0.7114	0.7760	0.6987	0.8134	0.4180	0.3846	0.5844	0.6472	
	5D	0.7726	0.6603	0.7433	0.6416	0.8037	0.3545	0.3369	0.5404	0.6067	
	7D	0.7195	0.6370	0.7368	0.6266	0.7624	0.2961	0.3282	0.5247	0.5789	
	Avg.	0.7807	0.6960	0.7662	0.6777	0.8107	0.3818	0.3795	0.5808	0.6342	
GNN	FourierGNN	1D	0.8842	0.8339	0.8446	0.8034	0.8794	0.4924	0.4661	0.5660	0.7213
		3D	0.8071	0.7862	0.7878	0.6988	0.8383	0.3740	0.3840	0.5319	0.6510
		5D	0.7275	0.6700	0.7674	0.6396	0.8200	0.3434	0.3735	0.5033	0.6056
		7D	0.7208	0.6275	0.7505	0.6695	0.7939	0.3157	0.3493	0.4627	0.5863
		Avg.	0.7849	0.7294	0.7876	0.7028	0.8329	0.3814	0.3932	0.5160	0.6410
StemGNN	1D	0.8646	0.7785	0.7954	0.7146	0.8769	0.4439	0.4379	0.4597	0.6714	
	3D	0.7744	0.6132	0.7012	0.6006	0.7568	0.2438	0.2951	0.3788	0.5455	
	5D	0.6814	0.6045	0.6612	0.5584	0.5665	0.2083	0.2529	0.3115	0.4806	
	7D	0.6648	0.5171	0.6013	0.4415	0.5642	0.1738	0.2100	0.2598	0.4291	
	Avg.	0.7463	0.6283	0.6898	0.5788	0.6911	0.2674	0.2990	0.3525	0.5316	
LLM	GPT4TS	1D	0.8216	0.7340	0.7701	0.7331	0.8414	0.3715	0.3565	0.6124	0.6551
		3D	0.7389	0.6277	0.6603	0.6071	0.7518	0.2678	0.2655	0.4756	0.5493
		5D	0.6746	0.5605	0.5939	0.5327	0.6894	0.2279	0.2255	0.3964	0.4876
		7D	0.6139	0.5161	0.5541	0.4812	0.6419	0.1929	0.1982	0.3275	0.4407
		Avg.	0.7123	0.6096	0.6446	0.5885	0.7311	0.2650	0.2614	0.4530	0.5332
AutoTimes	1D	0.8574	0.7975	0.8157	0.7598	0.8726	0.4113	0.4080	0.7040	0.7033	
	3D	0.7652	0.6858	0.7175	0.6466	0.8032	0.3067	0.3215	0.5668	0.6017	
	5D	0.7002	0.6266	0.6601	0.5837	0.7408	0.2576	0.2676	0.4668	0.5379	
	7D	0.6453	0.5683	0.6149	0.5348	0.6967	0.2213	0.2319	0.4040	0.4896	
	Avg.	0.7420	0.6695	0.7021	0.6312	0.7783	0.2992	0.3073	0.5354	0.5831	

Table 23: Benchmark(SED15) on part 2(Miami area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.8203	0.8018	0.8453	0.6927	0.8843	0.3386	0.3853	0.6361	0.6755
	3D	0.7427	0.7459	0.8080	0.6251	0.8496	0.2768	0.3320	0.5376	0.6147
	5D	0.6747	0.7152	0.7456	0.5607	0.8084	0.2658	0.3134	0.5075	0.5739
	7D	0.6180	0.6833	0.7520	0.5447	0.7774	0.2555	0.3284	0.4584	0.5522
	Avg.	0.7139	0.7366	0.7878	0.6058	0.8299	0.2842	0.3398	0.5349	0.6041
MLP	1D	0.7829	0.7231	0.7853	0.6187	0.8603	0.2823	0.3224	0.4521	0.6034
	3D	0.6518	0.5942	0.6704	0.5094	0.7743	0.2085	0.2223	0.3548	0.4982
	5D	0.5559	0.5210	0.5950	0.4429	0.7069	0.1762	0.1753	0.2891	0.4328
	7D	0.4809	0.4667	0.5379	0.3967	0.6516	0.1424	0.1484	0.2437	0.3835
	Avg.	0.6179	0.5763	0.6471	0.4919	0.7483	0.2023	0.2171	0.3349	0.4795
NLinear	1D	0.7842	0.7341	0.7857	0.6272	0.8583	0.3155	0.3398	0.5837	0.6286
	3D	0.6549	0.6065	0.6750	0.5297	0.7740	0.2379	0.2452	0.4378	0.5201
	5D	0.5606	0.5291	0.6000	0.4519	0.7081	0.2114	0.1966	0.3530	0.4513
	7D	0.4881	0.4734	0.5439	0.4077	0.6537	0.1884	0.1690	0.2964	0.4026
	Avg.	0.6220	0.5858	0.6512	0.5041	0.7485	0.2383	0.2377	0.4177	0.5007
TCN	1D	0.6732	0.7264	0.8085	0.4636	0.6769	0.1780	0.3211	0.1701	0.5022
	3D	0.6003	0.6072	0.7341	0.3332	0.6076	0.1251	0.2746	0.1106	0.4241
	5D	0.3759	0.4967	0.6449	0.3185	0.5094	0.0967	0.2207	0.1031	0.3457
	7D	0.3713	0.4688	0.6172	0.2808	0.4117	0.0658	0.1851	0.0710	0.3089
	Avg.	0.5052	0.5748	0.7012	0.3490	0.5514	0.1164	0.2504	0.1137	0.3952
CNN	1D	0.7753	0.7152	0.7470	0.5845	0.8521	0.2784	0.3290	0.5653	0.6058
	3D	0.6425	0.5891	0.6141	0.4597	0.7544	0.1988	0.2123	0.4098	0.4851
	5D	0.5456	0.4885	0.5393	0.3821	0.6921	0.1645	0.1691	0.3149	0.4120
	7D	0.4749	0.4345	0.4788	0.3437	0.6387	0.1410	0.1413	0.2653	0.3648
	Avg.	0.6096	0.5568	0.5948	0.4425	0.7343	0.1957	0.2129	0.3888	0.4669
TimesNet	1D	0.7487	0.6768	0.7300	0.5653	0.8128	0.2235	0.2528	0.4910	0.5626
	3D	0.6236	0.5582	0.6225	0.4521	0.7272	0.1473	0.1642	0.3592	0.4568
	5D	0.5369	0.4827	0.5380	0.3935	0.6590	0.1184	0.1226	0.2851	0.3920
	7D	0.4596	0.4353	0.4911	0.3486	0.5869	0.1043	0.0991	0.2359	0.3451
	Avg.	0.5922	0.5383	0.5954	0.4399	0.6965	0.1484	0.1597	0.3428	0.4391
Transformer	1D	0.7985	0.7670	0.8047	0.6444	0.8568	0.2787	0.3549	0.6033	0.6385
	3D	0.6767	0.6518	0.6871	0.5192	0.7699	0.1998	0.2785	0.4604	0.5304
	5D	0.5748	0.5559	0.6193	0.4492	0.6981	0.1652	0.1791	0.3529	0.4493
	7D	0.5036	0.4938	0.5666	0.4040	0.6433	0.1402	0.1705	0.2874	0.4012
	Avg.	0.6384	0.6171	0.6694	0.5042	0.7420	0.1960	0.2457	0.4260	0.5049
PatchTST	1D	0.7948	0.7663	0.8131	0.6364	0.8546	0.2859	0.3648	0.6001	0.6395
	3D	0.6706	0.6408	0.6990	0.5158	0.7641	0.2161	0.2558	0.4454	0.5260
	5D	0.5757	0.5538	0.6241	0.4486	0.6977	0.1833	0.2059	0.3516	0.4551
	7D	0.5016	0.4906	0.5665	0.4029	0.6450	0.1581	0.1751	0.2979	0.4047
	Avg.	0.6357	0.6129	0.6757	0.5009	0.7404	0.2108	0.2504	0.4238	0.5063

Table 24: Benchmark(SED15) on part 2(Miami area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
LSTM	1D	0.7782	0.7372	0.7975	0.6088	0.8707	0.2226	0.2661	0.5371	0.6023	
	3D	0.6193	0.6102	0.6635	0.4559	0.7919	0.1437	0.1772	0.4233	0.4856	
	5D	0.5323	0.5690	0.6237	0.3969	0.7194	0.1253	0.1418	0.3040	0.4266	
	7D	0.4947	0.4718	0.5666	0.3301	0.6305	0.1009	0.1329	0.2593	0.3734	
	Avg.	0.6061	0.5971	0.6628	0.4479	0.7531	0.1481	0.1795	0.3809	0.4720	
RNN	DeepAR	1D	0.7506	0.7444	0.7702	0.5797	0.8298	0.2470	0.2639	0.5466	0.5915
		3D	0.6252	0.6021	0.6902	0.4647	0.7750	0.1426	0.1678	0.4273	0.4869
		5D	0.5361	0.5584	0.5951	0.3839	0.7114	0.1120	0.1259	0.3603	0.4229
		7D	0.4436	0.5078	0.5647	0.3391	0.6956	0.0967	0.1104	0.2599	0.3772
		Avg.	0.5889	0.6032	0.6551	0.4419	0.7530	0.1496	0.1670	0.3985	0.4696
DilatedRNN	1D	0.8054	0.7393	0.8027	0.6504	0.8471	0.3023	0.3125	0.5724	0.6290	
	3D	0.6871	0.6505	0.7110	0.5126	0.7126	0.1749	0.2102	0.4345	0.5117	
	5D	0.5834	0.5993	0.6388	0.4463	0.6757	0.1499	0.1713	0.3450	0.4512	
	7D	0.5413	0.4900	0.5612	0.4197	0.6188	0.1372	0.1388	0.2847	0.3990	
	Avg.	0.6543	0.6198	0.6784	0.5072	0.7136	0.1911	0.2082	0.4092	0.4977	
GCN	1D	0.7502	0.7320	0.8148	0.6321	0.8439	0.3394	0.3757	0.5786	0.6333	
	3D	0.6899	0.6528	0.7953	0.5605	0.7889	0.2891	0.3176	0.5118	0.5757	
	5D	0.6957	0.5923	0.7658	0.5245	0.7818	0.2388	0.2643	0.4678	0.5414	
	7D	0.6440	0.5577	0.7625	0.4949	0.7402	0.2244	0.2587	0.4532	0.5169	
	Avg.	0.6949	0.6337	0.7846	0.5530	0.7887	0.2729	0.3041	0.5029	0.5668	
GNN	FourierGNN	1D	0.8279	0.7998	0.8378	0.6750	0.8669	0.3249	0.3500	0.4899	0.6465
		3D	0.7233	0.7394	0.7814	0.5680	0.8201	0.2450	0.2873	0.4436	0.5760
		5D	0.6198	0.6098	0.7593	0.5051	0.7966	0.2324	0.2902	0.4006	0.5267
		7D	0.6588	0.5544	0.7447	0.5131	0.7637	0.2154	0.2739	0.4022	0.5158
		Avg.	0.7074	0.6758	0.7808	0.5653	0.8118	0.2544	0.3003	0.4340	0.5662
StemGNN	1D	0.8015	0.7321	0.7884	0.5807	0.8594	0.2851	0.3385	0.3684	0.5943	
	3D	0.6608	0.5265	0.7032	0.4503	0.7169	0.1409	0.1868	0.2969	0.4603	
	5D	0.5775	0.5009	0.6416	0.4189	0.3915	0.1244	0.1569	0.2450	0.3821	
	7D	0.4858	0.4275	0.5752	0.2864	0.4690	0.0999	0.1371	0.1949	0.3345	
	Avg.	0.6314	0.5468	0.6771	0.4341	0.6092	0.1626	0.2048	0.2763	0.4428	
LLM	GPT4TS	1D	0.7424	0.6726	0.7407	0.5721	0.8175	0.2057	0.2580	0.4738	0.5604
		3D	0.6382	0.5471	0.6227	0.4534	0.7177	0.1509	0.1622	0.3482	0.4550
		5D	0.5484	0.4662	0.5447	0.3914	0.6360	0.1230	0.1265	0.2813	0.3897
		7D	0.4653	0.4102	0.4972	0.3462	0.5821	0.1040	0.1057	0.2184	0.3411
		Avg.	0.5986	0.5240	0.6013	0.4408	0.6883	0.1459	0.1631	0.3304	0.4366
AutoTimes	1D	0.7923	0.7578	0.8113	0.6238	0.8590	0.2414	0.3024	0.5828	0.6213	
	3D	0.6587	0.6310	0.6989	0.5032	0.7737	0.1776	0.2137	0.4392	0.5120	
	5D	0.5670	0.5597	0.6312	0.4430	0.7028	0.1544	0.1690	0.3459	0.4466	
	7D	0.4857	0.4895	0.5722	0.3936	0.6512	0.1312	0.1424	0.2904	0.3945	
	Avg.	0.6259	0.6095	0.6784	0.4909	0.7467	0.1761	0.2069	0.4146	0.4936	

Table 25: Benchmark(SED11) on part 2(Miami area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.6392	0.7486	0.7968	0.4636	0.7523	0.1510	0.1849	0.4622	0.5248
	3D	0.5460	0.6948	0.7434	0.4294	0.6635	0.1429	0.1752	0.3631	0.4698
	5D	0.5736	0.6829	0.6508	0.3551	0.6734	0.1527	0.1643	0.3667	0.4524
	7D	0.5641	0.6625	0.7139	0.4331	0.6920	0.1401	0.1617	0.3547	0.4653
	Avg.	0.5807	0.6972	0.7263	0.4203	0.6953	0.1467	0.1715	0.3867	0.4781
MLP	1D	0.5899	0.6011	0.6559	0.3407	0.6954	0.1186	0.1495	0.2268	0.4222
	3D	0.4351	0.4299	0.4696	0.2279	0.5486	0.0778	0.0867	0.1369	0.3016
	5D	0.3255	0.3379	0.3531	0.1776	0.4431	0.0599	0.0690	0.1006	0.2333
	7D	0.2522	0.2693	0.2787	0.1494	0.3661	0.0524	0.0697	0.0791	0.1896
	Avg.	0.4007	0.4096	0.4393	0.2239	0.5133	0.0772	0.0937	0.1359	0.2867
NLinear	1D	0.6032	0.6324	0.6721	0.3915	0.7185	0.1412	0.1508	0.3811	0.4613
	3D	0.4433	0.4556	0.4838	0.2745	0.5680	0.1052	0.1023	0.2296	0.3328
	5D	0.3408	0.3553	0.3639	0.2278	0.4626	0.0956	0.0801	0.1734	0.2624
	7D	0.2637	0.2851	0.2905	0.1992	0.3820	0.0851	0.0702	0.1435	0.2149
	Avg.	0.4128	0.4321	0.4526	0.2732	0.5328	0.1068	0.1008	0.2319	0.3179
TCN	1D	0.1324	0.4480	0.7226	0.2066	0.2128	0.0640	0.1211	0.0506	0.2448
	3D	0.0330	0.3732	0.5677	0.1723	0.1441	0.0387	0.1129	0.0265	0.1835
	5D	0.0258	0.2706	0.4580	0.1274	0.1506	0.0312	0.1130	0.0283	0.1506
	7D	0.0571	0.2566	0.4746	0.1193	0.1263	0.0128	0.0770	0.0104	0.1418
	Avg.	0.0621	0.3371	0.5557	0.1564	0.1584	0.0367	0.1060	0.0289	0.1802
CNN	1D	0.6055	0.5939	0.6605	0.3129	0.7173	0.1196	0.1522	0.3788	0.4426
	3D	0.4336	0.4005	0.4627	0.1837	0.5687	0.0721	0.0722	0.2224	0.3020
	5D	0.3390	0.2918	0.3464	0.1267	0.4670	0.0555	0.0590	0.1442	0.2287
	7D	0.2672	0.2162	0.2717	0.1025	0.3878	0.0519	0.0415	0.1163	0.1819
	Avg.	0.4113	0.3756	0.4353	0.1814	0.5352	0.0748	0.0812	0.2154	0.2888
TimesNet	1D	0.5404	0.5278	0.6337	0.2851	0.6289	0.0906	0.0988	0.2989	0.3880
	3D	0.4211	0.3841	0.4394	0.1624	0.5016	0.0434	0.0522	0.1626	0.2708
	5D	0.3299	0.2943	0.3291	0.1157	0.3647	0.0295	0.0269	0.1104	0.2001
	7D	0.2249	0.2508	0.2616	0.0851	0.3262	0.0234	0.0230	0.0944	0.1612
	Avg.	0.3791	0.3643	0.4160	0.1621	0.4554	0.0467	0.0502	0.1666	0.2550
Transformer	1D	0.6163	0.6752	0.7437	0.3883	0.7228	0.1268	0.1668	0.4121	0.4815
	3D	0.4566	0.5015	0.5470	0.2322	0.5716	0.0745	0.0972	0.2676	0.3435
	5D	0.3503	0.3820	0.4225	0.1779	0.4512	0.0560	0.0565	0.1561	0.2566
	7D	0.2829	0.2964	0.3415	0.1450	0.3636	0.0446	0.0921	0.1163	0.2103
	Avg.	0.4265	0.4638	0.5137	0.2359	0.5273	0.0755	0.1031	0.2380	0.3230
PatchTST	1D	0.6112	0.6690	0.7429	0.3695	0.7238	0.1261	0.1691	0.4179	0.4787
	3D	0.4618	0.4940	0.5613	0.2286	0.5741	0.0822	0.0949	0.2473	0.3430
	5D	0.3572	0.3782	0.4444	0.1734	0.4708	0.0668	0.0695	0.1708	0.2664
	7D	0.2791	0.3030	0.3527	0.1388	0.3865	0.0589	0.0600	0.1358	0.2143
	Avg.	0.4273	0.4610	0.5253	0.2276	0.5388	0.0835	0.0984	0.2429	0.3256

Table 26: Benchmark(SED11) on part 2(Miami area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
RNN	LSTM	1D	0.5173	0.5951	0.6803	0.2523	0.4421	0.0731	0.0896	0.3249	0.3718
		3D	0.3565	0.3467	0.4334	0.1171	0.2033	0.0380	0.0398	0.1530	0.2110
		5D	0.3267	0.2902	0.3336	0.0565	0.2339	0.0143	0.0343	0.1030	0.1740
		7D	0.2477	0.2608	0.4109	0.0553	0.0981	0.0078	0.0051	0.0914	0.1471
		Avg.	0.3621	0.3732	0.4646	0.1203	0.2443	0.0333	0.0422	0.1681	0.2260
	DeepAR	1D	0.4712	0.6332	0.6489	0.2446	0.4433	0.0756	0.1161	0.3283	0.3701
		3D	0.3647	0.3977	0.4097	0.0820	0.1804	0.0310	0.0387	0.1553	0.2074
		5D	0.2339	0.2331	0.2853	0.0355	0.1330	0.0220	0.0048	0.1158	0.1329
		7D	0.1074	0.2155	0.2837	0.0609	0.0813	0.0165	0.0063	0.1074	0.1099
		Avg.	0.2943	0.3699	0.4069	0.1058	0.2095	0.0363	0.0415	0.1767	0.2051
	DilatedRNN	1D	0.6372	0.5916	0.7119	0.4065	0.7022	0.1088	0.1333	0.3922	0.4605
		3D	0.5222	0.4714	0.6236	0.2306	0.4730	0.0741	0.0689	0.2576	0.3402
		5D	0.3632	0.4556	0.5025	0.1754	0.4427	0.0560	0.0513	0.1841	0.2789
		7D	0.3681	0.3293	0.4419	0.1630	0.4273	0.0491	0.0411	0.1600	0.2475
		Avg.	0.4727	0.4620	0.5700	0.2439	0.5113	0.0720	0.0737	0.2485	0.3317
GNN	GCN	1D	0.5814	0.5680	0.7359	0.3744	0.6712	0.1554	0.1852	0.4307	0.4628
		3D	0.5052	0.5117	0.7286	0.2892	0.5979	0.1691	0.1555	0.3875	0.4181
		5D	0.4462	0.4254	0.6934	0.2584	0.5692	0.1573	0.1310	0.3559	0.3796
		7D	0.4309	0.3783	0.6980	0.2593	0.5114	0.1639	0.1560	0.3369	0.3668
		Avg.	0.4909	0.4709	0.7140	0.2953	0.5874	0.1614	0.1569	0.3777	0.4068
	FourierGNN	1D	0.6570	0.6977	0.7677	0.4359	0.7147	0.1492	0.1707	0.3647	0.4947
		3D	0.5526	0.6366	0.6920	0.3083	0.6153	0.1285	0.1482	0.2883	0.4212
		5D	0.4619	0.4865	0.6791	0.2633	0.5934	0.1286	0.1450	0.2778	0.3794
		7D	0.6103	0.4415	0.7151	0.3320	0.6158	0.1261	0.1514	0.2791	0.4089
		Avg.	0.5705	0.5656	0.7135	0.3349	0.6348	0.1331	0.1538	0.3025	0.4261
	StemGNN	1D	0.5891	0.5290	0.7186	0.3042	0.6594	0.1308	0.1546	0.2231	0.4136
		3D	0.3107	0.2996	0.6110	0.1650	0.4861	0.0484	0.0712	0.1342	0.2658
		5D	0.3170	0.2849	0.5104	0.1400	0.1212	0.0405	0.0469	0.1600	0.2026
		7D	0.2300	0.2427	0.3921	0.0846	0.1728	0.0315	0.0513	0.0838	0.1611
		Avg.	0.3617	0.3391	0.5580	0.1735	0.3599	0.0628	0.0810	0.1503	0.2608
LLM	GPT4TS	1D	0.5577	0.5380	0.6412	0.2952	0.6627	0.0716	0.0904	0.2589	0.3895
		3D	0.4106	0.3726	0.4585	0.1692	0.4833	0.0402	0.0416	0.1690	0.2681
		5D	0.3439	0.2834	0.3346	0.1284	0.3308	0.0317	0.0291	0.1154	0.1997
		7D	0.2543	0.2254	0.2716	0.0894	0.2808	0.0250	0.0233	0.0790	0.1561
		Avg.	0.3917	0.3548	0.4265	0.1706	0.4394	0.0421	0.0461	0.1556	0.2533
	AutoTimes	1D	0.5904	0.6595	0.7269	0.3583	0.6964	0.1035	0.1381	0.3728	0.4557
		3D	0.4307	0.4794	0.5365	0.2296	0.5553	0.0609	0.0708	0.2425	0.3257
		5D	0.3386	0.3883	0.4334	0.2038	0.4503	0.0524	0.0534	0.1652	0.2607
		7D	0.2539	0.2968	0.3465	0.1403	0.3743	0.0457	0.0386	0.1488	0.2056
		Avg.	0.4034	0.4560	0.5108	0.2330	0.5191	0.0656	0.0752	0.2323	0.3119

Table 27: Benchmark(MAE) on part 3(Fort Myers area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
MLP	1D	0.0251	0.0246	0.0207	0.0463	0.0302	0.0628	0.0505	0.0380	0.0373	
	3D	0.0526	0.0607	0.0523	0.0777	0.0603	0.1228	0.0977	0.0757	0.0750	
	5D	0.0839	0.0940	0.0814	0.1102	0.0958	0.1610	0.1305	0.1105	0.1084	
	7D	0.1005	0.1202	0.1013	0.1327	0.1143	0.1891	0.1558	0.1281	0.1303	
	Avg.	0.0655	0.0749	0.0639	0.0917	0.0752	0.1339	0.1086	0.0881	0.0877	
MLP	TSMixer	1D	0.0247	0.0327	0.0265	0.0477	0.0353	0.0684	0.0548	0.0461	0.0420
	3D	0.0503	0.0683	0.0560	0.0821	0.0665	0.1226	0.0987	0.0811	0.0782	
	5D	0.0734	0.0989	0.0827	0.1093	0.0899	0.1596	0.1307	0.1108	0.1069	
	7D	0.0946	0.1254	0.1081	0.1330	0.1100	0.1873	0.1561	0.1368	0.1314	
	Avg.	0.0608	0.0813	0.0683	0.0930	0.0754	0.1345	0.1101	0.0937	0.0896	
NLinear	1D	0.0184	0.0243	0.0200	0.0427	0.0307	0.0615	0.0480	0.0377	0.0354	
	3D	0.0463	0.0613	0.0516	0.0798	0.0636	0.1196	0.0947	0.0752	0.0740	
	5D	0.0705	0.0931	0.0796	0.1089	0.0880	0.1577	0.1279	0.1055	0.1039	
	7D	0.0933	0.1207	0.1053	0.1334	0.1078	0.1861	0.1545	0.1322	0.1292	
	Avg.	0.0571	0.0749	0.0641	0.0912	0.0725	0.1312	0.1063	0.0877	0.0856	
CNN	TCN	1D	0.0577	0.0505	0.0318	0.0970	0.1129	0.0920	0.2015	0.1758	0.1024
	3D	0.0968	0.1382	0.0689	0.1048	0.1373	0.1937	0.1485	0.2512	0.1424	
	5D	0.1406	0.1002	0.0931	0.1558	0.1816	0.2363	0.1885	0.2557	0.1690	
	7D	0.2256	0.1780	0.1241	0.1633	0.2116	0.2240	0.2846	0.2676	0.2098	
	Avg.	0.1302	0.1167	0.0795	0.1302	0.1608	0.1865	0.2058	0.2376	0.1559	
CNN	ModernTCN	1D	0.0221	0.0265	0.0205	0.0443	0.0278	0.0654	0.0491	0.0357	0.0364
	3D	0.0567	0.0702	0.0510	0.0861	0.0603	0.1332	0.1010	0.0719	0.0788	
	5D	0.0871	0.1048	0.0834	0.1215	0.0883	0.1774	0.1384	0.1015	0.1128	
	7D	0.1202	0.1362	0.1056	0.1499	0.1094	0.2057	0.1686	0.1280	0.1405	
	Avg.	0.0715	0.0845	0.0651	0.1005	0.0714	0.1454	0.1143	0.0843	0.0921	
TimesNet	1D	0.0199	0.0288	0.0242	0.0513	0.0445	0.0809	0.0651	0.0528	0.0459	
	3D	0.0467	0.0651	0.0558	0.0875	0.0724	0.1426	0.1164	0.0866	0.0841	
	5D	0.0703	0.0970	0.0838	0.1148	0.0988	0.1852	0.1484	0.1127	0.1139	
	7D	0.0923	0.1284	0.1121	0.1438	0.1158	0.2145	0.1759	0.1417	0.1406	
	Avg.	0.0573	0.0798	0.0690	0.0993	0.0829	0.1558	0.1264	0.0984	0.0961	
Transformer	iTransformer	1D	0.0190	0.0253	0.0192	0.0383	0.0278	0.0641	0.0482	0.0345	0.0346
	3D	0.0469	0.0617	0.0487	0.0726	0.0576	0.1244	0.0959	0.0706	0.0723	
	5D	0.0709	0.0948	0.0766	0.1023	0.0834	0.1633	0.1331	0.0977	0.1028	
	7D	0.0914	0.1226	0.1041	0.1275	0.1048	0.1889	0.1596	0.1214	0.1275	
	Avg.	0.0571	0.0761	0.0621	0.0852	0.0684	0.1352	0.1092	0.0811	0.0843	
	PatchTST	1D	0.0183	0.0241	0.0177	0.0381	0.0288	0.0637	0.0481	0.0349	0.0342
	3D	0.0464	0.0654	0.0545	0.0765	0.0588	0.1242	0.1007	0.0714	0.0747	
	5D	0.0762	0.0951	0.0761	0.1057	0.0835	0.1651	0.1354	0.1014	0.1048	
	7D	0.0984	0.1266	0.1032	0.1319	0.1053	0.1955	0.1608	0.1257	0.1309	
	Avg.	0.0598	0.0778	0.0629	0.0881	0.0691	0.1371	0.1112	0.0833	0.0862	

Table 28: Benchmark(MAE) on part 3(Fort Myers area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.0486	0.0447	0.0393	0.0640	0.0465	0.1120	0.0839	0.0606	0.0625
	3D	0.0895	0.0878	0.0788	0.1048	0.0828	0.1775	0.1299	0.1091	0.1075
	5D	0.1246	0.1247	0.1167	0.1344	0.1172	0.2057	0.1590	0.1281	0.1388
	7D	0.1621	0.1584	0.1496	0.1725	0.1389	0.2203	0.1823	0.1567	0.1676
	Avg.	0.1062	0.1039	0.0961	0.1189	0.0964	0.1789	0.1388	0.1136	0.1191
RNN DeepAR	1D	0.0569	0.0461	0.0381	0.0632	0.0486	0.1023	0.0763	0.0694	0.0626
	3D	0.0859	0.0950	0.0851	0.1180	0.0949	0.1640	0.1403	0.0993	0.1103
	5D	0.1086	0.1219	0.1137	0.1424	0.1388	0.2005	0.1742	0.1466	0.1433
	7D	0.1496	0.1542	0.1549	0.1745	0.1467	0.2247	0.2012	0.1485	0.1693
	Avg.	0.1003	0.1043	0.0980	0.1245	0.1073	0.1729	0.1480	0.1159	0.1214
DilatedRNN	1D	0.0394	0.0268	0.0277	0.0441	0.0472	0.0691	0.0596	0.0372	0.0439
	3D	0.0726	0.0714	0.0679	0.0978	0.0648	0.1390	0.1186	0.0837	0.0895
	5D	0.1100	0.1028	0.0887	0.1257	0.0966	0.2000	0.1578	0.1128	0.1243
	7D	0.1312	0.1500	0.1175	0.1499	0.1234	0.2363	0.1870	0.1478	0.1554
	Avg.	0.0883	0.0877	0.0754	0.1043	0.0830	0.1611	0.1308	0.0954	0.1033
GCN	1D	0.0543	0.0596	0.0264	0.3354	0.1422	0.0789	0.0946	0.1151	0.1133
	3D	0.0858	0.0942	0.0598	0.3643	0.1756	0.1443	0.1374	0.1593	0.1526
	5D	0.1006	0.1192	0.0851	0.3859	0.2002	0.1767	0.1713	0.1952	0.1793
	7D	0.1196	0.1408	0.1141	0.4079	0.2197	0.2033	0.2068	0.2315	0.2055
	Avg.	0.0901	0.1034	0.0713	0.3734	0.1845	0.1508	0.1525	0.1753	0.1627
GNN FourierGNN	1D	10.0224	0.0295	0.0262	0.0462	0.0370	0.0732	0.0543	0.0436	0.0415
	3D	0.0548	0.0673	0.0538	0.0829	0.0784	0.1336	0.1087	0.0841	0.0830
	5D	0.0830	0.0978	0.0814	0.1153	0.0967	0.1668	0.1511	0.1109	0.1129
	7D	0.1064	0.1236	0.1114	0.1452	0.1335	0.2015	0.1823	0.1354	0.1424
	Avg.	0.0666	0.0796	0.0682	0.0974	0.0864	0.1438	0.1241	0.0935	0.0949
StemGNN	1D	0.0636	0.0445	0.0335	0.0635	0.0586	0.1001	0.0599	0.0489	0.0591
	3D	0.1114	0.1001	0.0788	0.1273	0.1028	0.1524	0.1421	0.1408	0.1195
	5D	0.1690	0.1688	0.1022	0.1698	0.1352	0.1980	0.2366	0.2161	0.1745
	7D	0.2094	0.2090	0.1719	0.2016	0.1625	0.2325	0.2514	0.2730	0.2139
	Avg.	0.1384	0.1306	0.0966	0.1406	0.1148	0.1707	0.1725	0.1697	0.1417
LLM GPT4TS	1D	0.0209	0.0287	0.0261	0.0482	0.0407	0.0740	0.0596	0.7614	0.1324
	3D	0.0467	0.0638	0.0553	0.0845	0.0730	0.1431	0.1105	0.0869	0.0830
	5D	0.0702	0.0946	0.0832	0.1141	0.0980	0.1823	0.1455	0.1120	0.1125
	7D	0.0909	0.1231	0.1105	0.1392	0.1160	0.2087	0.1715	0.1397	0.1375
	Avg.	0.0572	0.0776	0.0688	0.0965	0.0819	0.1520	0.1218	0.2750	0.1163
AutoTimes	1D	0.0228	0.0282	0.0199	0.0429	0.0321	0.0680	0.0542	0.0407	0.0386
	3D	0.0475	0.0615	0.0486	0.0760	0.0638	0.1240	0.0993	0.0751	0.0745
	5D	0.0692	0.0965	0.0737	0.1075	0.0892	0.1632	0.1315	0.1056	0.1045
	7D	0.0921	0.1218	0.1038	0.1304	0.1083	0.1900	0.1573	0.1285	0.1290
	Avg.	0.0579	0.0770	0.0615	0.0892	0.0734	0.1363	0.1106	0.0875	0.0867

Table 29: Benchmark(MSE) on part 3(Fort Myers area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
MLP	1D	0.0036	0.0046	0.0052	0.0165	0.0137	0.0368	0.0139	0.0095	0.0130	
	3D	0.0156	0.0159	0.0199	0.0359	0.0318	0.0888	0.0374	0.0252	0.0338	
	5D	0.0295	0.0300	0.0345	0.0538	0.0486	0.1253	0.0552	0.0426	0.0524	
	7D	0.0461	0.0442	0.0494	0.0729	0.0635	0.1537	0.0701	0.0562	0.0695	
	Avg.	0.0237	0.0237	0.0272	0.0448	0.0394	0.1011	0.0441	0.0334	0.0422	
MLP	1D	0.0047	0.0056	0.0070	0.0182	0.0163	0.0420	0.0159	0.0116	0.0152	
	3D	0.0176	0.0174	0.0217	0.0381	0.0353	0.0983	0.0403	0.0280	0.0371	
	5D	0.0348	0.0315	0.0372	0.0561	0.0522	0.1410	0.0605	0.0452	0.0573	
	7D	0.0548	0.0465	0.0531	0.0740	0.0679	0.1735	0.0778	0.0624	0.0763	
	Avg.	0.0280	0.0253	0.0297	0.0466	0.0429	0.1137	0.0486	0.0368	0.0465	
NLinear	1D	0.0031	0.0044	0.0050	0.0174	0.0158	0.0392	0.0138	0.0098	0.0136	
	3D	0.0153	0.0159	0.0202	0.0378	0.0349	0.0970	0.0383	0.0263	0.0357	
	5D	0.0321	0.0298	0.0362	0.0561	0.0518	0.1392	0.0581	0.0433	0.0558	
	7D	0.0520	0.0449	0.0526	0.0742	0.0677	0.1724	0.0754	0.0607	0.0750	
	Avg.	0.0256	0.0238	0.0285	0.0464	0.0426	0.1120	0.0464	0.0350	0.0450	
TCN	1D	0.0925	0.0196	0.0078	0.0383	0.0743	0.0452	0.0900	0.1310	0.0623	
	3D	0.1029	0.1063	0.0244	0.0440	0.0804	0.1231	0.0514	0.1998	0.0916	
	5D	0.1186	0.0304	0.0379	0.0681	0.1249	0.1691	0.0767	0.2183	0.1055	
	7D	0.1905	0.1158	0.0575	0.0808	0.1354	0.1655	0.1613	0.1934	0.1375	
	Avg.	0.1261	0.0680	0.0319	0.0578	0.1038	0.1257	0.0949	0.1856	0.0992	
CNN	1D	0.0048	0.0050	0.0045	0.0208	0.0142	0.0451	0.0148	0.0100	0.0149	
	3D	0.0219	0.0210	0.0187	0.0451	0.0345	0.1297	0.0455	0.0276	0.0430	
	5D	0.0466	0.0406	0.0363	0.0757	0.0548	0.1949	0.0722	0.0451	0.0708	
	7D	0.0710	0.0626	0.0527	0.1014	0.0705	0.2231	0.0981	0.0619	0.0927	
	Avg.	0.0361	0.0323	0.0280	0.0607	0.0435	0.1482	0.0576	0.0362	0.0553	
TimesNet	1D	0.0035	0.0052	0.0066	0.0215	0.0262	0.0586	0.0206	0.0163	0.0198	
	3D	0.0159	0.0181	0.0214	0.0461	0.0445	0.1318	0.0553	0.0331	0.0458	
	5D	0.0315	0.0351	0.0369	0.0654	0.0684	0.2087	0.0811	0.0509	0.0723	
	7D	0.0514	0.0543	0.0537	0.0919	0.0871	0.2343	0.1012	0.0706	0.0931	
	Avg.	0.0256	0.0282	0.0297	0.0562	0.0565	0.1584	0.0645	0.0427	0.0577	
Transformer	iTransformer	1D	0.0035	0.0047	0.0050	0.0177	0.0157	0.0436	0.0140	0.0094	0.0142
		3D	0.0171	0.0171	0.0209	0.0403	0.0352	0.1103	0.0410	0.0255	0.0384
		5D	0.0343	0.0335	0.0356	0.0623	0.0527	0.1582	0.0674	0.0423	0.0608
		7D	0.0518	0.0522	0.0539	0.0807	0.0714	0.1897	0.0847	0.0590	0.0804
		Avg.	0.0267	0.0269	0.0288	0.0503	0.0437	0.1254	0.0518	0.0341	0.0485
	PatchTST	1D	0.0032	0.0048	0.0044	0.0179	0.0154	0.0444	0.0144	0.0096	0.0143
		3D	0.0161	0.0205	0.0212	0.0420	0.0353	0.1100	0.0432	0.0266	0.0393
		5D	0.0385	0.0364	0.0374	0.0608	0.0531	0.1632	0.0668	0.0438	0.0625
		7D	0.0564	0.0578	0.0514	0.0832	0.0700	0.2020	0.0825	0.0598	0.0829
		Avg.	0.0285	0.0299	0.0286	0.0510	0.0435	0.1299	0.0517	0.0350	0.0498

Table 30: Benchmark(MSE) on part 3(Fort Myers area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.0183	0.0087	0.0109	0.0287	0.0217	0.0748	0.0274	0.0181	0.0261
	3D	0.0420	0.0271	0.0312	0.0588	0.0444	0.1455	0.0540	0.0381	0.0551
	5D	0.0624	0.0427	0.0506	0.0774	0.0630	0.1761	0.0717	0.0552	0.0749
	7D	0.0949	0.0621	0.0773	0.0985	0.0797	0.2013	0.0880	0.0714	0.0967
	Avg.	0.0544	0.0352	0.0425	0.0659	0.0522	0.1494	0.0603	0.0457	0.0632
DeepAR	1D	0.0408	0.0090	0.0108	0.0298	0.0224	0.0750	0.0266	0.0173	0.0290
	3D	0.0634	0.0251	0.0316	0.0568	0.0463	0.1353	0.0580	0.0370	0.0567
	5D	0.0682	0.0414	0.0506	0.0743	0.0642	0.1775	0.0754	0.0564	0.0760
	7D	0.0910	0.0569	0.0728	0.0969	0.0799	0.1947	0.0899	0.0703	0.0941
	Avg.	0.0659	0.0331	0.0414	0.0644	0.0532	0.1457	0.0625	0.0453	0.0639
DilatedRNN	1D	0.0172	0.0048	0.0056	0.0194	0.0162	0.0408	0.0175	0.0104	0.0165
	3D	0.0395	0.0193	0.0225	0.0442	0.0356	0.1128	0.0489	0.0300	0.0441
	5D	0.0583	0.0347	0.0401	0.0675	0.0560	0.1787	0.0786	0.0509	0.0706
	7D	0.0808	0.0583	0.0578	0.0958	0.0747	0.2167	0.0990	0.0714	0.0943
	Avg.	0.0489	0.0293	0.0315	0.0567	0.0457	0.1372	0.0610	0.0407	0.0564
GCN	1D	0.0108	0.0094	0.0060	0.3682	0.0982	0.0392	0.0227	0.0923	0.0809
	3D	0.0267	0.0246	0.0196	0.3854	0.1184	0.0918	0.0472	0.1217	0.1044
	5D	0.0434	0.0359	0.0339	0.4006	0.1337	0.1285	0.0674	0.1440	0.1234
	7D	0.0569	0.0499	0.0486	0.4154	0.1489	0.1565	0.0868	0.1794	0.1428
	Avg.	0.0344	0.0300	0.0270	0.3924	0.1248	0.1040	0.0560	0.1343	0.1129
FourierGNN	1D	0.0037	0.0049	0.0063	0.0178	0.0153	0.0430	0.0145	0.0108	0.0145
	3D	0.0154	0.0166	0.0207	0.0379	0.0364	0.0938	0.0399	0.0268	0.0359
	5D	0.0298	0.0302	0.0359	0.0555	0.0496	0.1312	0.0609	0.0432	0.0545
	7D	0.0477	0.0434	0.0542	0.0769	0.0691	0.1594	0.0787	0.0591	0.0736
	Avg.	0.0241	0.0237	0.0293	0.0471	0.0426	0.1069	0.0485	0.0350	0.0446
StemGNN	1D	0.0390	0.0081	0.0070	0.0221	0.0281	0.0535	0.0164	0.0145	0.0236
	3D	0.0816	0.0405	0.0270	0.0609	0.0599	0.1179	0.0722	0.0861	0.0683
	5D	0.1100	0.0850	0.0433	0.1015	0.0809	0.1587	0.1540	0.1312	0.1081
	7D	0.1554	0.1173	0.1096	0.1232	0.1079	0.1982	0.1746	0.2072	0.1492
	Avg.	0.0965	0.0628	0.0467	0.0769	0.0692	0.1321	0.1043	0.1098	0.0873
GPT4TS	1D	0.0038	0.0050	0.0060	0.0202	0.0223	0.0542	0.0183	0.6054	0.0919
	3D	0.0151	0.0174	0.0205	0.0442	0.0474	0.1530	0.0496	0.0339	0.0476
	5D	0.0321	0.0319	0.0371	0.0648	0.0716	0.2071	0.0829	0.0526	0.0725
	7D	0.0494	0.0489	0.0528	0.0872	0.0841	0.2397	0.0996	0.0734	0.0919
	Avg.	0.0251	0.0258	0.0291	0.0541	0.0564	0.1635	0.0626	0.1913	0.0760
AutoTimes	1D	0.0037	0.0052	0.0048	0.0183	0.0159	0.0447	0.0161	0.0106	0.0149
	3D	0.0159	0.0167	0.0189	0.0396	0.0342	0.1026	0.0409	0.0263	0.0369
	5D	0.0321	0.0314	0.0337	0.0564	0.0519	0.1485	0.0614	0.0429	0.0573
	7D	0.0522	0.0466	0.0508	0.0745	0.0661	0.1807	0.0781	0.0586	0.0760
	Avg.	0.0260	0.0250	0.0271	0.0472	0.0420	0.1191	0.0491	0.0346	0.0463

Table 31: Benchmark(SED10) on part 3(Fort Myers area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.6714	0.9580	0.6989	0.7088	0.4910	0.6202	0.5661	0.8442	0.6948
	3D	0.6107	0.8594	0.6242	0.7260	0.4937	0.4783	0.5238	0.8326	0.6436
	5D	0.5833	0.7752	0.6230	0.6262	0.4892	0.5590	0.5791	0.7968	0.6290
	7D	0.5649	0.7550	0.5619	0.5997	0.5046	0.4772	0.4864	0.6886	0.5798
	Avg.	0.6076	0.8369	0.6270	0.6652	0.4946	0.5337	0.5389	0.7906	0.6368
MLP	1D	0.6675	0.9189	0.6900	0.7345	0.5080	0.6620	0.6030	0.8160	0.7000
	3D	0.5919	0.8285	0.6258	0.6505	0.4595	0.5203	0.4941	0.7342	0.6131
	5D	0.5567	0.7866	0.5754	0.6006	0.4316	0.4412	0.4199	0.6803	0.5615
	7D	0.5350	0.7498	0.5379	0.5651	0.4129	0.3812	0.3557	0.6375	0.5219
	Avg.	0.5878	0.8209	0.6073	0.6377	0.4530	0.5012	0.4682	0.7170	0.5991
NLinear	1D	0.6653	0.9466	0.7012	0.7387	0.5234	0.6204	0.6081	0.8296	0.7041
	3D	0.5963	0.8470	0.6314	0.6528	0.4532	0.5048	0.4892	0.7448	0.6149
	5D	0.5604	0.7925	0.5800	0.6023	0.4297	0.4342	0.4069	0.6914	0.5622
	7D	0.5393	0.7539	0.5417	0.5660	0.4113	0.3787	0.3420	0.6494	0.5228
	Avg.	0.5903	0.8350	0.6136	0.6399	0.4544	0.4845	0.4615	0.7288	0.6010
TCN	1D	0.6607	0.9349	0.7126	0.6214	0.3251	0.6057	0.3817	0.4229	0.5831
	3D	0.5849	0.7254	0.6607	0.6150	0.3200	0.3803	0.4778	0.3677	0.5165
	5D	0.4883	0.8096	0.5672	0.5006	0.2824	0.3337	0.3708	0.4034	0.4695
	7D	0.2490	0.6355	0.5764	0.6034	0.2939	0.3018	0.2916	0.3836	0.4169
	Avg.	0.4957	0.7763	0.6292	0.5851	0.3054	0.4054	0.3805	0.3944	0.4965
CNN	1D	0.6643	0.9457	0.6975	0.7506	0.5325	0.6309	0.5645	0.8346	0.7026
	3D	0.5833	0.8013	0.6528	0.6401	0.4782	0.4488	0.4296	0.7353	0.5962
	5D	0.5274	0.7259	0.5842	0.6140	0.4395	0.3732	0.3519	0.6687	0.5356
	7D	0.4967	0.6819	0.5463	0.5468	0.4399	0.3177	0.2592	0.6290	0.4897
	Avg.	0.5679	0.7887	0.6202	0.6379	0.4725	0.4426	0.4013	0.7169	0.5810
TimesNet	1D	0.6628	0.8907	0.6824	0.7392	0.4722	0.5926	0.5199	0.7606	0.6651
	3D	0.5922	0.8103	0.6122	0.6461	0.4452	0.4391	0.3654	0.6902	0.5751
	5D	0.5666	0.7517	0.5491	0.6090	0.4134	0.3682	0.3255	0.6374	0.5276
	7D	0.5199	0.7043	0.5105	0.5579	0.3969	0.2931	0.2576	0.5950	0.4794
	Avg.	0.5854	0.7892	0.5885	0.6380	0.4319	0.4233	0.3671	0.6708	0.5618
Transformer	iTransformer	1D	0.6753	0.9308	0.6948	0.7630	0.5149	0.6342	0.5745	0.7018
		3D	0.5920	0.8284	0.6263	0.6781	0.4784	0.4697	0.4621	0.6092
		5D	0.5688	0.7619	0.5898	0.6217	0.4517	0.4084	0.3802	0.5581
		7D	0.5440	0.7203	0.5514	0.5852	0.4287	0.3532	0.3130	0.5171
		Avg.	0.5950	0.8103	0.6156	0.6620	0.4684	0.4664	0.4325	0.5965
	PatchTST	1D	0.6779	0.9328	0.7007	0.7626	0.5236	0.6109	0.5828	0.7035
		3D	0.6072	0.8306	0.6327	0.6674	0.4839	0.4845	0.4406	0.6113
		5D	0.5537	0.7707	0.5892	0.6232	0.4570	0.4017	0.3710	0.5569
		7D	0.5374	0.7078	0.5492	0.5853	0.4358	0.3455	0.3136	0.5152
		Avg.	0.5941	0.8105	0.6180	0.6596	0.4751	0.4606	0.4270	0.5967

Table 32: Benchmark(SED110) on part 3(Fort Myers area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
LSTM	1D	0.6375	0.8919	0.6771	0.7127	0.5130	0.6141	0.4802	0.6979	0.6530	
	3D	0.5603	0.7366	0.5969	0.7031	0.4666	0.4219	0.4553	0.6716	0.5765	
	5D	0.5230	0.7399	0.5582	0.6459	0.4201	0.3505	0.4925	0.6755	0.5507	
	7D	0.5015	0.7195	0.5452	0.5964	0.4224	0.3320	0.4663	0.6533	0.5296	
	Avg.	0.5556	0.7720	0.5943	0.6645	0.4555	0.4296	0.4736	0.6746	0.5775	
RNN	DeepAR	1D	0.6113	0.9191	0.5886	0.7494	0.5485	0.4949	0.4766	0.7039	0.6365
		3D	0.5456	0.7545	0.6393	0.6426	0.4689	0.3235	0.3871	0.7007	0.5578
		5D	0.5199	0.7361	0.5829	0.6406	0.3841	0.2802	0.2172	0.5790	0.4925
		7D	0.5267	0.7149	0.4846	0.5765	0.4471	0.2646	0.3546	0.5960	0.4956
		Avg.	0.5509	0.7811	0.5738	0.6523	0.4621	0.3408	0.3589	0.6449	0.5456
DilatedRNN	1D	0.6157	0.9270	0.7114	0.8285	0.4265	0.5347	0.4866	0.7628	0.6616	
	3D	0.5809	0.7952	0.6612	0.5798	0.4546	0.3865	0.3726	0.7252	0.5695	
	5D	0.5291	0.8040	0.5665	0.6098	0.4765	0.2496	0.2638	0.6925	0.5239	
	7D	0.5392	0.6853	0.5523	0.5684	0.4365	0.2247	0.3043	0.5970	0.4885	
	Avg.	0.5662	0.8029	0.6228	0.6466	0.4485	0.3489	0.3568	0.6944	0.5609	
GCN	1D	0.6792	0.8309	0.7292	0.4914	0.4288	0.5120	0.4721	0.7129	0.6071	
	3D	0.6144	0.8262	0.6910	0.4325	0.3749	0.4319	0.5107	0.6594	0.5676	
	5D	0.5949	0.7921	0.6016	0.4126	0.3364	0.3577	0.3270	0.6403	0.5078	
	7D	0.5728	0.7445	0.6076	0.3891	0.3311	0.2809	0.3782	0.6840	0.4985	
	Avg.	0.6153	0.7984	0.6574	0.4314	0.3678	0.3956	0.4220	0.6741	0.5453	
GNN	FourierGNN	1D	0.6606	0.9310	0.7011	0.7420	0.5204	0.5268	0.6103	0.7723	0.6831
		3D	0.6044	0.8487	0.6062	0.6820	0.4898	0.5028	0.5337	0.7566	0.6281
		5D	0.5545	0.8082	0.6223	0.6375	0.4579	0.3712	0.5406	0.7085	0.5876
		7D	0.5737	0.7776	0.6044	0.6699	0.4793	0.4239	0.5228	0.6609	0.5891
		Avg.	0.5983	0.8414	0.6335	0.6828	0.4869	0.4562	0.5519	0.7246	0.6219
StemGNN	1D	0.6429	0.9027	0.6334	0.6619	0.4429	0.4493	0.5562	0.7738	0.6329	
	3D	0.5120	0.8540	0.6633	0.6193	0.4460	0.3921	0.3833	0.6762	0.5683	
	5D	0.4908	0.7262	0.6228	0.5595	0.3974	0.3235	0.3157	0.6449	0.5101	
	7D	0.4009	0.6838	0.4941	0.5233	0.3831	0.2672	0.2345	0.5653	0.4440	
	Avg.	0.5116	0.7917	0.6034	0.5910	0.4173	0.3580	0.3724	0.6650	0.5388	
LLM	GPT4TS	1D	0.6635	0.9232	0.6942	0.7230	0.4875	0.6188	0.5335	0.7614	0.6757
		3D	0.6057	0.8277	0.6177	0.6298	0.4394	0.4481	0.4085	0.6951	0.5840
		5D	0.5684	0.7776	0.5751	0.5932	0.4164	0.3858	0.3212	0.6346	0.5340
		7D	0.5243	0.7301	0.5333	0.5432	0.3953	0.3318	0.2623	0.5767	0.4871
		Avg.	0.5905	0.8147	0.6051	0.6223	0.4347	0.4462	0.3814	0.6670	0.5702
AutoTimes	1D	0.6599	0.9250	0.6921	0.7510	0.5256	0.6094	0.5451	0.8114	0.6899	
	3D	0.6089	0.8376	0.6380	0.6568	0.4840	0.4850	0.4504	0.7400	0.6126	
	5D	0.5746	0.7828	0.5837	0.6040	0.4538	0.4243	0.3857	0.6793	0.5610	
	7D	0.5422	0.7350	0.5560	0.5717	0.4415	0.3649	0.3277	0.6459	0.5231	
	Avg.	0.5964	0.8201	0.6175	0.6459	0.4762	0.4709	0.4272	0.7191	0.5967	

Table 33: Benchmark(SED15) on part 3(Fort Myers area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
MLP	1D	0.4894	0.8078	0.7149	0.4916	0.3294	0.5618	0.3728	0.6667	0.5543	
	3D	0.4773	0.7190	0.6372	0.5178	0.3379	0.4137	0.3164	0.6467	0.5083	
	5D	0.4737	0.6523	0.5924	0.4466	0.3030	0.4697	0.3772	0.6306	0.4932	
	7D	0.4645	0.6408	0.5297	0.4251	0.3127	0.3881	0.2769	0.5457	0.4479	
	Avg.	0.4762	0.7050	0.6186	0.4703	0.3208	0.4583	0.3358	0.6224	0.5009	
MLP	1D	0.4885	0.7318	0.7059	0.5263	0.3722	0.5500	0.3410	0.6631	0.5474	
	3D	0.4713	0.6731	0.6384	0.4570	0.3260	0.3770	0.2068	0.5727	0.4653	
	5D	0.4577	0.6479	0.5783	0.4103	0.3053	0.2971	0.1542	0.5068	0.4197	
	7D	0.4474	0.6274	0.5311	0.3795	0.2879	0.2336	0.1456	0.4626	0.3894	
	Avg.	0.4662	0.6700	0.6134	0.4433	0.3229	0.3644	0.2119	0.5513	0.4554	
NLinear	1D	0.4908	0.8016	0.7131	0.5444	0.3801	0.5330	0.3531	0.6808	0.5621	
	3D	0.4729	0.6957	0.6459	0.4658	0.3114	0.3720	0.2127	0.5928	0.4711	
	5D	0.4581	0.6564	0.5865	0.4188	0.2919	0.2901	0.1535	0.5293	0.4231	
	7D	0.4475	0.6307	0.5373	0.3836	0.2755	0.2271	0.1539	0.4864	0.3927	
	Avg.	0.4673	0.6961	0.6207	0.4532	0.3147	0.3555	0.2183	0.5724	0.4623	
TCN	1D	0.4885	0.5418	0.6754	0.4250	0.0903	0.5215	0.1972	0.1387	0.3848	
	3D	0.2287	0.4575	0.4992	0.3864	0.0903	0.2572	0.2046	0.1727	0.2871	
	5D	0.2222	0.6698	0.3806	0.3737	0.1045	0.2205	0.1504	0.1179	0.2799	
	7D	0.2372	0.4134	0.4001	0.3923	0.1494	0.2020	0.1318	0.1100	0.2545	
	Avg.	0.2941	0.5206	0.4888	0.3943	0.1086	0.3003	0.1710	0.1348	0.3016	
CNN	1D	0.4916	0.8000	0.7168	0.5206	0.3806	0.5178	0.3296	0.6500	0.5509	
	3D	0.4736	0.6654	0.6446	0.4359	0.3385	0.2951	0.1874	0.5540	0.4493	
	5D	0.4546	0.6087	0.5675	0.3828	0.3163	0.2291	0.1094	0.4870	0.3944	
	7D	0.4437	0.5747	0.5238	0.3442	0.2964	0.1796	0.0746	0.4491	0.3607	
	Avg.	0.4659	0.6622	0.6132	0.4209	0.3329	0.3054	0.1752	0.5350	0.4388	
TimesNet	1D	0.4887	0.7914	0.6931	0.5047	0.3416	0.4730	0.2862	0.5783	0.5196	
	3D	0.4738	0.6845	0.6056	0.4265	0.3145	0.3008	0.1060	0.4963	0.4260	
	5D	0.4572	0.6307	0.5365	0.3854	0.2857	0.2354	0.0782	0.4423	0.3814	
	7D	0.4477	0.6095	0.4801	0.3425	0.2637	0.1745	0.0602	0.3952	0.3467	
	Avg.	0.4669	0.6790	0.5788	0.4148	0.3013	0.2959	0.1327	0.4780	0.4184	
Transformer	iTransformer	1D	0.4861	0.7835	0.7049	0.5526	0.3745	0.5381	0.3514	0.6485	0.5549
		3D	0.4722	0.6740	0.6211	0.4677	0.3346	0.3229	0.2136	0.5557	0.4577
		5D	0.4566	0.6520	0.5766	0.4145	0.3157	0.2654	0.1486	0.5018	0.4164
		7D	0.4463	0.6171	0.5237	0.3832	0.2963	0.2088	0.1013	0.4623	0.3799
		Avg.	0.4653	0.6817	0.6066	0.4545	0.3303	0.3338	0.2037	0.5421	0.4522
	PatchTST	1D	0.4924	0.8199	0.7104	0.5501	0.3796	0.4983	0.3188	0.6643	0.5542
		3D	0.4747	0.6933	0.6454	0.4688	0.3416	0.3528	0.1760	0.5698	0.4653
		5D	0.4567	0.6503	0.5773	0.4337	0.3168	0.2647	0.1300	0.5138	0.4179
		7D	0.4544	0.6030	0.5288	0.3971	0.2896	0.2039	0.1030	0.4685	0.3810
		Avg.	0.4695	0.6916	0.6154	0.4624	0.3319	0.3299	0.1820	0.5541	0.4546

Table 34: Benchmark(SED15) on part 3(Fort Myers area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
LSTM	1D	0.4820	0.6970	0.6832	0.5265	0.3526	0.4093	0.2297	0.5224	0.4878
	3D	0.4696	0.6260	0.5874	0.4875	0.2963	0.1749	0.2572	0.4429	0.4177
	5D	0.4635	0.6474	0.3289	0.4312	0.2716	0.1569	0.0442	0.4783	0.3528
	7D	0.4340	0.6733	0.3075	0.3608	0.2654	0.1364	0.0542	0.4455	0.3347
	Avg.	0.4623	0.6609	0.4768	0.4515	0.2965	0.2194	0.1463	0.4723	0.3982
DeepAR	1D	0.4587	0.7348	0.5771	0.5112	0.3805	0.3081	0.1743	0.4682	0.4516
	3D	0.4104	0.6217	0.3898	0.4468	0.3524	0.1630	0.1703	0.4901	0.3806
	5D	0.3463	0.6626	0.3252	0.4434	0.2483	0.1350	0.0699	0.3937	0.3280
	7D	0.4636	0.6772	0.2954	0.4170	0.2511	0.1049	0.1605	0.4080	0.3472
	Avg.	0.4197	0.6741	0.3969	0.4546	0.3081	0.1778	0.1437	0.4400	0.3769
DilatedRNN	1D	0.4427	0.7950	0.7256	0.6408	0.2791	0.4482	0.2522	0.5895	0.5216
	3D	0.4861	0.6492	0.6471	0.3783	0.3198	0.2445	0.1764	0.5450	0.4308
	5D	0.3852	0.6652	0.5548	0.4024	0.3185	0.1632	0.1136	0.5060	0.3886
	7D	0.4685	0.5954	0.5216	0.3734	0.2855	0.1483	0.1444	0.4271	0.3705
	Avg.	0.4456	0.6762	0.6123	0.4487	0.3007	0.2510	0.1716	0.5169	0.4279
GCN	1D	0.4893	0.7551	0.7258	0.3366	0.3070	0.3908	0.2399	0.5583	0.4754
	3D	0.4817	0.7295	0.6715	0.2813	0.2968	0.2954	0.2091	0.5118	0.4346
	5D	0.4785	0.7154	0.5946	0.2677	0.2595	0.2635	0.0498	0.4793	0.3885
	7D	0.4697	0.6812	0.5678	0.2610	0.2722	0.2174	0.1484	0.5102	0.3910
	Avg.	0.4798	0.7203	0.6399	0.2867	0.2839	0.2918	0.1618	0.5149	0.4224
FourierGNN	1D	0.4873	0.7843	0.7085	0.5126	0.3541	0.4430	0.4340	0.6063	0.5412
	3D	0.4800	0.7305	0.6125	0.4674	0.3419	0.3558	0.3325	0.5709	0.4864
	5D	0.4773	0.6881	0.5911	0.4507	0.3064	0.2782	0.3179	0.5243	0.4543
	7D	0.4705	0.6634	0.5482	0.4566	0.3343	0.3072	0.3234	0.4842	0.4485
	Avg.	0.4788	0.7166	0.6151	0.4718	0.3342	0.3460	0.3519	0.5464	0.4826
StemGNN	1D	0.4950	0.7993	0.6503	0.4581	0.3070	0.3161	0.3138	0.5819	0.4902
	3D	0.4878	0.6894	0.6336	0.4333	0.2762	0.2694	0.1468	0.4612	0.4247
	5D	0.4762	0.6203	0.6074	0.4065	0.2529	0.2048	0.1256	0.4329	0.3908
	7D	0.2821	0.5487	0.4545	0.3398	0.2205	0.1658	0.0749	0.3691	0.3069
	Avg.	0.4353	0.6644	0.5865	0.4094	0.2642	0.2390	0.1653	0.4613	0.4032
GPT4TS	1D	0.4909	0.7803	0.6876	0.4807	0.3624	0.4659	0.2754	0.6054	0.5186
	3D	0.4731	0.6916	0.6056	0.4087	0.3166	0.3103	0.1481	0.4977	0.4315
	5D	0.4595	0.6523	0.5701	0.3712	0.2935	0.2494	0.0887	0.4326	0.3896
	7D	0.4498	0.6217	0.5153	0.3324	0.2688	0.2017	0.0738	0.3889	0.3566
	Avg.	0.4683	0.6865	0.5947	0.3983	0.3103	0.3068	0.1465	0.4811	0.4241
AutoTimes	1D	0.4910	0.7669	0.6932	0.5324	0.3777	0.5069	0.3207	0.6462	0.5419
	3D	0.4693	0.6760	0.6316	0.4674	0.3322	0.3641	0.2025	0.5692	0.4640
	5D	0.4547	0.6498	0.5702	0.4229	0.3073	0.2926	0.1335	0.5071	0.4173
	7D	0.4421	0.6195	0.5349	0.3920	0.2933	0.2368	0.1115	0.4639	0.3868
	Avg.	0.4643	0.6781	0.6075	0.4537	0.3277	0.3501	0.1920	0.5466	0.4525

Table 35: Benchmark(SED11) on part 3(Fort Myers area) stations(MLP,CNN,Transformer)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
MLP	1D	0.4747	0.4387	0.5051	0.2137	0.1295	0.3272	0.0017	0.2836	0.2968	
	3D	0.4444	0.4520	0.2868	0.2392	0.1628	0.2860	0.0006	0.3115	0.2729	
	5D	0.4302	0.2950	0.1017	0.1766	0.1696	0.3612	0.0005	0.3690	0.2380	
	7D	0.4116	0.2920	0.2236	0.1575	0.1613	0.2435	0.0004	0.2357	0.2157	
	Avg.	0.4402	0.3694	0.2793	0.1968	0.1558	0.3045	0.0008	0.3000	0.2558	
MLP	TSMixer	1D	0.4450	0.5180	0.4305	0.2132	0.1929	0.2698	0.0000	0.3516	0.3026
	3D	0.3848	0.3874	0.2667	0.1432	0.1651	0.1587	0.0000	0.2378	0.2180	
	5D	0.3283	0.3044	0.2146	0.1247	0.1417	0.1347	0.0000	0.1959	0.1805	
	7D	0.2849	0.2479	0.1902	0.1069	0.1197	0.1177	0.0000	0.1482	0.1519	
	Avg.	0.3607	0.3644	0.2755	0.1470	0.1548	0.1702	0.0000	0.2334	0.2133	
NLinear	1D	0.4657	0.5401	0.4711	0.2360	0.2096	0.3693	0.0017	0.3968	0.3363	
	3D	0.4022	0.3957	0.2858	0.1465	0.1677	0.2077	0.0006	0.2691	0.2344	
	5D	0.3454	0.3299	0.2238	0.1311	0.1556	0.1897	0.0003	0.2083	0.1980	
	7D	0.3003	0.2835	0.1990	0.1112	0.1377	0.1638	0.0002	0.1777	0.1717	
	Avg.	0.3784	0.3873	0.2949	0.1562	0.1676	0.2326	0.0007	0.2630	0.2351	
TCN	1D	0.3817	0.3979	0.5681	0.1620	0.1078	0.2254	0.0001	0.2565	0.2624	
	3D	0.3269	0.2261	0.2620	0.1106	0.1482	0.1507	0.0000	0.1577	0.1728	
	5D	0.3281	0.3077	0.2576	0.1217	0.1144	0.1199	0.0000	0.1687	0.1773	
	7D	0.2895	0.2433	0.2785	0.0881	0.0582	0.1289	0.0000	0.1664	0.1566	
	Avg.	0.3315	0.2938	0.3415	0.1206	0.1072	0.1562	0.0000	0.1873	0.1923	
CNN	ModernTCN	1D	0.4700	0.3796	0.4606	0.1984	0.1834	0.2864	0.0003	0.3521	0.2913
	3D	0.4134	0.2972	0.3154	0.1244	0.1610	0.1089	0.0000	0.2540	0.2093	
	5D	0.3802	0.2393	0.2666	0.0989	0.1319	0.0585	0.0000	0.1811	0.1696	
	7D	0.3497	0.2258	0.2511	0.0857	0.1173	0.0431	0.0000	0.1418	0.1518	
	Avg.	0.4033	0.2855	0.3234	0.1269	0.1484	0.1242	0.0001	0.2322	0.2055	
TimesNet	1D	0.4591	0.3666	0.4525	0.1548	0.1686	0.1891	0.0000	0.2756	0.2583	
	3D	0.4283	0.2930	0.2820	0.1182	0.1470	0.0856	0.0000	0.1680	0.1902	
	5D	0.3777	0.2554	0.2337	0.1004	0.1092	0.0487	0.0000	0.1362	0.1577	
	7D	0.3632	0.2254	0.1776	0.0801	0.0995	0.0378	0.0000	0.0923	0.1345	
	Avg.	0.4071	0.2851	0.2865	0.1134	0.1311	0.0903	0.0000	0.1680	0.1852	
Transformer	iTransformer	1D	0.4584	0.4340	0.4950	0.2389	0.1775	0.3983	0.0005	0.3726	0.3219
	3D	0.4305	0.2699	0.2939	0.1542	0.1563	0.1485	0.0001	0.2423	0.2120	
	5D	0.3914	0.2541	0.2244	0.1336	0.1363	0.1145	0.0001	0.1856	0.1800	
	7D	0.3506	0.2224	0.1918	0.1115	0.1161	0.0841	0.0000	0.1842	0.1576	
	Avg.	0.4077	0.2951	0.3013	0.1595	0.1465	0.1863	0.0002	0.2462	0.2179	
PatchTST	1D	0.4734	0.4614	0.4548	0.2418	0.1918	0.3446	0.0003	0.3847	0.3191	
	3D	0.4278	0.3069	0.2981	0.1757	0.1645	0.1653	0.0001	0.2403	0.2223	
	5D	0.3990	0.2060	0.2292	0.1506	0.1415	0.0919	0.0000	0.1697	0.1735	
	7D	0.3603	0.1898	0.1968	0.1058	0.1200	0.0701	0.0000	0.1344	0.1472	
	Avg.	0.4151	0.2910	0.2947	0.1685	0.1544	0.1680	0.0001	0.2323	0.2155	

Table 36: Benchmark(SED11) on part 3(Fort Myers area) stations(RNN,GNN,LLM)

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
LSTM	1D	0.2500	0.3899	0.0246	0.1762	0.1810	0.1240	0.0000	0.2035	0.1686	
	3D	0.2375	0.0599	0.0035	0.1100	0.1246	0.0055	0.0000	0.1008	0.0802	
	5D	0.0000	0.0617	0.0000	0.0920	0.0927	0.0081	0.0000	0.0967	0.0439	
	7D	0.2466	0.0486	0.0230	0.0938	0.0791	0.0256	0.0000	0.0555	0.0715	
	Avg.	0.1835	0.1400	0.0128	0.1180	0.1194	0.0408	0.0000	0.1141	0.0911	
RNN	DeepAR	1D	0.1603	0.0000	0.0998	0.1648	0.1890	0.1125	0.0000	0.1389	0.1082
		3D	0.2332	0.1744	0.0000	0.0798	0.0822	0.0015	0.0000	0.0887	0.0825
		5D	0.2488	0.0519	0.0000	0.1098	0.0928	0.0190	0.0000	0.0000	0.0653
		7D	0.0000	0.0091	0.0056	0.0657	0.0243	0.0088	0.0000	0.0961	0.0262
		Avg.	0.1606	0.0589	0.0263	0.1050	0.0971	0.0355	0.0000	0.0809	0.0705
DilatedRNN	1D	0.1049	0.3690	0.3560	0.3224	0.0969	0.3043	0.0003	0.2794	0.2291	
	3D	0.2490	0.2811	0.0161	0.1159	0.1277	0.0744	0.0001	0.1527	0.1271	
	5D	0.0949	0.2827	0.2367	0.1348	0.1246	0.0415	0.0000	0.2265	0.1427	
	7D	0.2413	0.1618	0.2236	0.1230	0.1192	0.0497	0.0000	0.0805	0.1249	
	Avg.	0.1725	0.2736	0.2081	0.1740	0.1171	0.1175	0.0001	0.1848	0.1560	
GCN	1D	0.3817	0.3979	0.5681	0.1620	0.1078	0.2254	0.0001	0.2565	0.2624	
	3D	0.3269	0.2261	0.2620	0.1106	0.1482	0.1507	0.0000	0.1577	0.1728	
	5D	0.3281	0.3077	0.2576	0.1217	0.1144	0.1199	0.0000	0.1687	0.1773	
	7D	0.2895	0.2433	0.2785	0.0881	0.0582	0.1289	0.0000	0.1664	0.1566	
	Avg.	0.3315	0.2938	0.3415	0.1206	0.1072	0.1562	0.0000	0.1873	0.1923	
GNN	FourierGNN	1D	0.4363	0.3351	0.3967	0.2117	0.1928	0.2551	0.0018	0.2885	0.2647
		3D	0.4293	0.2745	0.2772	0.1733	0.2145	0.1698	0.0000	0.2810	0.2274
		5D	0.4124	0.3231	0.2661	0.1731	0.1441	0.2090	0.0000	0.2253	0.2191
		7D	0.3564	0.2544	0.2314	0.1561	0.2023	0.1808	0.0000	0.1790	0.1951
		Avg.	0.4086	0.2968	0.2928	0.1785	0.1884	0.2037	0.0005	0.2434	0.2266
StemGNN	1D	0.1616	0.4190	0.1862	0.1781	0.1094	0.1578	0.0002	0.2462	0.1823	
	3D	0.2372	0.2514	0.1446	0.1443	0.0443	0.1068	0.0000	0.1309	0.1324	
	5D	0.0956	0.1130	0.0178	0.1026	0.0790	0.0844	0.0000	0.1162	0.0761	
	7D	0.0360	0.0911	0.0384	0.1245	0.0247	0.1175	0.0000	0.0726	0.0631	
	Avg.	0.1326	0.2186	0.0968	0.1374	0.0644	0.1167	0.0001	0.1415	0.1135	
LLM	GPT4TS	1D	0.4586	0.4502	0.4421	0.1424	0.1876	0.2122	0.0000	0.2792	0.2715
		3D	0.4088	0.3268	0.2713	0.1127	0.1530	0.0738	0.0000	0.1810	0.1909
		5D	0.3916	0.2650	0.2210	0.1011	0.1201	0.0412	0.0000	0.1263	0.1583
		7D	0.3606	0.2190	0.1939	0.0847	0.1091	0.0466	0.0000	0.1000	0.1392
		Avg.	0.4049	0.3153	0.2821	0.1102	0.1424	0.0935	0.0000	0.1716	0.1900
	AutoTimes	1D	0.4710	0.3713	0.4458	0.2398	0.1955	0.3082	0.0000	0.3205	0.2940
		3D	0.4297	0.2009	0.2890	0.1551	0.1653	0.1697	0.0000	0.2165	0.2033
		5D	0.3880	0.1895	0.2212	0.1363	0.1375	0.1271	0.0000	0.1561	0.1695
		7D	0.3494	0.1505	0.1984	0.1012	0.1175	0.0965	0.0000	0.1250	0.1423
		Avg.	0.4095	0.2280	0.2886	0.1581	0.1539	0.1753	0.0000	0.2045	0.2023

Table 37: Benchmark(MAE) on all stations

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.1324	0.1424	0.1441	0.1043	0.1098	0.1192	0.1423	0.1310	0.1282
	3D	0.2158	0.2352	0.2471	0.1838	0.1875	0.1903	0.2516	0.2082	0.2150
	5D	0.2810	0.2902	0.3178	0.2277	0.2437	0.2354	0.3228	0.2565	0.2719
	7D	0.3329	0.3374	0.3666	0.2661	0.2837	0.2719	0.3793	0.2915	0.3162
	Avg.	0.2405	0.2513	0.2689	0.1955	0.2062	0.2042	0.2740	0.2218	0.2328
TCN	1D	0.3079	0.2613	0.2634	0.2414	0.2079	0.1885	0.2957	0.2740	0.2550
	3D	0.4332	0.3215	0.3490	0.3061	0.2900	0.2554	0.3944	0.3465	0.3370
	5D	0.4324	0.3788	0.4040	0.3563	0.3457	0.3070	0.4547	0.3992	0.3848
	7D	0.4842	0.3951	0.4429	0.3915	0.3803	0.3397	0.4979	0.4242	0.4195
	Avg.	0.4144	0.3392	0.3648	0.3238	0.3060	0.2727	0.4107	0.3610	0.3491
LSTM	1D	0.1867	0.2613	0.2736	0.1595	0.2496	0.1697	0.3087	0.1836	0.2241
	3D	0.2862	0.3516	0.3751	0.2391	0.3291	0.2446	0.4267	0.2704	0.3153
	5D	0.3475	0.4012	0.4475	0.2892	0.3753	0.2944	0.4867	0.3172	0.3699
	7D	0.4174	0.4312	0.5125	0.3261	0.4193	0.3261	0.5173	0.3429	0.4116
	Avg.	0.3094	0.3613	0.4022	<u>0.2535</u>	0.3433	0.2587	0.4348	0.2785	0.3302
GCN	1D	0.1850	0.2317	0.2365	0.2061	0.1953	0.1681	0.1944	0.2152	0.2040
	3D	0.2668	0.3177	0.3426	0.2735	0.2782	0.2385	0.3119	0.2914	0.2901
	5D	0.3329	0.3673	0.4137	0.3184	0.3256	0.2796	0.3977	0.3336	0.3461
	7D	0.3863	0.3860	0.4624	0.3484	0.3646	0.3102	0.4265	0.3620	0.3808
	Avg.	<u>0.2928</u>	<u>0.3257</u>	<u>0.3638</u>	0.2866	<u>0.2909</u>	<u>0.2491</u>	<u>0.3326</u>	<u>0.3005</u>	<u>0.3053</u>

Table 38: Benchmark(MSE) on all stations

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.1144	0.1360	0.1284	0.0684	0.1124	0.0747	0.1674	0.1256	0.1159
	3D	0.2226	0.3080	0.2875	0.1478	0.3256	0.1424	0.5486	0.2405	0.2779
	5D	0.3117	0.4687	0.4318	0.2081	0.6883	0.1928	1.2099	0.3317	0.4804
	7D	0.3808	0.6967	0.6094	0.2648	1.0917	0.2369	1.8233	0.3899	0.6867
	Avg.	0.2574	0.4024	0.3643	0.1723	0.5545	0.1617	0.9373	0.2719	0.3902
TCN	1D	0.3964	0.5800	0.4714	0.2575	0.3783	0.2180	1.3093	0.5552	0.5208
	3D	0.6306	0.6849	0.6180	0.3337	0.5102	0.2900	1.5333	0.6517	0.6566
	5D	0.5709	0.7997	0.7216	0.3930	0.6323	0.3534	1.6885	0.7298	0.7362
	7D	0.6880	0.8404	0.7966	0.4639	0.7027	0.3941	1.8442	0.7434	0.8092
	Avg.	0.5715	0.7263	<u>0.6519</u>	0.3620	<u>0.5559</u>	0.3139	1.5938	0.6700	<u>0.6807</u>
LSTM	1D	0.2550	1.4732	1.0367	0.1620	5.2575	0.1497	6.8509	0.4084	1.9492
	3D	0.3871	1.3765	1.0735	0.2642	5.7927	0.2476	8.5420	0.5052	2.2736
	5D	0.4911	1.5590	1.4324	0.3373	5.8335	0.3032	8.3886	0.5652	2.3638
	7D	0.6198	1.6812	1.8663	0.3982	6.0814	0.3494	8.5648	0.6176	2.5224
	Avg.	0.4383	1.5225	1.3522	<u>0.2904</u>	5.7413	0.2625	8.0866	<u>0.5241</u>	2.2772
GCN	1D	0.1994	0.2886	0.7728	0.3590	0.2534	0.1165	0.2171	6.2444	1.0564
	3D	0.3102	0.4645	1.0667	0.4061	0.5848	0.1883	0.9433	3.7576	0.9652
	5D	0.4273	0.5908	1.3546	0.4490	0.8592	0.2408	2.4139	2.7351	1.1338
	7D	0.5166	0.5968	1.5264	0.4728	1.3609	0.2752	1.8411	2.3212	1.1139
	Avg.	<u>0.3634</u>	<u>0.4852</u>	1.1801	0.4217	0.7646	<u>0.2052</u>	<u>1.3539</u>	3.7646	1.0673

Table 39: Benchmark(SED10) on all stations

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.7865	0.8159	0.8152	0.7883	0.7313	0.7192	0.7116	0.7581	0.7658
	3D	0.7108	0.7527	0.7714	0.7639	0.6709	0.5924	0.6336	0.6414	0.6921
	5D	0.6823	0.6972	0.7482	0.7092	0.6185	0.5809	0.5830	0.6343	0.6567
	7D	0.6455	0.6726	0.7134	0.6753	0.6142	0.5620	0.5496	0.5815	0.6268
	Avg.	0.7063	0.7346	0.7621	0.7342	0.6587	0.6136	0.6195	0.6538	0.6853
TCN	1D	0.6211	0.7314	0.7264	0.6545	0.5807	0.6232	0.5594	0.4628	0.6199
	3D	0.4990	0.6284	0.6505	0.5697	0.4678	0.5020	0.4852	0.3738	0.5221
	5D	0.4997	0.6025	0.6164	0.5319	0.4059	0.4549	0.4289	0.3187	0.4824
	7D	0.4366	0.5680	0.5829	0.4747	0.3637	0.4181	0.3906	0.3059	0.4426
	Avg.	0.5141	0.6326	0.6440	0.5577	0.4545	0.4996	0.4660	0.3653	0.5167
LSTM	1D	0.7478	0.7266	0.7410	0.7615	0.6781	0.6422	0.6255	0.6716	0.6993
	3D	0.6290	0.6424	0.6606	0.6671	0.5615	0.5070	0.5091	0.5369	0.5892
	5D	0.5632	0.5683	0.6139	0.6053	0.5035	0.4434	0.4307	0.4694	0.5247
	7D	0.5249	0.5386	0.5845	0.5758	0.4605	0.3974	0.3886	0.4256	0.4870
	Avg.	0.6162	0.6190	0.6500	0.6524	0.5509	0.4975	0.4885	0.5259	0.5751
GCN	1D	0.7461	0.7112	0.7543	0.7031	0.6442	0.6466	0.6565	0.6686	0.6913
	3D	0.6725	0.6669	0.6912	0.6308	0.5846	0.5725	0.5800	0.6044	0.6254
	5D	0.6331	0.6218	0.6668	0.5908	0.5515	0.5247	0.5314	0.5519	0.5840
	7D	0.5989	0.5811	0.6531	0.5765	0.5183	0.4903	0.5025	0.5109	0.5539
	Avg.	<u>0.6626</u>	<u>0.6452</u>	<u>0.6913</u>	<u>0.6253</u>	<u>0.5746</u>	<u>0.5585</u>	<u>0.5676</u>	<u>0.5840</u>	<u>0.6137</u>

Table 40: Benchmark(SED15) on all stations

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.7136	0.7660	0.7338	0.7105	0.6322	0.6125	0.6060	0.6216	0.6745
	3D	0.6163	0.7050	0.6930	0.6926	0.5791	0.4788	0.5176	0.5137	0.5995
	5D	0.5987	0.6437	0.6679	0.6416	0.5329	0.4810	0.4710	0.5052	0.5678
	7D	0.5702	0.6306	0.6431	0.6119	0.5267	0.4753	0.4457	0.4662	0.5462
	Avg.	0.6247	0.6863	0.6845	0.6642	0.5678	0.5119	0.5101	0.5267	0.5970
TCN	1D	0.4196	0.6603	0.6184	0.5291	0.4095	0.4960	0.4138	0.3313	0.4848
	3D	0.3450	0.5523	0.5307	0.4549	0.2944	0.3870	0.3365	0.2404	0.3926
	5D	0.3236	0.5302	0.5064	0.4307	0.2343	0.3585	0.2869	0.2074	0.3597
	7D	0.2981	0.4912	0.4486	0.3805	0.1940	0.3273	0.2666	0.1886	0.3243
	Avg.	0.3466	0.5585	0.5260	0.4488	0.2831	0.3922	0.3259	0.2419	0.3904
LSTM	1D	0.6546	0.6720	0.6402	0.6737	0.5568	0.5009	0.4883	0.5126	0.5874
	3D	0.5112	0.5662	0.5342	0.5557	0.4398	0.3529	0.3555	0.3751	0.4613
	5D	0.4395	0.4974	0.4877	0.4993	0.3820	0.3008	0.2920	0.2929	0.3990
	7D	0.3879	0.4701	0.4458	0.4642	0.3109	0.2694	0.2556	0.2664	0.3588
	Avg.	0.4983	0.5514	0.5270	0.5482	0.4224	0.3560	0.3479	0.3618	0.4516
GCN	1D	0.6660	0.6406	0.6734	0.6192	0.5498	0.5445	0.5430	0.5278	0.5955
	3D	0.5885	0.6013	0.6062	0.5427	0.4961	0.4673	0.4629	0.4690	0.5292
	5D	0.5593	0.5528	0.5793	0.5059	0.4684	0.4301	0.4192	0.4291	0.4930
	7D	0.5471	0.5177	0.5610	0.4934	0.4320	0.4081	0.3855	0.3945	0.4674
	Avg.	<u>0.5902</u>	<u>0.5781</u>	<u>0.6050</u>	<u>0.5403</u>	<u>0.4866</u>	<u>0.4625</u>	<u>0.4526</u>	<u>0.4551</u>	<u>0.5213</u>

Table 41: Benchmark(SEDI1) on all stations

Method	T	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
MLP	1D	0.5452	0.6313	0.5586	0.5222	0.3956	0.3222	0.3557	0.3511	0.4602
	3D	0.4265	0.5790	0.5332	0.5326	0.3359	0.2318	0.2916	0.2890	0.4025
	5D	0.4535	0.5114	0.5228	0.4968	0.3349	0.2440	0.2813	0.3036	0.3935
	7D	0.4738	0.5377	0.5065	0.4550	0.3195	0.2545	0.2705	0.2740	0.3864
	Avg.	0.4748	0.5649	0.5303	0.5017	0.3465	0.2631	0.2998	0.3044	0.4107
TCN	1D	0.1333	0.4546	0.4028	0.3001	0.0736	0.2119	0.1582	0.1412	0.2345
	3D	0.1118	0.3684	0.3182	0.2284	0.0440	0.1646	0.1282	0.0833	0.1809
	5D	0.1033	0.3531	0.3050	0.2328	0.0305	0.1499	0.1147	0.0757	0.1706
	7D	0.0802	0.3176	0.2611	0.2063	0.0249	0.1433	0.0978	0.0746	0.1507
	Avg.	0.1071	0.3734	0.3218	0.2419	0.0432	0.1674	0.1247	0.0937	0.1842
LSTM	1D	0.4140	0.4712	0.4004	0.4026	0.2813	0.1812	0.2226	0.2254	0.3249
	3D	0.2247	0.2719	0.2113	0.2167	0.1419	0.0936	0.1123	0.1012	0.1717
	5D	0.1839	0.1920	0.1746	0.1489	0.1184	0.0703	0.0868	0.0744	0.1312
	7D	0.1173	0.1752	0.1429	0.1642	0.0394	0.0573	0.0715	0.0608	0.1036
	Avg.	0.2350	0.2776	0.2323	0.2331	0.1453	0.1006	0.1233	0.1155	0.1828
GCN	1D	0.4770	0.4553	0.4873	0.4167	0.3108	0.2708	0.3098	0.2742	0.3752
	3D	0.4046	0.4419	0.3886	0.3418	0.2585	0.2237	0.2572	0.2331	0.3187
	5D	0.3840	0.3823	0.3712	0.3168	0.2323	0.1991	0.2144	0.2147	0.2894
	7D	0.3641	0.3583	0.3773	0.2930	0.2013	0.1843	0.1942	0.1889	0.2702
	Avg.	0.4074	0.4095	0.4061	0.3421	0.2507	0.2195	0.2439	0.2277	0.3134

Table 42: MAE results of input factors ablation study on S6. All, G, R, and C represent all factors: groundwater, rainfall, and human control(pump and gate).

Method	T	w/ All	w/o G	w/o R	w/o C	w/o GR	w/o RC	w/o GC	w/o GRC
iTransformer	1D	0.0738	0.0735	0.0729	0.0733	0.0739	0.0737	0.0739	0.0742
	3D	0.1275	0.1279	0.1278	0.1275	0.1290	0.1280	0.1265	0.1276
	5D	0.1660	0.1662	0.1664	0.1662	0.1660	0.1660	0.1658	0.1671
	7D	0.1953	0.1953	0.1953	0.1948	0.1954	0.1961	0.1947	0.1955
	Avg.	0.1406	0.1407	0.1406	0.1405	0.1411	0.1410	0.1402	0.1411
PatchTST	1D	0.0726	0.0731	0.0713	0.0719	0.0732	0.0722	0.0727	0.0772
	3D	0.1252	0.1262	0.1252	0.1255	0.1273	0.1273	0.1254	0.1295
	5D	0.1615	0.1639	0.1624	0.1618	0.1649	0.1649	0.1634	0.1667
	7D	0.1910	0.1933	0.1923	0.1916	0.1938	0.1941	0.1927	0.1951
	Avg.	0.1376	0.1391	0.1378	0.1377	0.1398	0.1396	0.1385	0.1421
TSMixer	1D	0.1004	0.0781	0.1599	0.1443	0.0778	0.1789	0.0782	0.0782
	3D	0.1471	0.1296	0.2611	0.2216	0.1297	0.2506	0.1299	0.1299
	5D	0.1802	0.1658	0.3104	0.2350	0.1657	0.3018	0.1664	0.1663
	7D	0.2107	0.1941	0.2776	0.2435	0.1939	0.3914	0.1947	0.1944
	Avg.	0.1596	0.1419	0.2523	0.2111	0.1418	0.2807	0.1423	0.1422
NLinear	1D	0.0937	0.0977	0.0854	0.0933	0.0850	0.0804	0.0983	0.0780
	3D	0.1429	0.1461	0.1363	0.1422	0.1366	0.1328	0.1462	0.1314
	5D	0.1769	0.1796	0.1713	0.1764	0.1717	0.1689	0.1797	0.1678
	7D	0.2050	0.2074	0.2003	0.2043	0.2008	0.1978	0.2072	0.1970
	Avg.	0.1546	0.1577	0.1483	0.1540	0.1485	0.1450	0.1579	0.1435
TimesNet	1D	0.0986	0.0936	0.0986	0.0986	0.0928	0.0970	0.0924	0.0930
	3D	0.1513	0.1446	0.1511	0.1525	0.1465	0.1529	0.1448	0.1468
	5D	0.1883	0.1838	0.1889	0.1890	0.1843	0.1908	0.1830	0.1814
	7D	0.2187	0.2177	0.2262	0.2200	0.2131	0.2219	0.2111	0.2109
	Avg.	0.1642	0.1599	0.1662	0.1650	0.1592	0.1656	0.1578	0.1580

Table 43: MSE results of input factors ablation study on S6. All, G, R, and C represent all factors: groundwater, rainfall, and human control(pump and gate).

Method	T	w/ All	w/o G	w/o R	w/o C	w/o GR	w/o RC	w/o GC	w/o GRC
iTransformer	1D	0.0400	0.0407	0.0398	0.0402	0.0405	0.0400	0.0405	0.0409
	3D	0.0817	0.0828	0.0821	0.0818	0.0827	0.0825	0.0818	0.0832
	5D	0.1175	0.1172	0.1170	0.1167	0.1166	0.1169	0.1164	0.1184
	7D	0.1419	0.1430	0.1407	0.1430	0.1429	0.1431	0.1407	0.1424
	Avg.	0.0953	0.0960	0.0949	0.0954	0.0957	0.0956	0.0949	0.0962
PatchTST	1D	0.0395	0.0393	0.0392	0.0393	0.0405	0.0399	0.0389	0.0432
	3D	0.0798	0.0807	0.0806	0.0797	0.0831	0.0819	0.0797	0.0850
	5D	0.1107	0.1128	0.1122	0.1105	0.1154	0.1140	0.1123	0.1173
	7D	0.1368	0.1400	0.1388	0.1367	0.1412	0.1393	0.1385	0.1428
	Avg.	0.0917	0.0932	0.0927	0.0916	0.0950	0.0938	0.0923	0.0971
TSMixer	1D	0.0562	0.0415	0.1891	0.2590	0.0414	0.4396	0.0419	0.0416
	3D	0.0957	0.0827	0.4979	0.2580	0.0830	0.4833	0.0825	0.0826
	5D	0.1243	0.1145	0.5722	0.3232	0.1145	0.4855	0.1139	0.1141
	7D	0.1557	0.1396	0.3653	0.2389	0.1394	1.2701	0.1389	0.1382
	Avg.	0.1080	0.0946	0.4061	0.2698	0.0946	0.6697	0.0943	0.0941
NLinear	1D	0.0477	0.0497	0.0450	0.0471	0.0459	0.0434	0.0494	0.0426
	3D	0.0878	0.0898	0.0848	0.0871	0.0856	0.0837	0.0895	0.0829
	5D	0.1178	0.1196	0.1154	0.1171	0.1160	0.1145	0.1191	0.1139
	7D	0.1434	0.1451	0.1411	0.1425	0.1417	0.1400	0.1444	0.1395
	Avg.	0.0992	0.1011	0.0966	0.0984	0.0973	0.0954	0.1006	0.0947
TimesNet	1D	0.0564	0.0518	0.0580	0.0573	0.0521	0.0550	0.0515	0.0517
	3D	0.1049	0.0970	0.1043	0.1049	0.0985	0.1086	0.0966	0.1015
	5D	0.1409	0.1357	0.1419	0.1416	0.1397	0.1485	0.1338	0.1338
	7D	0.1729	0.1770	0.1863	0.1770	0.1677	0.1826	0.1624	0.1624
	Avg.	0.1188	0.1154	0.1226	0.1202	0.1145	0.1237	0.1111	0.1123

Table 44: SEDI(10%) results of input factors ablation study on S6. All, G, R, and C represent all factors: groundwater, rainfall, and human control(pump and gate).

Method	T	w/ All	w/o G	w/o R	w/o C	w/o GR	w/o RC	w/o GC	w/o GRC
iTransformer	1D	0.6508	0.6479	0.6510	0.6551	0.6525	0.6548	0.6512	0.6520
	3D	0.5526	0.5429	0.5485	0.5510	0.5393	0.5485	0.5513	0.5533
	5D	0.4856	0.4793	0.4831	0.4877	0.4786	0.4864	0.4824	0.4779
	7D	0.4384	0.4323	0.4408	0.4399	0.4301	0.4380	0.4320	0.4326
	Avg.	0.5318	0.5256	0.5308	0.5334	0.5251	0.5319	0.5292	0.5289
PatchTST	1D	0.6632	0.6629	0.6665	0.6644	0.6686	0.6618	0.6621	0.6561
	3D	0.5696	0.5614	0.5704	0.5717	0.5616	0.5586	0.5658	0.5455
	5D	0.5028	0.5013	0.4997	0.5043	0.4926	0.4945	0.4993	0.4857
	7D	0.4562	0.4518	0.4498	0.4544	0.4442	0.4465	0.4476	0.4364
	Avg.	0.5480	0.5443	0.5466	0.5487	0.5418	0.5404	0.5437	0.5309
TSMixer	1D	0.6052	0.6538	0.5202	0.5722	0.6475	0.5493	0.6499	0.6525
	3D	0.5267	0.5571	0.4251	0.4290	0.5580	0.4402	0.5581	0.5596
	5D	0.4732	0.4959	0.3523	0.4067	0.4963	0.3709	0.4932	0.4973
	7D	0.4245	0.4487	0.3516	0.3703	0.4497	0.3206	0.4454	0.4474
	Avg.	0.5074	0.5389	0.4123	0.4446	0.5379	0.4202	0.5367	0.5392
NLinear	1D	0.6477	0.6485	0.6485	0.6481	0.6506	0.6529	0.6602	0.6561
	3D	0.5476	0.5556	0.5521	0.5472	0.5521	0.5566	0.5517	0.5582
	5D	0.4868	0.4928	0.4905	0.4865	0.4904	0.4936	0.4960	0.4950
	7D	0.4356	0.4355	0.4383	0.4345	0.4381	0.4440	0.4345	0.4450
	Avg.	0.5294	0.5331	0.5324	0.5291	0.5328	0.5368	0.5356	0.5386
TimesNet	1D	0.5936	0.6093	0.5855	0.5884	0.6067	0.5939	0.6086	0.6089
	3D	0.4772	0.5059	0.4986	0.4925	0.5117	0.4854	0.5093	0.5046
	5D	0.4304	0.4489	0.4264	0.4305	0.4460	0.4202	0.4456	0.4396
	7D	0.3878	0.3959	0.3696	0.3837	0.4095	0.3839	0.4091	0.4082
	Avg.	0.4723	0.4900	0.4700	0.4738	0.4935	0.4709	0.4931	0.4903

Table 45: MAE results of temporal information ablation study on S6.

Method	T	6H	12H	1D	2D	3D	4D	5D	6D
iTransformer	1D	0.1136	0.1123	0.1097	0.1123	0.1133	0.1198	0.1214	0.1209
	3D	0.1999	0.1978	0.1980	0.2025	0.2071	0.2121	0.2168	0.2137
	5D	0.2571	0.2537	0.2547	0.2586	0.2725	0.2776	0.2815	0.2779
	7D	0.3036	0.3007	0.3015	0.3059	0.3179	0.3242	0.3221	0.3261
	Avg.	0.2185	0.2161	0.2160	0.2198	0.2277	0.2334	0.2354	0.2346
PatchTST	1D	0.1266	0.1164	0.1135	0.1127	0.1094	0.1099	0.1106	0.1159
	3D	0.2123	0.2067	0.1988	0.1979	0.1963	0.1940	0.1979	0.2011
	5D	0.2693	0.2650	0.2566	0.2530	0.2554	0.2518	0.2522	0.2559
	7D	0.3143	0.3116	0.3039	0.2988	0.2984	0.2983	0.3011	0.3069
	Avg.	0.2306	0.2249	0.2182	0.2156	0.2149	0.2135	0.2155	0.2199
TSMixer	1D	0.1197	0.1216	0.1237	0.1269	0.1258	0.1269	0.1289	0.1266
	3D	0.2021	0.2033	0.2058	0.2087	0.2051	0.2073	0.2074	0.2052
	5D	0.2586	0.2592	0.2618	0.2639	0.2576	0.2604	0.2585	0.2582
	7D	0.3046	0.3049	0.3070	0.3092	0.3007	0.3029	0.2997	0.3009
	Avg.	0.2213	0.2222	0.2246	0.2272	0.2223	0.2244	0.2236	0.2227
NLinear	1D	0.1159	0.1180	0.1211	0.1462	0.1505	0.1508	0.1587	0.1586
	3D	0.1998	0.2011	0.2047	0.2220	0.2289	0.2305	0.2388	0.2382
	5D	0.2568	0.2577	0.2605	0.2743	0.2796	0.2870	0.2871	0.2925
	7D	0.3031	0.3036	0.3064	0.3186	0.3234	0.3293	0.3323	0.3355
	Avg.	0.2189	0.2201	0.2232	0.2403	0.2456	0.2494	0.2542	0.2562
TimesNet	1D	0.1175	0.1234	0.1357	0.1565	0.1750	0.1903	0.2066	0.2184
	3D	0.2017	0.2084	0.2235	0.2423	0.2618	0.2775	0.2880	0.3012
	5D	0.2602	0.2635	0.2757	0.2887	0.2989	0.3235	0.3386	0.3657
	7D	0.3086	0.3097	0.3254	0.3272	0.3570	0.3636	0.3804	0.4016
	Avg.	0.2220	0.2262	0.2401	0.2537	0.2732	0.2887	0.3034	0.3217

Table 46: MSE results of temporal information ablation study on S6.

Method	T	6H	12H	1D	2D	3D	4D	5D	6D
iTransformer	1D	0.0829	0.0815	0.0793	0.0825	0.0844	0.0878	0.0899	0.0899
	3D	0.2093	0.2068	0.2082	0.2152	0.2218	0.2281	0.2379	0.2306
	5D	0.3186	0.3162	0.3174	0.3219	0.3472	0.3551	0.3556	0.3594
	7D	0.4170	0.4149	0.4139	0.4220	0.4499	0.4614	0.4614	0.4757
	Avg.	0.2569	0.2549	0.2547	0.2604	0.2758	0.2831	0.2862	0.2889
PatchTST	1D	0.0888	0.0860	0.0851	0.0827	0.0795	0.0787	0.0788	0.0835
	3D	0.2151	0.2104	0.2107	0.2078	0.2064	0.2050	0.2072	0.2089
	5D	0.3243	0.3214	0.3210	0.3156	0.3130	0.3256	0.3408	0.3187
	7D	0.4236	0.4187	0.4270	0.4193	0.4275	0.4548	0.4255	0.4348
	Avg.	0.2630	0.2592	0.2609	0.2563	0.2566	0.2660	0.2631	0.2615
TSMixer	1D	0.0899	0.0899	0.0888	0.0887	0.0860	0.0853	0.0855	0.0837
	3D	0.2152	0.2145	0.2141	0.2121	0.2047	0.2042	0.2034	0.1992
	5D	0.3241	0.3228	0.3226	0.3170	0.3060	0.3050	0.3035	0.3024
	7D	0.4235	0.4222	0.4214	0.4142	0.3999	0.3982	0.3968	0.3985
	Avg.	0.2632	0.2623	0.2617	0.2580	0.2491	0.2482	0.2473	0.2460
NLinear	1D	0.0852	0.0858	0.0873	0.1111	0.1135	0.1122	0.1196	0.1176
	3D	0.2111	0.2115	0.2148	0.2370	0.2444	0.2436	0.2535	0.2509
	5D	0.3201	0.3206	0.3236	0.3439	0.3490	0.3587	0.3563	0.3629
	7D	0.4199	0.4200	0.4235	0.4427	0.4480	0.4555	0.4586	0.4623
	Avg.	0.2591	0.2595	0.2623	0.2836	0.2887	0.2925	0.2970	0.2984
TimesNet	1D	0.0866	0.0937	0.1104	0.1357	0.1603	0.1750	0.2039	0.2297
	3D	0.2177	0.2333	0.2594	0.2953	0.3156	0.3607	0.3811	0.4114
	5D	0.3316	0.3372	0.3781	0.3997	0.4169	0.4597	0.5154	0.5691
	7D	0.4363	0.4407	0.4918	0.4891	0.5910	0.5837	0.5953	0.6566
	Avg.	0.2681	0.2762	0.3100	0.3299	0.3710	0.3948	0.4239	0.4667

Table 47: SEDI(10%) results of temporal information ablation study on S6.

Method	T	6H	12H	1D	2D	3D	4D	5D	6D
iTransformer	1D	0.7825	0.7945	0.7941	0.7898	0.7703	0.7857	0.7728	0.7632
	3D	0.6792	0.6945	0.6987	0.6957	0.6847	0.6801	0.6691	0.6627
	5D	0.6220	0.6392	0.6420	0.6318	0.6201	0.6189	0.6178	0.6022
	7D	0.5797	0.5984	0.6006	0.5992	0.5867	0.5841	0.5738	0.5665
	Avg.	0.6659	0.6816	0.6838	0.6791	0.6654	0.6672	0.6584	0.6487
PatchTST	1D	0.7522	0.7946	0.8004	0.8033	0.8100	0.8118	0.8058	0.8099
	3D	0.6324	0.6425	0.6977	0.7010	0.7046	0.7030	0.7033	0.7032
	5D	0.5820	0.5796	0.6389	0.6427	0.6404	0.6467	0.6362	0.6326
	7D	0.5415	0.5314	0.5959	0.6024	0.5986	0.5957	0.5929	0.5982
	Avg.	0.6270	0.6370	0.6832	0.6874	0.6884	0.6893	0.6846	0.6860
TSMixer	1D	0.7812	0.7886	0.7954	0.7912	0.7931	0.7965	0.7905	0.7952
	3D	0.6861	0.6906	0.6971	0.7005	0.7016	0.7067	0.6946	0.6983
	5D	0.6307	0.6351	0.6397	0.6463	0.6445	0.6503	0.6400	0.6385
	7D	0.5917	0.5942	0.5984	0.6054	0.6003	0.6038	0.5959	0.5900
	Avg.	0.6724	0.6771	0.6827	0.6859	0.6849	0.6893	0.6803	0.6805
NLinear	1D	0.7915	0.8002	0.8029	0.7843	0.7891	0.7933	0.7929	0.7904
	3D	0.6956	0.6989	0.7019	0.6857	0.6958	0.6974	0.6925	0.6940
	5D	0.6359	0.6383	0.6447	0.6320	0.6329	0.6373	0.6422	0.6373
	7D	0.5944	0.5995	0.6016	0.5875	0.5889	0.5951	0.5973	0.5994
	Avg.	0.6794	0.6842	0.6878	0.6724	0.6767	0.6808	0.6812	0.6803
TimesNet	1D	0.7590	0.7610	0.7482	0.7130	0.6908	0.6854	0.6548	0.6392
	3D	0.6582	0.6606	0.6499	0.6148	0.5993	0.5889	0.5790	0.5731
	5D	0.5976	0.6096	0.5988	0.5779	0.5631	0.5461	0.5458	0.5246
	7D	0.5647	0.5692	0.5451	0.5447	0.5269	0.5263	0.5062	0.5060
	Avg.	0.6449	0.6501	0.6355	0.6126	0.5950	0.5867	0.5714	0.5607

Table 48: MAE results of spatial information ablation study on S6.

Method	T	MAE					MSE					SEDI(10%)				
		1.0R	1.2R	1.4R	1.6R	1.8R	1.0R	1.2R	1.4R	1.6R	1.8R	1.0R	1.2R	1.4R	1.6R	1.8R
iTransformer	1D	0.1123	0.1135	0.1125	0.1099	0.1119	0.0825	0.0825	0.0818	0.0822	0.0807	0.7898	0.7810	0.7894	0.7863	0.7939
	3D	0.2025	0.2038	0.2020	0.2024	0.2021	0.2152	0.2144	0.2117	0.2136	0.2119	0.6957	0.7011	0.6952	0.7022	0.7036
	5D	0.2586	0.2650	0.2584	0.2538	0.2513	0.3219	0.3314	0.3241	0.3257	0.3186	0.6318	0.6294	0.6329	0.6353	0.6361
	7D	0.3059	0.3040	0.3017	0.3006	0.2995	0.4220	0.4239	0.4181	0.4176	0.4197	0.5992	0.5973	0.5989	0.5972	0.5965
	Avg.	0.2198	0.2216	0.2187	0.2167	0.2162	0.2604	0.2631	0.2589	0.2598	0.2577	0.6791	0.6772	0.6791	0.6803	0.6825
PatchTST	1D	0.1127	0.1131	0.1139	0.1113	0.1118	0.0827	0.0833	0.0829	0.0825	0.0821	0.8033	0.8030	0.8066	0.8068	0.8095
	3D	0.1979	0.1967	0.1990	0.1976	0.1976	0.2078	0.2060	0.2087	0.2082	0.2083	0.7010	0.7036	0.7048	0.7040	0.7073
	5D	0.2530	0.2531	0.2548	0.2556	0.2550	0.3156	0.3133	0.3151	0.3182	0.3160	0.6427	0.6440	0.6433	0.6437	0.6483
	7D	0.2988	0.2971	0.2994	0.3000	0.2991	0.4193	0.4102	0.4102	0.4115	0.4097	0.6024	0.6035	0.6049	0.6056	0.6065
	Avg.	0.2156	0.2150	0.2168	0.2161	0.2159	0.2563	0.2532	0.2542	0.2551	0.2540	0.6874	0.6885	0.6899	0.6900	0.6929
TSMixer	1D	0.1269	0.1259	0.1255	0.1271	0.1256	0.0887	0.0882	0.0876	0.0888	0.0879	0.7912	0.7896	0.7918	0.7895	0.7891
	3D	0.2087	0.2082	0.2072	0.2075	0.2064	0.2121	0.2125	0.2096	0.2093	0.2096	0.7005	0.6996	0.7026	0.7011	0.7001
	5D	0.2639	0.2638	0.2621	0.2632	0.2625	0.3170	0.3182	0.3136	0.3147	0.3146	0.6463	0.6451	0.6503	0.6484	0.6492
	7D	0.3092	0.3086	0.3074	0.3082	0.3062	0.4142	0.4151	0.4117	0.4114	0.4097	0.6054	0.6042	0.6078	0.6066	0.6055
	Avg.	0.2272	0.2266	0.2255	0.2265	0.2252	0.2580	0.2585	0.2556	0.2560	0.2554	0.6859	0.6846	0.6881	0.6864	0.6860
NLinear	1D	0.1462	0.1482	0.1473	0.1464	0.1449	0.1111	0.1133	0.1123	0.1113	0.1097	0.7843	0.7816	0.7807	0.7811	0.7821
	3D	0.2220	0.2238	0.2227	0.2213	0.2205	0.2370	0.2397	0.2381	0.2358	0.2347	0.6857	0.6836	0.6850	0.6865	0.6876
	5D	0.2743	0.2756	0.2746	0.2744	0.2735	0.3439	0.3461	0.3443	0.3437	0.3423	0.6320	0.6300	0.6315	0.6317	0.6310
	7D	0.3186	0.3199	0.3191	0.3181	0.3173	0.4427	0.4450	0.4435	0.4415	0.4402	0.5875	0.5888	0.5904	0.5930	0.5932
	Avg.	0.2403	0.2419	0.2409	0.2400	0.2390	0.2836	0.2860	0.2845	0.2831	0.2817	0.6724	0.6710	0.6719	0.6731	0.6735
TimesNet	1D	0.1565	0.1582	0.1597	0.1655	0.1661	0.1357	0.1423	0.1392	0.1458	0.1492	0.7130	0.7099	0.7155	0.7093	0.7007
	3D	0.2423	0.2451	0.2408	0.2424	0.2454	0.2953	0.3075	0.2847	0.2893	0.3104	0.6148	0.6168	0.6174	0.6202	0.6131
	5D	0.2887	0.2935	0.3038	0.2924	0.2985	0.3997	0.4087	0.4531	0.4028	0.4239	0.5779	0.5812	0.5651	0.5824	0.5674
	7D	0.3272	0.3399	0.3380	0.3422	0.3413	0.4891	0.5223	0.5136	0.5219	0.5451	0.5447	0.5377	0.5359	0.5426	0.5410
	Avg.	0.2537	0.2592	0.2606	0.2606	0.2628	0.3299	0.3452	0.3476	0.3400	0.3571	0.6126	0.6114	0.6085	0.6136	0.6056

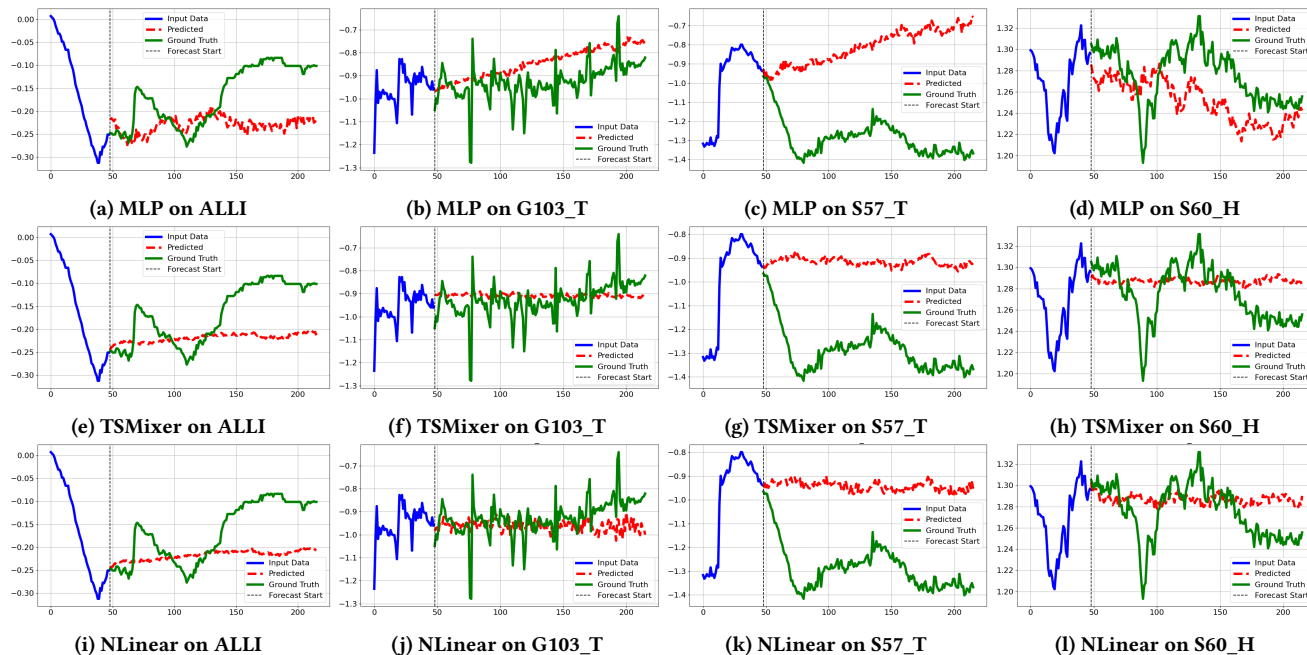


Figure 11: Cases visualization of MLP-based architecture methods on S7(Orlando area).

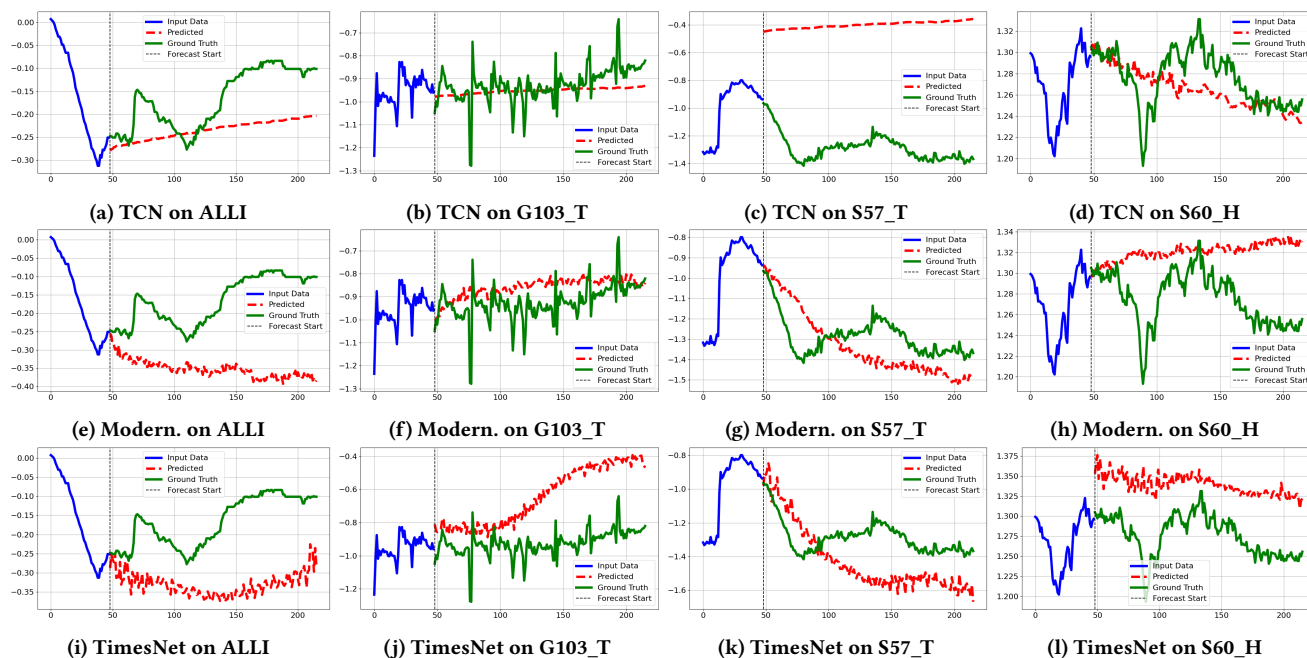


Figure 12: Cases visualization of CNN-based architecture methods on S7(Orlando area). Modern. means ModernTCN.

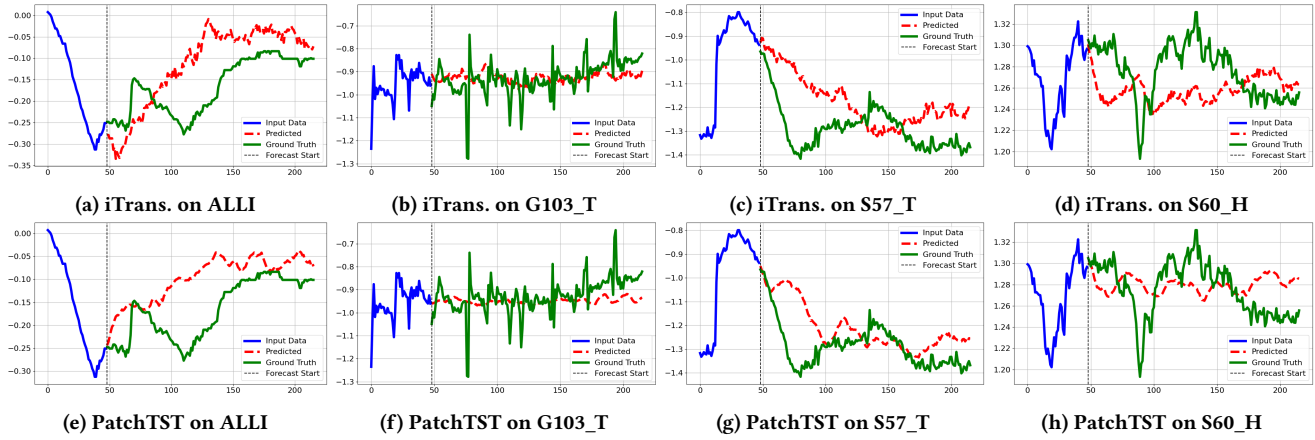


Figure 13: Cases visualization of Transformer-based architecture methods on S7(Orlando area). iTrans. means iTransformer.

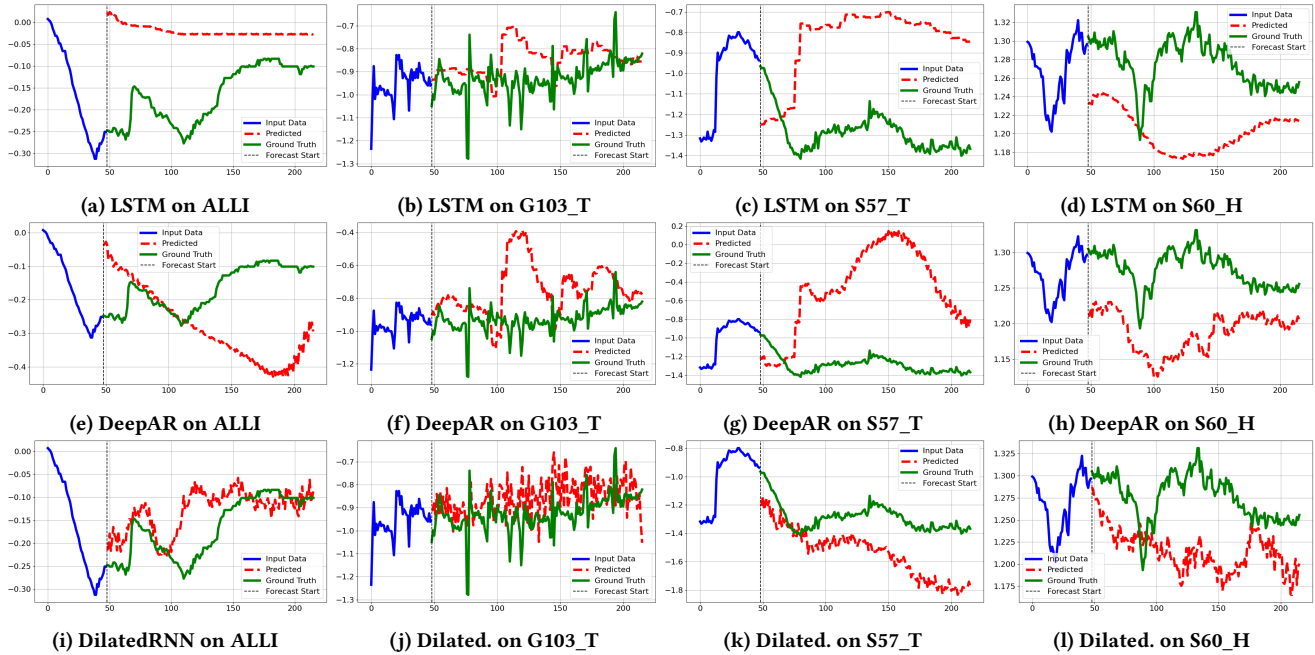


Figure 14: Cases visualization of RNN-based architecture methods on S7(Orlando area). Dilated. means DilatedRNN.

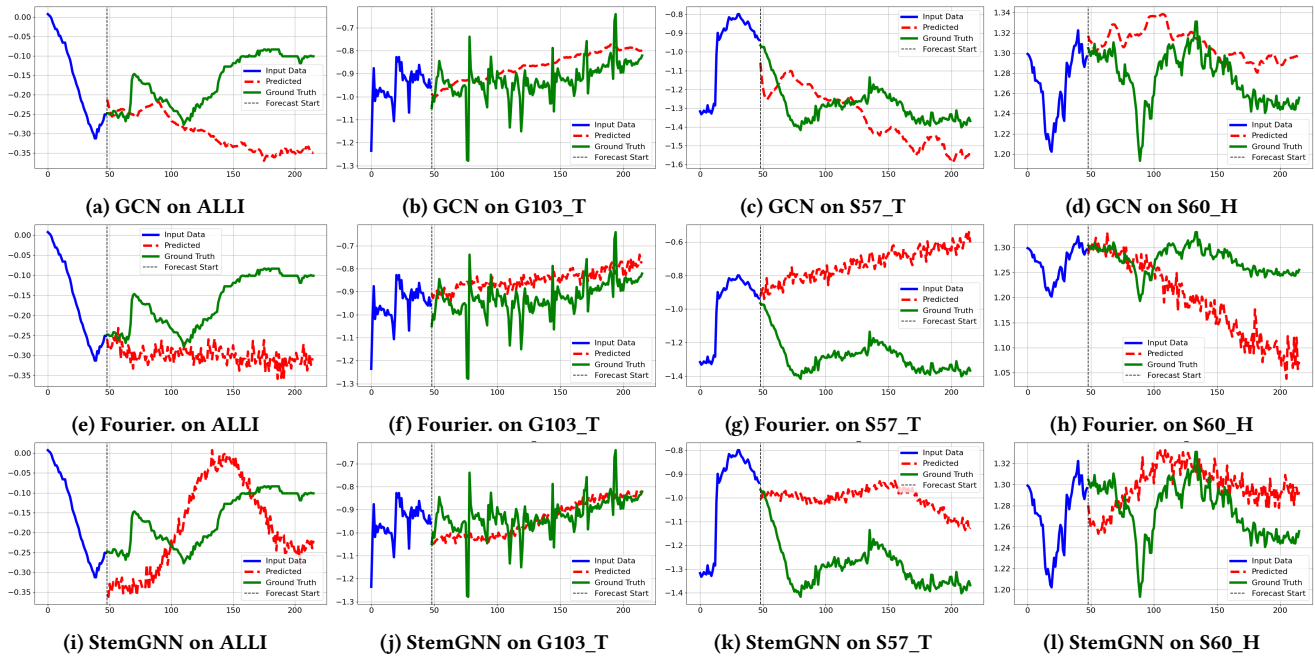


Figure 15: Cases visualization of GNN-based architecture methods on S7(Orlando area). Fourier. means FourierGNN.

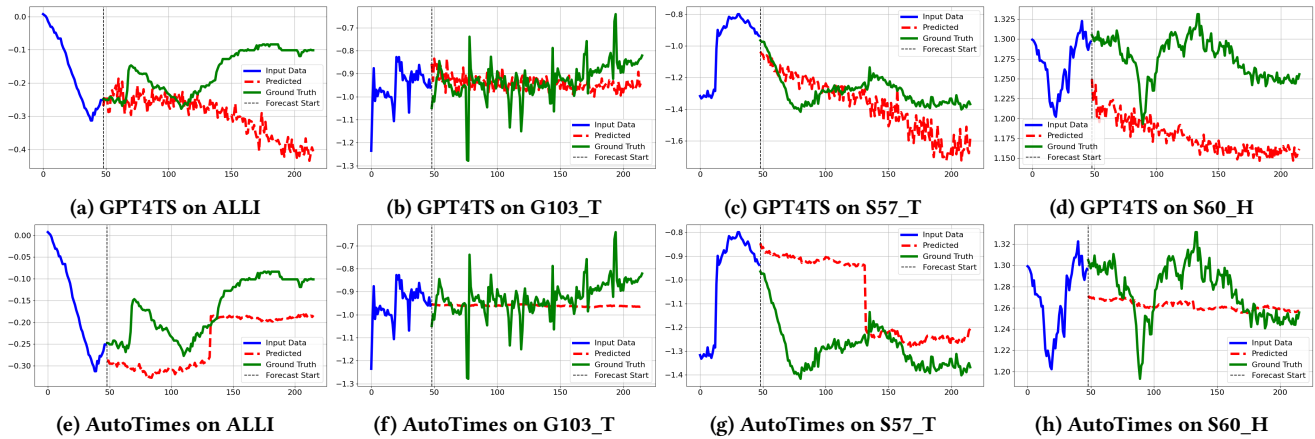


Figure 16: Cases visualization of LLM-based architecture methods on S7(Orlando area). Dilated. means DilatedRNN.